

SUSTAINABLE SEAS

Managing the marine environment



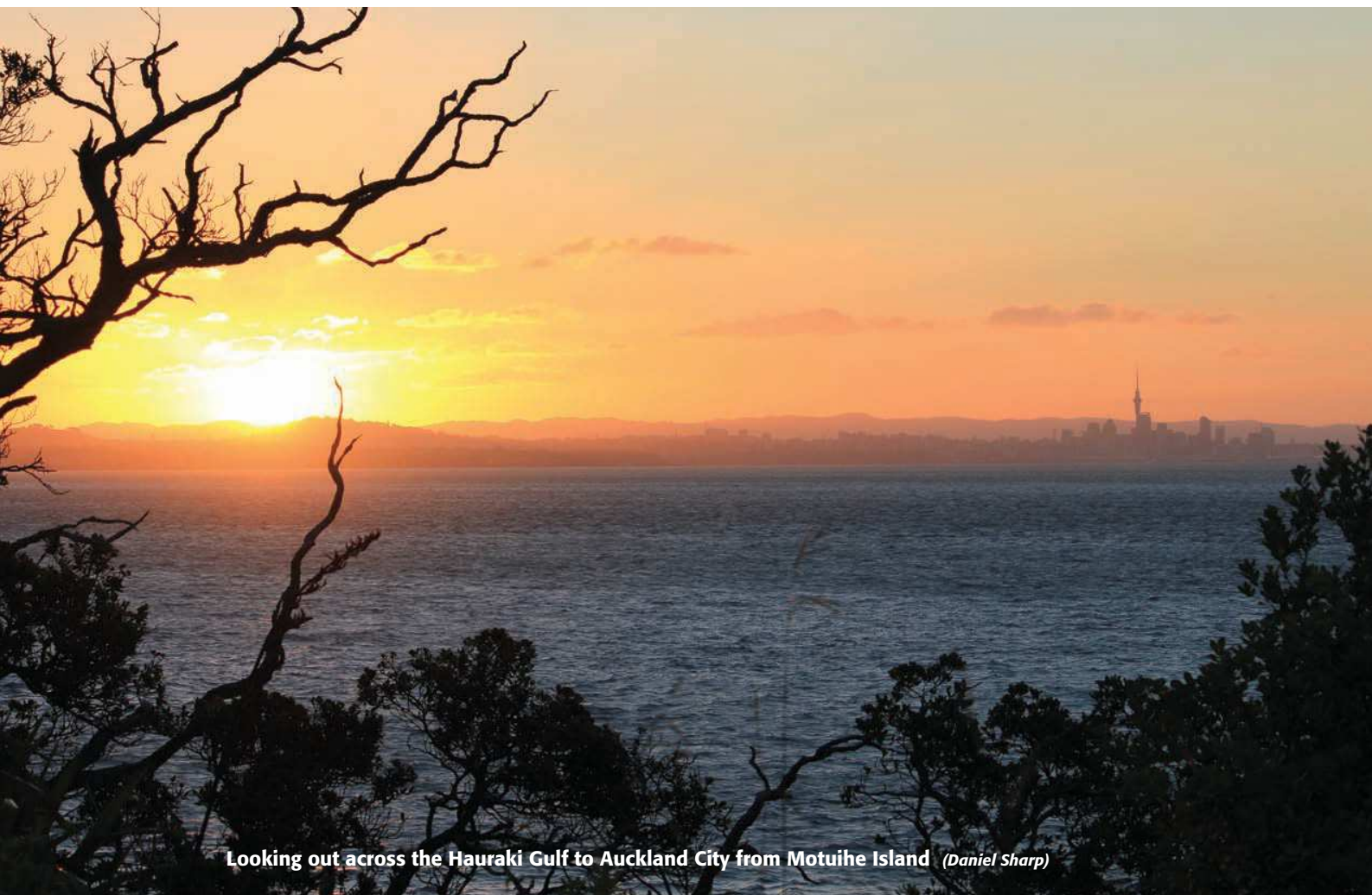
Lucy Brake and Raewyn Peart

eds
ENVIRONMENTAL DEFENCE SOCIETY

SUSTAINABLE SEAS

Managing the marine environment

Lucy Brake and Raewyn Peart



Looking out across the Hauraki Gulf to Auckland City from Motuihe Island *(Daniel Sharp)*

For Tyler and Zach

First published 2015

Published by:

Environmental Defence Society Incorporated

PO Box 91736 Victoria St West,

Auckland 1142, New Zealand

Phone +64 9 480 2565

manager@eds.org.nz

www.eds.org.nz

ISBN 978-0-9876660-5-5

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Artwork: Neale Wills – Wilsy Design & Production

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Foreword

I would like to congratulate the Environmental Defence Society on the latest in a series of first-class environmental guides that the Society publishes.

Sustainable Seas: Managing the Marine Environment is a comprehensive examination of New Zealand's vast coastal and oceanic environment. For a relatively small country, New Zealand has jurisdiction over a very large area of sea. More than 20,000 kilometres of coastline borders a marine area extending over 5.8 million square kilometres. This is fifteen times larger than our land mass. This huge area has a rich biodiversity ranging from the sub-tropical to the sub-antarctic and has as much as 80 per cent of the country's biodiversity.

New Zealand should aspire to be a world leader in ocean management. We have the advantage of being a developed nation with a low population and a vast ocean. We have a strong cultural connection with the sea in both our Māori and European heritage. We also have a strong marine science sector.

We can be proud of our leadership in developing a sound quota management system for fisheries and for being one of the first countries in the world to create no-take marine reserves, but we have more to do. Improving New Zealand's system for marine protection is an important and current priority for the Government.

I am therefore happy to acknowledge the extensive effort EDS has undertaken to explain some ways to apply best practice management to our oceans. This guide provides individuals, communities and users of the marine environment with knowledge and insight into decision-making processes, and how they can participate constructively and effectively in them, either as an applicant or a submitter.

I welcome EDS's effort to bring together environmental and development interests in oceans management and to promote the approach that any development within the marine area is well considered and undertaken in a way that minimises effects on our environment.

Hon Dr Nick Smith

Minister for the Environment



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List of abbreviations and acronyms

EEZ	Exclusive Economic Zone
EEZ Act	Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012
EEZ Amendment Act	Exclusive Economic Zone and Continental Shelf (Environmental Effects) Amendment Act 2013
EEZ Permitted Activities Regulations	Exclusive Economic Zone and Continental Shelf (Environmental Effects – Permitted Activities) Regulations 2013
EPA	Environmental Protection Authority
Heritage NZ	Heritage New Zealand Pouhere Taonga
NIWA	National Institute for Water and Atmospheric Research
NZCPS	New Zealand Coastal Policy Statement 2010
MARPOL	International Convention for the Prevention of Pollution from Ships
PADI	Professional Association of Diving Instructors
QMS	Quota Management System
RMA	Resource Management Act 1991
SOPAC	Secretariat of the Pacific Community
UNCLOS	United Nations Convention on the Law of the Sea

Section One

Overview of marine management in New Zealand



John Dory, Alderman Islands (Daniel Sharp)

New Zealand's marine environment

'The sea, the great unifier, is man's only hope. Now, as never before, the old phrase has a literal meaning: we are all in the same boat.'

JACQUES YVES COUSTEAU

Snapper and demoiselle, Poor Knights Islands, Northland (Daniel Sharp)

Introduction

Oceans are enormously important to human lives. Water covers more than 70 per cent, or just over 365 million square kilometres, of the globe. The Pacific Ocean alone is 25 per cent larger than all the world's land areas combined. By volume, oceans account for more than 99 per cent of all living space on earth. All water on earth is ultimately connected and interlinked. The oceans drive our climate and rainfall, replenishing lakes and rivers, which in turn drain back into the sea.¹

New Zealand is fortunate to have a very extensive and diverse marine area. It is home to numerous plant, invertebrate, fish, marine mammal and seabird species. The identity of many of these species, and the interactions between them, are still not fully understood. This makes the effective management of the marine environment a considerable challenge.

Physical marine area

Size

For a relatively small country, New Zealand has a long coastline, and jurisdiction over a very large area of sea. There is approximately 19,000 kilometres of coastline for the mainland, increasing to 20,500 kilometres when the offshore islands and the Chatham Islands are included. This coastline borders a marine area extending over 5.8 million square kilometres (see Figure 1.1). New Zealand's Exclusive Economic Zone (EEZ) is estimated at 4 million square kilometres and its Extended Continental Shelf covers around 1.7 million square kilometres. The EEZ is 15 times larger than the country's land area, and the EEZ combined with the continental shelf is 21 times as large.²

But the marine environment is considerably larger than this two-dimensional figure indicates. It also has a substantial third dimension, extending down from the surface of the sea, to the seabed. Although the sea is relatively shallow in coastal areas, where the land abuts the continental shelf, 75 per cent of New Zealand's marine area has depths of more than 1 kilometre. The deepest place is within the Kermadec Trench, some 10 kilometres from the surface.

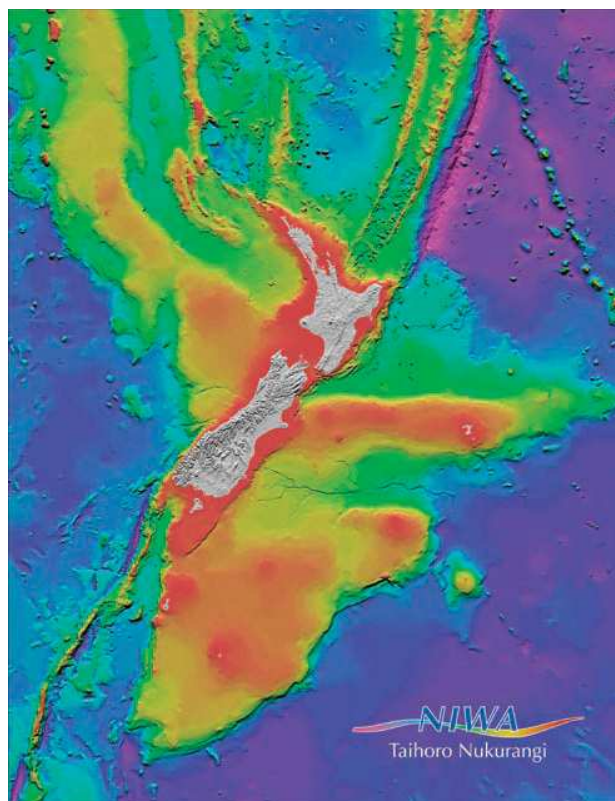


Figure 1.1 Topography of New Zealand's marine area (NIWA)

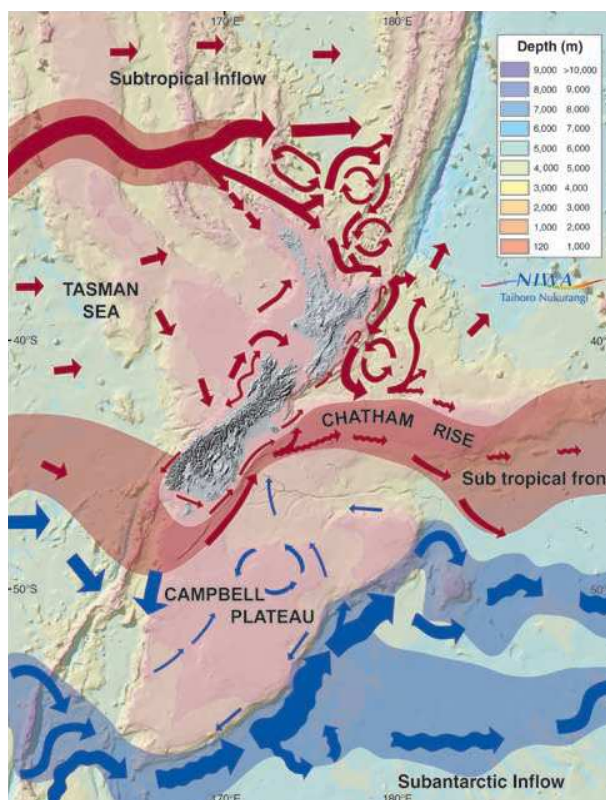


Figure 1.2 Ocean currents around New Zealand (NIWA)

marine habitats. New Zealand has around 300 estuarine systems interspersed along the coastline. They cover over 100,000 hectares, and provide a home for a wide range of species, including mangroves, seagrasses, saltmarsh plants, invertebrates, algae and phytoplankton.

There are many different habitat types within an estuary, such as unvegetated sand and mudflats, as well as vegetated habitats like mangrove and seagrass. They provide spawning and nursery areas for fish and shellfish species. In addition, estuaries are a critical habitat and food source for migratory wading birds, some which travel to New Zealand from as far afield as Alaska each year.

Rocky reefs

Rocky reefs are common in nearshore areas around New Zealand's coast. Intertidal reefs are rocky areas of the coastline which extend between spring low water mark and the spring high tide mark. They include a variety of habitats, differentiated by exposure and aspect, and host a wide range of organisms including algae, crustaceans, molluscs (shellfish) and polychaete worms (small worms).

The Hauraki Gulf, the northeast coast of Northland, the east coast of the North Island (near Gisborne), the southwest tip of the South Island and the west coast of the Chatham Islands are all important areas for intertidal rocky reefs. Shallow subtidal rocky reefs, to a maximum depth of 50 metres, are found around many parts of the New Zealand coastline. Fiordland, East Cape and the Chatham Islands stand out as being particularly important areas for subtidal rocky reef habitat.⁶

Many reefs are covered in rich algal forests, which can contain several large kelp species and smaller brown, red and green seaweeds. These provide important habitats for a wide range of sea creatures, including over 2,000 species of invertebrates, which are permanently attached to the rocks. Many of these species are only found in New Zealand.

The presence of lush seaweed beds on rocky reefs is generally associated with high biodiversity because the beds provide food and refuge for other small organisms. These creatures, which include crustaceans, molluscs and polychaetes, primarily live amongst kelp. They provide an important trophic (the feeding position of an organism in the food chain) link between the seaweed, as a primary producer, and predatory fishes, nearly all of which feed on the small rocky reef animals as juveniles.⁷



Intertidal rocky reefs, such as the one shown here at Pig Bay, Motutapu Island, are important habitats in the Hauraki Gulf (Raewyn Peart)



Marine algal forests, such as the one shown here at Tūhūa (Mayor Island) in the Bay of Plenty, are important habitats for a wide range of sea creatures (Daniel Sharp)

Fiords

A fiord is a steep-sided valley that has been carved out by glaciers and then flooded by the sea. The freshwater input comes from the surrounding landscapes, via waterfalls and runoff from the mountains and native forests. In New Zealand fiords, the copious rainwater leaches through the rich humus of the beech forest and results in a yellow-coloured layer of freshwater overlaying the denser seawater. This reduces the light levels within the fiords and has resulted in normally deep water species living closer to the surface.

Fiordland has 14 fiords of varying depths reaching up to over 400 metres. The unique climate, vegetation and topography in this area has resulted in some specialised underwater habitats. Species living within the fiords are tolerant to the freshwater layer and reduced light levels.¹³ Spectacular large tree-like black corals and sea pens, normally only present in deepwater, can be found growing there.

Seamounts

Being enormous underwater structures (usually defined as over 1,000 metres high), seamounts in the deep waters of New Zealand's EEZ are of considerable scientific interest, often hosting unusual or unique groups of organisms and a biodiversity disproportionate to their size and area.¹⁴ These environments are not only widely recognised as areas of high productivity, but are also regarded as a fragile habitat because of their poor ability to recover if damaged.

Seamounts can support a very diverse range of life. They provide a hard stable surface for sessile plankton feeders to attach to, and act as oases within large plains of seabed covered in low-lying soft mud. Animals attached to the seamounts receive their primary energy supply from nutrient-rich water, which wells up around the flanks of the large structures as they intercept the ocean currents. In addition, tiny invertebrate prey (zooplankton) which are transported along in the currents become trapped on the seamounts. The seamount communities are also fed by detritus and faecal pellets drifting down from organisms which live closer to the surface. The high productivity of these areas attracts large congregations of fish, which in turn attract other species to the surrounding waters, such as sperm whales and sea birds.¹⁵

Seamount ecosystems are complex and varied. Many harbour their own unique endemic species. Since the mid-1990s, National Institute of Water and Atmospheric Research (NIWA) scientists have studied a variety of seamount habitats in New Zealand, including those of the Chatham Rise and the southern Kermadec volcanic arc. They found species and taxonomic diversity to be high. A 2004 study of Northland Plateau seamounts recorded 396 species of macro-invertebrates on two seamounts. At least 17 per cent of the species (Bryozoa alone) recovered by the survey are currently undescribed for the New Zealand region, including six genera entirely new to science.¹⁶

Hydrothermal vents

Hydrothermal vents are commonly found near volcanically active places such as areas where tectonic plates are moving apart. As hydrothermal fluids pour out of the vent and react with cold, oxygenated seawater, a number of rapid chemical reactions take place. These ultimately form metal-rich chimneys which provide an important environmental niche for deep-sea marine life.



The unique marine habitats in Fiordland enable normally deepwater species, such as the black coral shown here (appearing white), to grow in shallow waters (Raewyn Peart)

Algae and seagrasses

Algae are an ancient group of primitive plants that support major marine food webs. In contrast, seagrasses are flowering plants (angiosperms) whose ancestors once lived on land. One prominent and important temperate zone seagrass is eelgrass which provides excellent habitat for grazing animals. These fast-growing plants form the base of a rich and productive food web.³² In New Zealand there is one main species of seagrass (*Zostera capricorni*) which primarily grows in intertidal areas but which can also grow in areas permanently submerged, where the water is sheltered and clear.



Subtidal seagrass beds, such as the one shown here at Slipper Island in the Bay of Plenty, provide important habitat for juvenile fish (Daniel Sharp)

The distribution of algae is generally clearly zoned within the marine environment. Yellow, white and grey lichens (a combination of fungi and bacteria) all live within the zone which is both wet and dry. Green seaweeds are common in the upper intertidal zone, whilst brown algae are found closer to the low-water line. Red seaweeds are found below the brown-seaweed zone and the large kelps are found in the subtidal zone.

Invertebrates

This is a highly diverse group of species, including sponges, worms, shellfish and echinoderms. The sessile animals living on rocky reefs harvest plankton and organic particles from passing seawater. They include sponges, hydroids, corals, anemones, bryozoans, tube worms and barnacles.

Hydroids, jellyfish, corals and sea anemones belong to the group Cnidaria and are among the simplest of marine invertebrates. Jellyfish may consist of individual organisms or colonies of individuals, some of them specialised for tasks such as feeding, stinging and reproduction.³³ The name 'coral' is given to a variety of animals which grow in colonies and have skeletons made of calcium carbonate or horny material. The largest invertebrate seafloor species on the planet, the bubblegum coral, lives in New Zealand waters. True (or 'stony') corals may be solitary (cup corals, to which most New Zealand species belong) or colonial (reef-building corals). Their skeletons are made up of calcium carbonate and are external to the body. Black corals, of which there are 58 species in New Zealand waters, belong to a different group than the true corals. They have hard, protein-based skeletons with tiny polyps. Mostly they live in deep water, but in Fiordland, the black coral *Antipathella fiordensis* can live in relatively shallow depths.³⁴

Bryozoans are fascinating creatures, which have been variously referred to as lace corals, moss animals or sea mats. These animals form coral-like groupings on rocky and gravel areas, which in turn provide shelter for other invertebrates and juvenile fish. Nearly 1,000 different species of bryozoans are found within New Zealand's marine area and most of these are unique to the country.

In contrast to the sessile animals, other invertebrates move around the rocky reef to feed. These include herbivores, such as kina and top shells, which graze on seaweed, and carnivores, such as octopus, starfish, crayfish and crabs. The reef is also home to



New Zealand has a vast array of fascinating marine invertebrates including anemones seen at Great Mercury Island (left and right), jellyfish seen at Tūhua (Mayor Island) (centre) (Daniel Sharp)

Marine mammals

New Zealand's seas are a marine mammal 'hotspot'. It is estimated that just under half of the world's total number of dolphin and whale species are found in New Zealand. There are five different species of dolphin that live year round in New Zealand's coastal waters, including the common dolphin, dusky dolphin, bottlenose dolphin, Hector's dolphin and orca. There are a few other species living in offshore waters. New Zealand is home to one of the smallest dolphins in the world, the Hector's dolphin. This species is threatened with just over 7,000 remaining. The Maui's dolphin, which is a subspecies of the Hector's, is only found along the west coast of the North Island. There are as few as 55 adults of this critically endangered dolphin remaining.

Species of whale, including the humpback and blue whale, migrate through New Zealand waters. Other species such as the Bryde's whale and southern right whale breed here. A small population of around 40 to 50 Bryde's whales resides in the Hauraki Gulf year round, with a further 150 visiting seasonally. The species is classified as nationally critical in New Zealand, although there are secure populations overseas.

New Zealand fur seals and sea lions are part of the other main group of marine mammals found in New Zealand's waters, the pinnipeds. It is estimated that up to two million fur seals lived around the New Zealand coastline before humans arrived. Their population was decimated by hunting but is now recovering. The endemic and threatened New Zealand sea lion breeds in the Auckland Islands and can be seen around southern coasts, including the Catlins and Otago Peninsula. Southern elephant seals and leopard seals are occasional visitors to New Zealand beaches, travelling up from subantarctic and antarctic waters.

Seabirds

New Zealand has been referred to as the 'seabird capital of the world'. This is because more species of seabirds breed here than in any other country. There are thought to be more species of seabirds in New Zealand than land birds. Thirty-five species of seabird only breed in New Zealand.

New Zealand's seabirds range from the large southern royal albatross, which can have a wingspan of over 3 metres and which spends most of its time at sea, to the small and critically endangered New Zealand fairy tern, which nests on Auckland and Northland beaches.

Seven species of penguin can be found in New Zealand, with four of these breeding on the mainland (yellow-eyed, white-flippered, Fiordland-crested and blue). The blue penguin is the smallest penguin species in the world and can be found along the coasts of the North and South Islands. The largest penguin in New Zealand is the distinctive yellow-eyed penguin, which nests along the south-east coast of the South Island, and on subantarctic islands further south.



New Zealand has a rich variety of marine mammals including the bottlenose dolphin (top), New Zealand sea lion (centre) and orca (bottom) (Raewyn Peart)



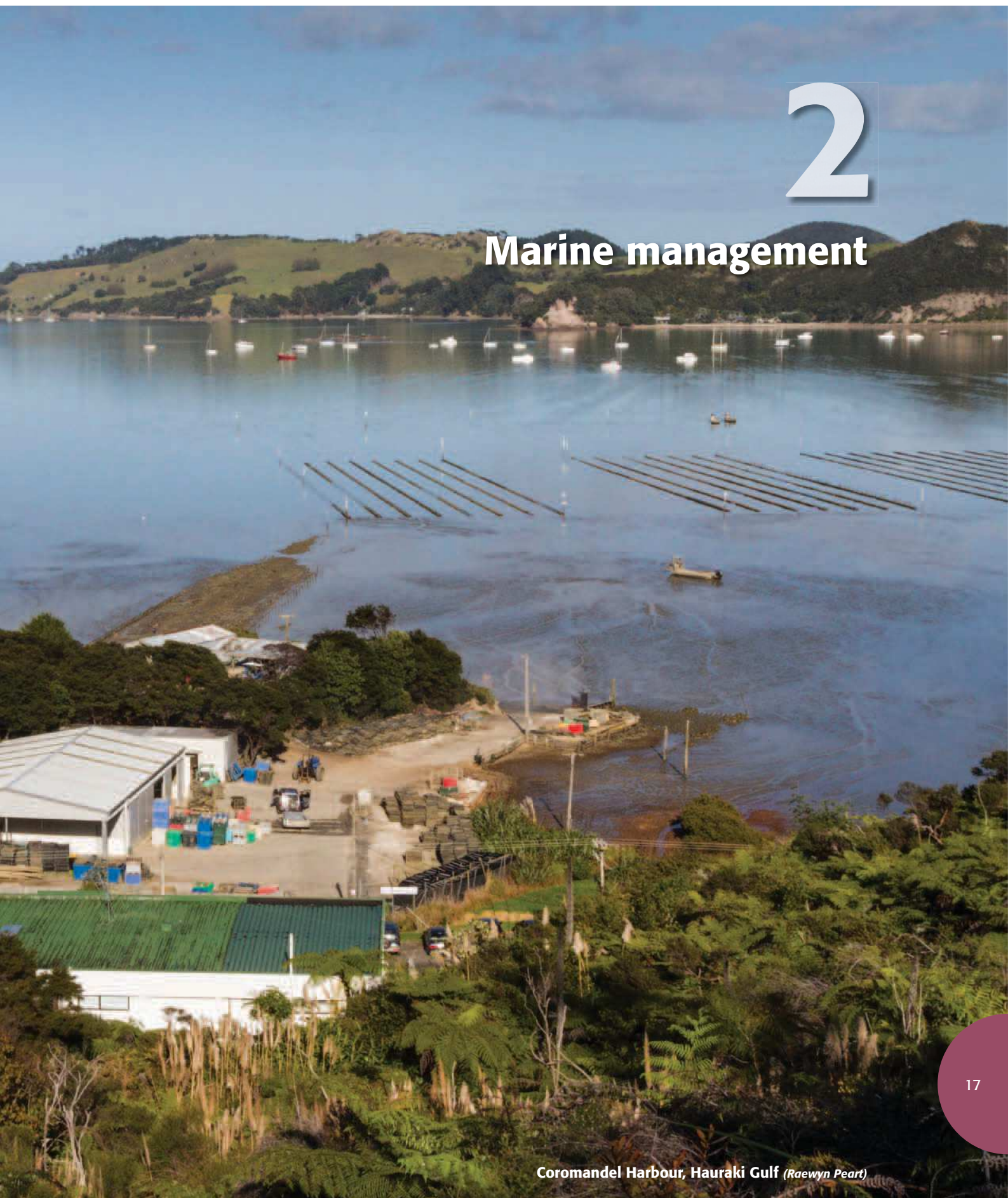
New Zealand is thought to have more species of seabird than land bird, including the flesh-footed shearwater shown here near the Mokohinau Islands in the Hauraki Gulf (Raewyn Peart)

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2

Marine management



Introduction

New Zealand's marine environment is managed within a complex and overlapping legislative framework which has developed over the past 50 or so years. This chapter provides an overview of the agencies currently involved in managing the marine environment and the legislation which is used to guide their activities.

Maritime areas

Management of the sea is governed internationally by multilateral treaties, the foremost being the United Nations Convention on the Law of the Sea (UNCLOS). Under UNCLOS, New Zealand exercises different jurisdictional powers over four defined maritime areas: the territorial sea, the contiguous zone, EEZ and the continental shelf – see Figure 2.1.

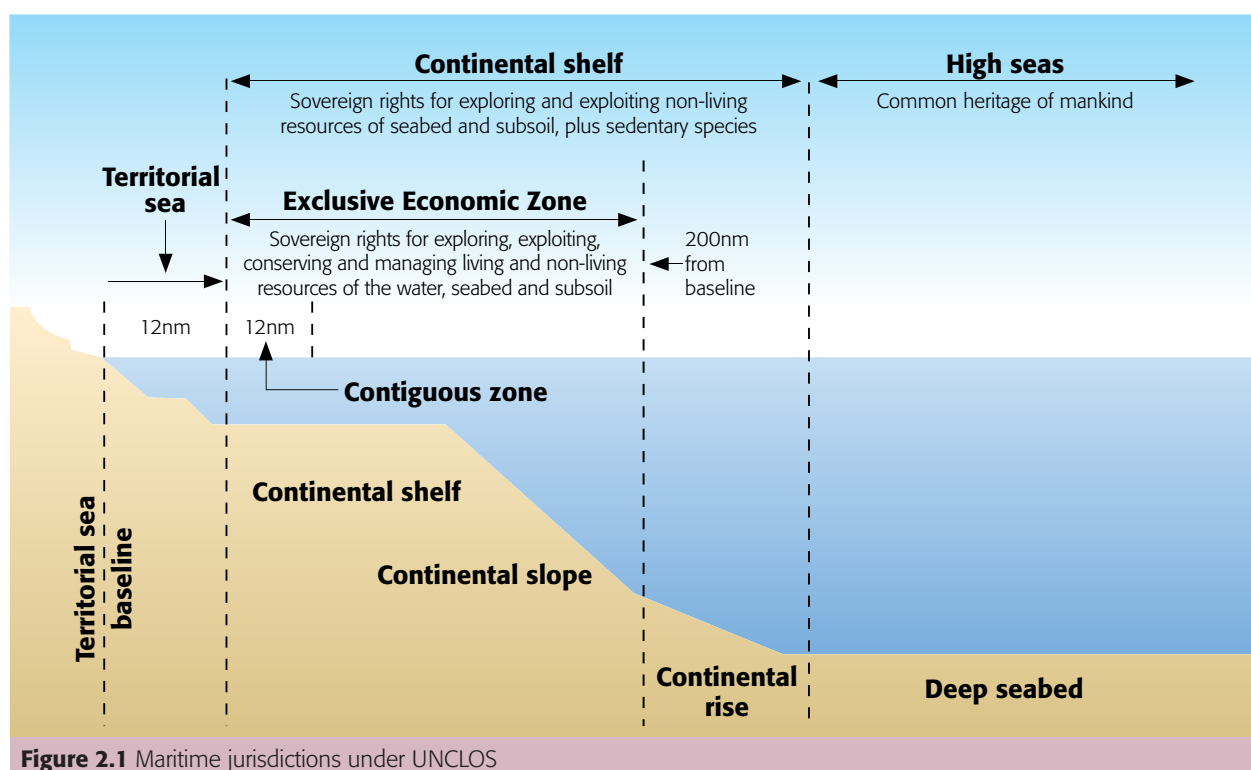


Figure 2.1 Maritime jurisdictions under UNCLOS

The territorial sea is part of New Zealand's sovereign territory. This means that the government's power to govern this marine area is the same as on land, although other states have rights established under international law, such as 'innocent passage' of their vessels. The territorial sea comprises the marine area within 12 nautical miles of low-water mark. In New Zealand, it encompasses 167,650 square kilometres.¹

The contiguous zone lies on the seaward side of the territorial sea, extending from 12 to 24 nautical miles from the low-water mark. It overlaps with the EEZ. This is not part of New Zealand's territory. But under UNCLOS, in addition to the rights New Zealand has in this area as part of the EEZ, it has the right to police the area to prevent or punish infringements (of customs, fiscal, immigration or sanitary laws and regulations) in its territory or territorial sea.

The EEZ is the largest of New Zealand's marine areas, covering an area which extends from the outer edge of the territorial sea to 200 nautical miles from the low-water mark. In some places, the EEZ extends over the edge of the continental shelf into deep ocean areas.

The EEZ is not part of New Zealand's territory, but under UNCLOS, New Zealand has sovereign rights for the purpose of exploring, exploiting, conserving and managing all of its living and non-living natural resources. These resources include marine life, oil, gas

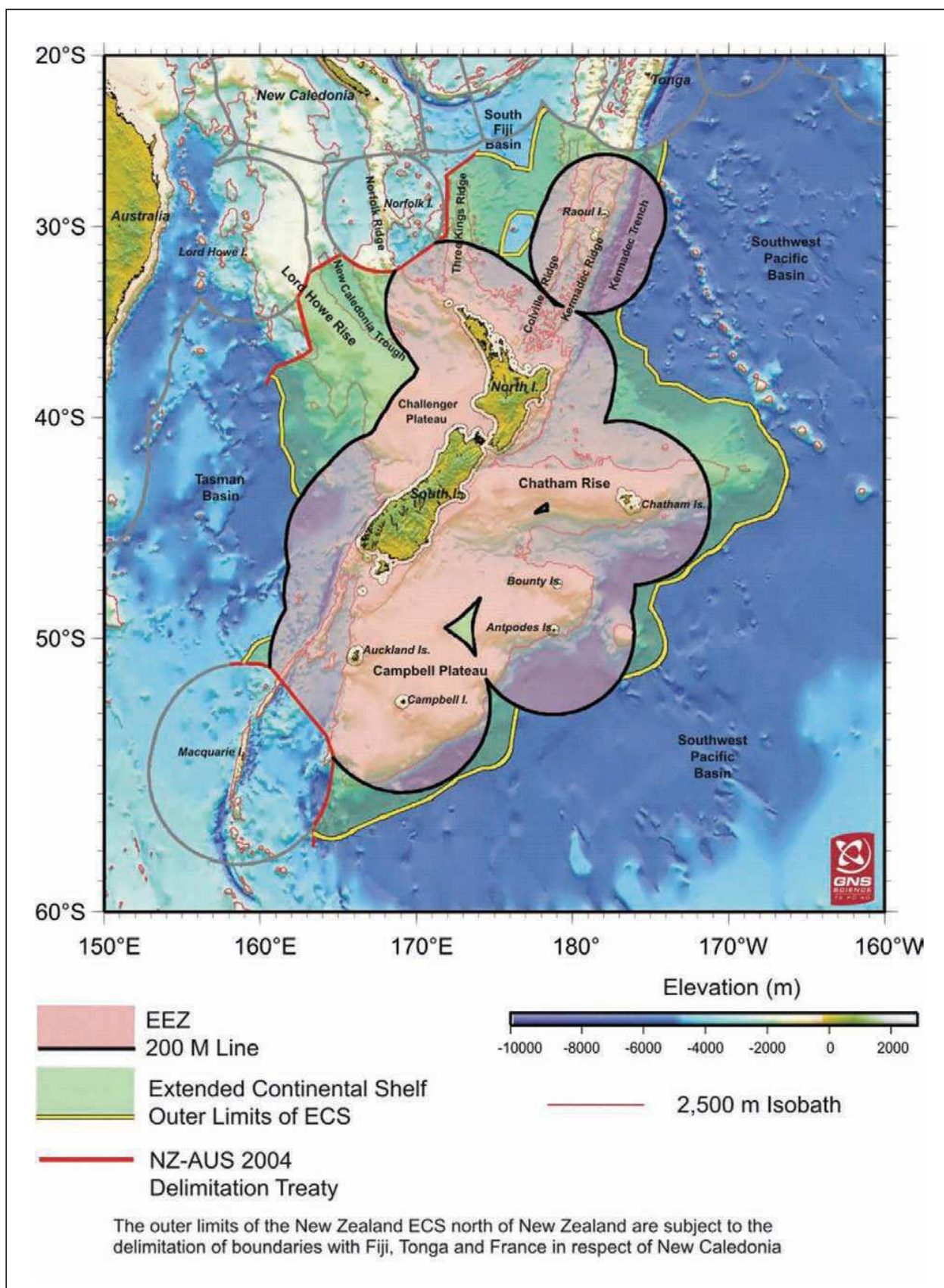


Figure 2.2 New Zealand's maritime areas (Land Information New Zealand)⁴

Agency/organisation	Areas of management focus	Key legislation
Territorial authorities	Land use and subdivision	Resource Management Act 1991 Local Government Act 2002 Reserves Act 1977
Heritage New Zealand Pouhere Taonga	Protection of historic heritage places	Heritage New Zealand Pouhere Taonga Act 2014
Iwi and hapū	All coastal and marine resources	All the above legislation Co-management agreements

Figure 2.3 New Zealand coastal and marine management agencies

Minister and Department of Conservation

The Department of Conservation was established by the Conservation Act 1987. Its prime role is to manage Crown-owned conservation land and to protect and rehabilitate endangered indigenous species and habitats. The Department is also tasked with advocating for the conservation of natural and historic resources generally, amongst other things. The definition of ‘natural resources’ in the Conservation Act is very wide and includes living organisms, habitats, ecosystems, landscapes, landform and geological features. The New Zealand Conservation Authority provides oversight of the Department’s work and gives advice to the Minister and the Director-General of Conservation.

The Minister and Department of Conservation play a significant role in the protection of marine habitats and species. The Minister approves the establishment of marine reserves (with the concurrence of the Ministers for Primary Industries and of Transport) under the Marine Reserves Act 1971 and marine mammal sanctuaries under the Marine Mammals Protection Act 1978. Once created, these areas are managed by the Department of Conservation. The Department administers the Wildlife Act 1953 which provides protection for seabirds and some marine species. The Department is also involved, in conjunction with the Ministry for Primary Industries, in managing the impacts of fishing activity on protected marine species such as dolphins, sea lions, seals and seabirds.

The Minister of Conservation has a specific role in the management of coastal land and the territorial sea under the Resource Management Act 1991 (RMA). The Minister developed the New Zealand Coastal Policy Statement 2010 (NZCPS) which guides, and in some cases directs, the planning and decision-making functions of councils within the ‘coastal environment’. This includes



The Department of Conservation plays a significant role in the protection of marine habitats and species including activity within marine reserves such as water taxis (Raewyn Peart)

In the marine area, the Minister is responsible for recommending regulations promulgated under the Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012 (EEZ Act) to the Governor-General.

The Ministry also has responsibilities for environmental reporting. The Environmental Reporting Bill, expected to be passed in 2015, requires the Ministry for the Environment and Statistics New Zealand to publish a domain report on one of five environmental domains (air, atmosphere and climate, freshwater, land and marine) every six months and a synthesis report (providing an analysis of cross-domain trends and interactions) every three years. The Bill does not require new information to be generated but relies on collation of existing data.

Environmental Protection Authority

The Environmental Protection Authority (EPA) is a crown entity established in 2011 under the Environmental Protection Authority Act 2011. It is governed by a Board whose members are appointed by the Minister for the Environment. The EPA is responsible for processing consents for matters of national significance under the RMA, the management of the Emissions Trading Scheme and the regulation of hazardous substances and new organisms. The EPA also has a key role in the marine area under the EEZ Act as the agency responsible for determining marine consent applications.

Maritime New Zealand

Maritime New Zealand is governed by an independent board appointed by the Governor-General on the recommendation of the Minister of Transport. It manages shipping and marine pollution under the Maritime Transport Act 1994. The management of waste dumping and discharge from ships and offshore installations in the EEZ and continental shelf, which was previously undertaken by Maritime New Zealand, has recently been transferred to the EPA.

Maritime New Zealand develops, on behalf of the Minister of Transport, and enforces marine protection rules aimed at preventing pollution incidents. It maintains the country's marine oil spill response strategy. It also supports New Zealand's engagement with the International Maritime Organization.



Maritime New Zealand is responsible for managing shipping within New Zealand waters, as shown here at the Port of Auckland (Raewyn Peart)

Regional councils

Regional councils are democratically elected bodies constituted under the Local Government Act 2002. There are 11 regional councils throughout the country whose boundaries broadly coincide with water catchment areas. In addition, there are six unitary authorities, which carry out the role of both regional and territorial authority.⁷

Regional councils have a broad role in managing the marine environment through their jurisdiction under the RMA over both catchments and activities in the coastal marine area (the area from mean high water springs to the edge of the territorial sea). The scope of the RMA is broad and it encompasses the environmental impacts of all activities except where they are specifically excluded. The main exclusion in the marine area is the taking, allocation or enhancement of fisheries resources for the purpose of managing fishing or fisheries resources controlled under the Fisheries Act.⁸ Council management of the marine environment under the RMA is overseen by the Minister of Conservation.

Regional councils prepare regional policy statements and plans under the RMA, which can have a major impact on the way the marine environment is managed. These include regional policy statements, which provide an overview of regionally significant

Marine management legislation

The key pieces of legislation relating to the marine environment are shown in Figure 2.4 and described in the following sections.

Management area	Legislation	Spatial area	Managing authorities
Resource management	Resource Management Act 1991 (including New Zealand Coastal Policy Statement 2010)	Catchments, islands and territorial sea	Minister of Conservation Regional councils Territorial authorities
	Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012	Exclusive economic zone and extended continental shelf	Minister for the Environment Environmental Protection Authority
	Hauraki Gulf Marine Park Act 2000	Hauraki Gulf territorial sea, islands and catchments	Hauraki Gulf Forum members
Fisheries	Fisheries Act 1996 Treaty of Waitangi (Fisheries Claims) Settlement Act 1992 Fisheries (Quota Operations Validation) Act 1997 Māori Fisheries Act 2004 Māori Commercial Aquaculture Claims Settlement Act 2004	Freshwater, territorial sea and exclusive economic zone	Minister for Primary Industries Supported by Ministry for Primary Industries
Mining	Crown Minerals Act 1991 Continental Shelf Act 1964	Land, territorial sea, exclusive economic zone and extended continental shelf	Minister of Energy and Resources Supported by Ministry of Business, Innovation and Employment through New Zealand Petroleum and Minerals
Marine pollution, except that managed by the EPA	Maritime Transport Act 1994	Territorial sea and exclusive economic zone	Maritime New Zealand
Biosecurity	Biosecurity Act 1993	Land and territorial sea	Minister for Primary Industries Supported by Ministry for Primary Industries
Marine protection	Marine Reserves Act 1971 Marine Mammals Protection Act 1978 Wildlife Act 1952	Territorial sea Territorial sea and exclusive economic zone Land, territorial sea and exclusive economic zone	Minister of Conservation Supported by Department of Conservation
	Fiordland (Te Moana o Atawhenua) Marine Management Act 2005	Fiordland (Te Moana o Atawhenua) Marine Area	Fiordland Marine Guardians (advisory)
Historic heritage	Heritage New Zealand Pouhere Taonga Act 2014	Land and territorial sea	Heritage New Zealand Pouhere Taonga

The Māori-related provisions in Part 2 of the RMA, which have been utilised extensively, are section 6(e) which relates to the relationship of Māori with the marine area; section 7(a) which requires decision-makers to have particular regard to kaitiakitanga; and section 8 which requires decision-makers to take into account the principles of the Treaty of Waitangi.

Management of the coastal environment at a regional and local level is carried out within the framework established by the NZCPS and other policy and planning documents within a planning hierarchy established under the RMA. The NZCPS is discussed further below. The National Policy Statement for Freshwater Management 2014 (Freshwater NPS) is discussed in Chapter 17: Catchment-based activities.

Regional policy statements, regional plans and district plans must give effect to the policies contained within the NZCPS. The phrase 'give effect to' means 'implement'. It is a strong directive that creates a firm obligation to take action. The NZCPS is more than a '[list] of potentially relevant considerations, which will have varying weight in different fact situations' and its specific and unqualified policies (such as Policies 11, 13, and 15) 'provide environmental bottom lines'.¹⁰

Regional policy statements provide an overview of resource management issues throughout the region and identify policies and methods to achieve the integrated management of natural and physical resources. They cover all the catchments and the territorial sea within a council's jurisdiction, so can embrace a mountains-to-the-sea management approach.

Regional coastal plans apply to activities in the marine area below mean high water springs. These usually include the identification and 'zoning' of some marine areas for protection as well as for specific activities such as ports, wharves, marinas, moorings and marine farms. Other activities not specifically provided for are usually assessed against a set of criteria that are laid out in the plan.

Several regional councils have prepared regional coastal environment plans that extend over the land and water portions of the coast. They typically contain objectives and policies which apply to the whole area, but the rules apply only to the coastal marine area, leaving district councils to develop rules to control land-based development within the guidance provided by the regional plan.

Regional councils may prepare other regional plans to assist them to carry out their functions under the RMA. These functions include the management of soil conservation, air quality, water quality and quantity, and biodiversity. The management of soil and freshwater, in particular, can have significant effects on the marine environment which is the ultimate receiving environment for most discharges from land.

Territorial authorities are charged with controlling the 'effects of the use, development or protection of land', including effects within the land portion of the coastal environment but not in the marine area. Provisions in the plans control activities such as subdivision and the building of houses and structures. Rules may determine the location, density and design of development that is likely to be permitted within the catchment and on the coast. They may also control activities such as vegetation clearance and earthworks which can generate sediment which ends up in the marine area.

The objectives, policies and rules in regional and district plans determine whether or not a consent is required before an activity can commence and the criteria that will apply when considering resource consent applications. Unlike the case with the use of land, most activities within the coastal marine area cannot be undertaken unless expressly authorised by a provision of a regional coastal plan or a resource consent.



The Resource Management Act controls most activities within catchments and the territorial sea including those shown in this image of Ngunguru in Whangarei District (Raewyn Peart)

NZCPS, Policy 11: Indigenous biological diversity (biodiversity)

To protect indigenous biological diversity in the coastal environment:

- a. avoid adverse effects of activities on:
 - i. indigenous taxa⁴ that are listed as threatened⁵ or at risk in the New Zealand Threat Classification System lists;
 - ii. taxa that are listed by the International Union for Conservation of Nature and Natural Resources as threatened;
 - iii. indigenous ecosystems and vegetation types that are threatened in the coastal environment, or are naturally rare⁶;
 - iv. habitats of indigenous species where the species are at the limit of their natural range, or are naturally rare;
 - v. areas containing nationally significant examples of indigenous community types; and
 - vi. areas set aside for full or partial protection of indigenous biological diversity under other legislation; and
- b. avoid significant adverse effects and avoid, remedy or mitigate other adverse effects of activities on:
 - i. areas of predominantly indigenous vegetation in the coastal environment;
 - ii. habitats in the coastal environment that are important during the vulnerable life stages of indigenous species;
 - iii. indigenous ecosystems and habitats that are only found in the coastal environment and are particularly vulnerable to modification, including estuaries, lagoons, coastal wetlands, dunelands, intertidal zones, rocky reef systems, eelgrass and saltmarsh;
 - iv. habitats of indigenous species in the coastal environment that are important for recreational, commercial, traditional or cultural purposes;
 - v. habitats, including areas and routes, important to migratory species; and
 - vi. ecological corridors, and areas important for linking or maintaining biological values identified under this policy.

⁴ Taxa – as defined in the Glossary.

⁵ Examples of taxa listed as threatened are – Māui dolphin, Hector's dolphin, New Zealand fairy tern, Southern New Zealand dotterel.

⁶ Naturally rare – as defined in the Glossary.



Hector's dolphins, such as the one shown here in Akaroa Harbour, Banks Peninsula, are specifically protected under the NZCPS and councils are required to avoid the adverse effects of activities on them (Raewyn Peart)



Poorly performing septic tanks can pollute the marine environment, and did so at Kawakawa Bay, Auckland (Raewyn Peart)

Natural character

Objective 2 of the NZCPS sets out a vision for the preservation of the natural character of New Zealand's coast. It recognises the 'characteristics and qualities' that contribute to natural character and the importance of identifying areas where development is inappropriate and where these characteristics and qualities should be protected. It also encourages restoration of the coastal environment. Preserving natural character requires maintaining natural processes, elements and patterns. In addition, it requires the integrity, functioning and resilience of the coastal environment to be preserved.

NZCPS, Objective 2

To preserve the natural character of the coastal environment and protect natural features and landscape values through:

- recognising the characteristics and qualities that contribute to natural character, natural features and landscape values and their location and distribution;
- identifying those areas where various forms of subdivision, use, and development would be inappropriate and protecting them from such activities; and
- encouraging restoration of the coastal environment.

There are a number of policies which set out how natural character should be preserved, for example effective catchment management will be required to effectively address the discharge of nutrients and sediment from land which can significantly reduce the natural character of the coastal marine environment. Policy 13(1)(a) includes the need to 'avoid adverse effects of activities on natural character in areas of the coastal environment with outstanding natural character' and Policy 13(1)(b) includes the need to 'avoid significant adverse effects and avoid, remedy or mitigate other adverse effects of activities on natural character in all other areas of the coastal environment'. This recognises that many activities in the marine environment can have an impact on natural character, including aquaculture, structures and mining. Policy 14 sets out the ways that restoration or rehabilitation of the natural character of the coastal environment should be achieved.

NZCPS, Policy 14: Restoration of natural character

Promote restoration or rehabilitation of the natural character of the coastal environment, including by:

- a. identifying areas and opportunities for restoration or rehabilitation;
- b. providing policies, rules and other methods directed at restoration or rehabilitation in regional policy statements, and plans;
- c. where practicable, imposing or reviewing restoration or rehabilitation conditions on resource consents and designations, including for the continuation of activities; and recognising that where degraded areas of the coastal environment require restoration or rehabilitation, possible approaches include:
 - i. restoring indigenous habitats and ecosystems, using local genetic stock where practicable; or
 - ii. encouraging natural regeneration of indigenous species, recognising the need for effective weed and animal pest management; or
 - iii. creating or enhancing habitat for indigenous species; or
 - iv. rehabilitating dunes and other natural coastal features or processes, including saline wetlands and intertidal saltmarsh; or
 - v. restoring and protecting riparian and intertidal margins; or
 - vi. reducing or eliminating discharges of contaminants; or
 - vii. removing redundant structures and materials that have been assessed to have minimal heritage or amenity values and when the removal is authorised by required permits, including an archaeological authority under the Historic Places Act 1993; or
 - viii. restoring cultural landscape features; or
 - ix. redesign of structures that interfere with ecosystem processes; or
 - x. decommissioning or restoring historic landfill and other contaminated sites which are, or have the potential to, leach material into the coastal marine area.



The NZCPS addresses the restoration of natural character including the decommissioning or restoring of historic landfills such as this one on the Kaiaua coast, Firth of Thames (Raewyn Peart)

Precautionary approach

Policy 3 of the NZCPS requires that a precautionary approach be taken when effects from proposed activities are uncertain, but could be significantly adverse. There is a particular focus on taking such an approach when dealing with resources that are vulnerable to climate change. This is a key approach that is recognised as part of the Rio Declaration on Environment and Development 1992. The Department of Conservation, through its document 'NZCPS 2010 Guidance Note Policy 3: Precautionary approach', has suggested that the application of the precautionary approach requires a risk management rather than risk assessment approach. It is when the risk of potential significant adverse or irreversible environmental effects cannot be adequately assessed (because of uncertainty about the nature and consequences of human activities or other processes) that a precautionary approach becomes appropriate.¹²



A precautionary approach is required to be taken under the NZCPS to the use and management of coastal resources potentially vulnerable to the effects of climate change, such as the dune systems shown here at Ocean Beach, Hastings (Raewyn Peart)

The implementation of Policy 3 requires a prudent avoidance approach to be taken where the individual or cumulative effects of an activity are uncertain. Adaptive management is one approach which can be applied in some cases of uncertainty but not all. It will not be appropriate, for example, where there is risk of irreversible change, or resources of very high value might be threatened. The New Zealand Biodiversity Strategy 2000 defines adaptive management to include 'structured learning by doing'.¹³ The Strategy notes that adaptive management is most useful when large complex ecological systems are being managed and management decisions cannot wait for final research results.

NZCPS, Policy 3: Precautionary approach

1. Adopt a precautionary approach towards proposed activities whose effects on the coastal environment are uncertain, unknown, or little understood, but potentially significantly adverse.
2. In particular, adopt a precautionary approach to use and management of coastal resources potentially vulnerable to effects from climate change, so that:
 - a. avoidable social and economic loss and harm to communities does not occur;
 - b. natural adjustments for coastal processes, natural defences, ecosystems, habitat and species are allowed to occur; and
 - c. the natural character, public access, amenity and other values of the coastal environment meet the needs of future generations.

Requirement to 'avoid' adverse effects

Where an area has outstanding natural character or is an outstanding natural landscape, Policies 13 and 15 require any adverse effects to be avoided. In addition Policy 11 requires adverse effects to be avoided on a range of species and habitats. In practice, this means that consent should not be granted for activities which impact on these species or values of these areas.¹⁴ This is supported by a recent Supreme Court decision on a salmon farming proposal, which determined that plan changes must give effect to Policies 13(1)(a) and 15(a) of the NZCPS and avoid adverse effects on outstanding natural landscapes.¹⁵ This case is discussed further in Chapter 10: Aquaculture.

Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act

The regulatory regime provided by the EEZ Act is designed to cover previously unregulated activities beyond the 12 nautical mile jurisdiction of the RMA. The EEZ Act was not intended to duplicate controls that regulate the fishing industry, maritime transport, the allocation of petroleum resources, and the response to oil spills.¹⁶ Most other activities that have an environmental impact on the EEZ or continental shelf are managed under the Act, including petroleum and minerals activities. The legislation is very new and at the time of writing there had been no appeal to a higher court to help interpret how the provisions should be applied.

The EEZ legislation, including examples, is discussed in detail in Chapter 3: Exclusive Economic Zone legislation.

When setting or varying any 'sustainability measure' within the Gulf under the Fisheries Act, the Minister for Primary Industries must 'have regard to' sections 7 and 8.¹⁷ Statutory bodies undertaking powers and functions affecting the Gulf under a range of other legislation must also 'have particular regard to' these provisions.¹⁸ This includes the Fisheries Act, the Conservation Act, the Biosecurity Act, the Heritage New Zealand Pouhere Taonga Act, the Local Government Act, the Marine Mammals Protection Act, the Marine Reserves Act, the Wildlife Act and the Reserves Act.

A marine spatial planning process, called SeaChange, is currently underway for the Hauraki Gulf. A Stakeholder Working Group is working collaboratively to deliver a marine spatial plan for the Hauraki Gulf by June 2015. This is discussed further in Chapter 7: Marine spatial planning.

Hauraki Gulf Marine Park Act, sections 7 and 8

7 Recognition of national significance of Hauraki Gulf

- (1) The interrelationship between the Hauraki Gulf, its islands, and catchments and the ability of that interrelationship to sustain the life-supporting capacity of the environment of the Hauraki Gulf and its islands are matters of national significance.
- (2) The life-supporting capacity of the environment of the Gulf and its islands includes the capacity—
 - (a) to provide for—
 - (i) the historic, traditional, cultural, and spiritual relationship of the tangata whenua of the Gulf with the Gulf and its islands; and
 - (ii) the social, economic, recreational, and cultural well-being of people and communities:
 - (b) to use the resources of the Gulf by the people and communities of the Gulf and New Zealand for economic activities and recreation:
 - (c) to maintain the soil, air, water, and ecosystems of the Gulf.

8 Management of Hauraki Gulf

To recognise the national significance of the Hauraki Gulf, its islands, and catchments, the objectives of the management of the Hauraki Gulf, its islands, and catchments are—

- (a) the protection and, where appropriate, the enhancement of the life-supporting capacity of the environment of the Hauraki Gulf, its islands, and catchments:
- (b) the protection and, where appropriate, the enhancement of the natural, historic, and physical resources of the Hauraki Gulf, its islands, and catchments:
- (c) the protection and, where appropriate, the enhancement of those natural, historic, and physical resources (including kaimoana) of the Hauraki Gulf, its islands, and catchments with which tangata whenua have an historic, traditional, cultural, and spiritual relationship:
- (d) the protection of the cultural and historic associations of people and communities in and around the Hauraki Gulf with its natural, historic, and physical resources:
- (e) the maintenance and, where appropriate, the enhancement of the contribution of the natural, historic, and physical resources of the Hauraki Gulf, its islands, and catchments to the social and economic well-being of the people and communities of the Hauraki Gulf and New Zealand:
- (f) the maintenance and, where appropriate, the enhancement of the natural, historic, and physical resources of the Hauraki Gulf, its islands, and catchments, which contribute to the recreation and enjoyment of the Hauraki Gulf for the people and communities of the Hauraki Gulf and New Zealand.

Fisheries Act

The Fisheries Act governs fisheries management throughout New Zealand's territorial sea and EEZ. The purpose of the Act is 'to provide for the utilisation of fisheries resources while ensuring sustainability'.¹⁹

'Ensuring sustainability' is defined in the Act as meaning:

- Maintaining the potential of the fisheries resources to meet the reasonably foreseeable needs of future generations; and

Maritime Transport Act

The Maritime Transport Act 1994 is primarily focused on the safety of shipping. The Act does not have an explicit purpose, but it does state that the objectives for the Minister of Transport under the Act are to contribute to an 'integrated, safe, responsive, and sustainable transport system' and to 'ensure New Zealand's obligations under conventions are implemented'.

The Act provides that 'harmful substances shall not be discharged or escape' into the sea or seabed within the EEZ or continental shelf except in accordance with marine protection rules. Marine protection rules made under this Act provide for both marine safety and pollution prevention within the coastal marine area and EEZ. Oil spill preparedness is funded by an industry levy, the Oil Pollution Levy, which is required under the Act. The Levy is collected to run New Zealand's maritime oil pollution preparedness and response system.

This Act is discussed further in a number of chapters in Section Two of this Guide where relevant to particular activities in the marine environment.

Biosecurity Act

The transport of invasive species into New Zealand waters by shipping, and their eradication or management after arrival, is primarily controlled under the Biosecurity Act. The jurisdiction of the Act was extended to apply to the EEZ in 2012. The Act is discussed further in Chapter 5: Marine biosecurity.

Marine Reserves Act

The Marine Reserves Act, administered by the Department of Conservation, provides for the establishment of marine reserves. These are spatially defined marine areas where fishing and other activities are excluded to maintain the areas in their natural state. There are currently over 30 marine reserves in New Zealand. The legislation is discussed further in Chapter 6: Marine protected areas.



The Marine Reserves Act enables the creation of no-take marine reserves where marine life is maintained in its natural state, such as in the Cape Rodney-Okakari Point marine reserve shown here (Raewyn Peart)

within the fiords covering an area of 103 square kilometres, or one per cent of the Fiordland marine biogeographic region. The Fiordland Marine Guardians have an ongoing role as government-appointed advisors to oversee monitoring and management of the area.



The Fiordland (Te Moana o Atawhenua) Marine Management Act provides a framework for the management of the Fiordland marine area including Doubtful Sound which is shown here (Raewyn Peart)

Heritage New Zealand Pouhere Taonga Act

Historic marine sites and places can be protected either under the Heritage New Zealand Pouhere Taonga Act or the RMA. Some sites and places are protected under both pieces of legislation. There are numerous marine heritage sites throughout New Zealand including wāhi tapu, submerged sites and structures, shipwrecks and historic anchorages. Most marine heritage sites have yet to be identified. For example, there are over 2,000 known shipwrecks around the coast of which only 150 have been located.²⁹

Archaeological sites are protected under the Heritage New Zealand Pouhere Taonga Act and can only be modified or destroyed if an authority (permit) is first obtained from Heritage NZ. Archaeological sites are defined in the legislation as:

... any place in New Zealand, including any building or structure ... that

- (i) was associated with human activity that occurred before 1900 or is the site of the wreck of any vessel where that wreck occurred before 1900; and*
- (ii) provides or may provide, through investigation by archaeological methods, evidence relating to the history of New Zealand.*

Other marine heritage sites can be protected under the RMA, through listing them in the regional coastal plan and including protective rules. Few plans, however, provide protection for marine cultural heritage. District plans can provide protection for land-based marine heritage (such as buildings that had connections to fishing or whaling) through scheduling and relevant protection rules. Heritage orders only take effect through district plans, which apply above mean high water springs, so such orders are not applicable to marine heritage.

Marine management and climate change

The potential impacts of climate change are well established. A recent report by the Intergovernmental Panel on Climate Change³³ outlined future risks and opportunities for adaptation to climate change. It identified key risks in relation to the ocean and the coast as including:³⁴

- the distributional shift of fish and invertebrate species
- decrease in fisheries abundance and catches at low latitudes and within coastal boundary systems
- reduction in biodiversity through coastal inundation and habitat loss due to sea level rise
- changes in precipitation and the frequency of extreme weather events
- increase in risks to infrastructure through increased frequency and intensity of flooding and coastal erosion due to sea level rise

These risks gave rise to a need to adapt to climate change, as well as to pursue measures to reduce or remove atmospheric greenhouse gas emissions.

The RMA and the EEZ Act allow for the consideration of climate change effects except in relation to the discharge of greenhouse gas emissions. The effect of climate change on marine systems is therefore something that councils and the EPA should consider when exercising their role in marine management.

The NZCPS addresses some aspects of climate change adaption (in Policies 24 to 26), requiring councils to identify coastal hazards, to discourage the use of hard protection structures and to enhance natural defences amongst other things.

Councils are required, under the RMA, to have particular regard to 'the benefits to be derived from the use and development of renewable energy' under section 7(j). This includes marine energy generation, and indicates that such projects should be given favourable consideration (although not necessarily approved if environmental impacts are unacceptable).

In addition, the National Policy Statement on Renewable Electricity Generation 2011 requires councils to provide for the development, operation, maintenance and upgrading of new and existing renewable energy generation activities which utilise tidal, wave and ocean current energy resources to the extent applicable within the region.³⁵

No marine energy projects have been built in New Zealand as yet, although consent has been granted to Crest Energy to install 200 underwater tidal turbines within the Kaipara Harbour.

Marine areas can contribute to the mitigation of climate change through capturing and storing carbon (carbon sequestration). Carbon can be stored in marine plants (such as mangroves, seaweeds and seagrass) and in the calcium carbonate shells and structures of invertebrates. In New Zealand the productive shelf waters of the Hauraki Gulf, oceanic frontal regions such as the Chatham Rise, and areas with dense shellfish, mussel and tube-worm beds have been noted as being particularly significant.³⁶ Impacts of activities on the ability of the marine area to store carbon is also something which decision-makers may consider.

Marine biodiversity offsetting

Marine biodiversity offsetting is a tool to address residual effects of development on marine biodiversity where they cannot be adequately avoided, minimised or remedied. A biodiversity offset aims to create at least equivalent biodiversity benefits on the same site or elsewhere, and to apply to similar types of biodiversity to that which is impacted. The goal is to achieve no net loss and ultimately a net gain of biodiversity in the marine area.³⁷ Biodiversity offsetting seeks to address the residual impacts of an activity once available avoidance, mitigation and remediation measures are taken.³⁸ Offsetting is not appropriate where irreplaceable ecological values might be lost at the impact site.

There's no specific requirement for marine biodiversity offsetting in New Zealand at present. However, it can potentially be applied under the RMA and the EEZ Act, as part of the consenting process. The Department of Conservation has recently prepared a non-statutory guidance document on biodiversity offsetting called 'Guidance on Good Practice Biodiversity Offsetting in New Zealand'.³⁹ The guidance does not consider offsetting in the offshore marine environment. However, lessons can be drawn from the approaches to offsetting in terrestrial ecosystems.

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3

Exclusive Economic Zone legislation



The EEZ Act governs a range of mining activities within the EEZ, including the sampling of manganese nodules which is shown here (*Secretariat of the Pacific Community, 2013b – Photo courtesy of BGR*)

Introduction

The EEZ Act came into force on 28 June 2013 and applies to certain activities taking place in the EEZ and extended continental shelf. The activities are primarily related to disturbance of the seabed, including petroleum and mineral exploration and production, aquaculture, marine energy generation, and carbon capture and storage. These activities and their effects are described in more detail in Section Two of this Guide. The Act does not apply to fishing activity or shipping.

The EEZ Act sets up a management and decision-making framework for managing the environmental effects of these activities. Within this framework, the Governor-General is empowered to make regulations to cover technical matters, including classifying activities as permitted, discretionary or prohibited. Permitted activities can proceed, subject to compliance with any relevant conditions. Prohibited activities are effectively banned and no consent can be issued for them. Discretionary activities are subject to a marine consent application process, which is administered by the EPA.

The EPA is responsible for deciding applications for marine consents, monitoring compliance with the EEZ Act, enforcement, promoting public awareness of the requirements of the Act, and performing any other function specified in the Act. The Decision-Making Committee members, who consider marine consent applications, are appointed by the EPA and must include one EPA Board member. Information about current marine consent applications is available on the EPA website.¹

The EEZ Act framework is closely based on that of the RMA. However, there are a number of important differences. For example, the classification of marine activities in the EEZ Act is defined in regulations which are promulgated with some (but limited) public involvement. In contrast, under the RMA, regional and district plans determine the classification of activities and the plans are developed through an open public process including appeal rights. Another difference is that the EEZ Act requires notification requirements to be defined in regulations instead of being determined on a case-by-case basis by the consent authority.

Purpose and principles

Purpose

The purpose of the EEZ Act, which is relevant to all decisions made under it, is to promote the 'sustainable management' of natural resources of the EEZ and continental shelf. Sustainable management is defined to mean:²

... managing the use, development, and protection of natural resources in a way, or at a rate, that enables people to provide for their economic well-being while—

- (a) sustaining the potential of natural resources (excluding minerals) to meet the reasonably foreseeable needs of future generations; and*
- (b) safeguarding the life-supporting capacity of the environment; and*
- (c) avoiding, remedying, or mitigating any adverse effects of activities on the environment.*

The EEZ Act purpose is deliberately very similar to that of the RMA. Parliament considered that utilising a well-understood purpose would provide certainty for stakeholders and provide some consistency of approach between decision-making within the territorial sea and outside of it.³ Similar to the RMA purpose, subsection (a) excludes minerals, thereby enabling exploitation without regard for the need to retain minerals for future use.

However, Parliament deliberately chose not to carry across the exact RMA purpose. Although the EEZ Act purpose has many similarities to that of the RMA, there are a number of differences including:

- The definition of 'environment'
- No references to 'physical resources', 'communities', 'social wellbeing', 'cultural wellbeing' or 'health and safety' in the EEZ Act purpose
- No equivalent to sections 6 (matters of national importance), 7 (other matters) and 8 (Treaty of Waitangi principles) of the RMA in the EEZ Act
- No equivalent to the EEZ Act information principles in the RMA

EEZ Act, section 61 Information principles applying to the consideration of marine consent applications

- (1) When considering an application for a marine consent, the Environmental Protection Authority must—
 - (a) make full use of its powers to request information from the applicant, obtain advice, and commission a review or a report; and
 - (b) base decisions on the best available information; and
 - (c) take into account any uncertainty or inadequacy in the information available.
- (2) If, in relation to making a decision under this Act, the information available is uncertain or inadequate, the EPA must favour caution and environmental protection.
- (3) If favouring caution and environmental protection means that an activity is likely to be refused, the EPA must first consider whether taking an adaptive management approach would allow the activity to be undertaken.
- (4) Subsection (3) does not limit section 63 or 64.
- (5) In this section, **best available information** means the best information that, in the particular circumstances, is available without unreasonable cost, effort, or time.

These provisions require the EPA to make full use of its powers to request information, obtain advice, and commission reviews or reports when considering an application for marine consent. This places a positive obligation on the EPA to seek out further information where there are gaps or inadequacies in the information provided in the marine consent application. The EPA is also required to base its decisions on the best available information. If information is uncertain or inadequate, the EPA must favour caution and environmental protection, and must consider whether an adaptive management approach⁷ would allow an activity to be undertaken.

Best available information

Best available information is defined to mean 'the best information that, in the particular circumstances, is available without unreasonable cost, effort, or time'.⁸ This is about ensuring the Decision-Making Committee has 'the right amount and quality of information to be able to make a good decision'. It means that the Committee may not be able to rely on the information presented by participants in the process and may need to obtain additional information if necessary.⁹ Gaps and uncertainties in the information do not present a difficulty in terms of the requirement to base a decision on the best available information as 'gaps and uncertainties arise because the necessary information to remedy these does not exist. It is not a case where important and relevant information is known but has not been shared'.¹⁰

Favouring caution and environmental protection

The requirement that the decision-maker must favour caution and environmental protection where the information available to the decision-maker is uncertain or inadequate is a strong direction. Oxford Dictionaries' definition of 'favoured' is 'preferred or recommended'.¹¹ This means that, when the caution requirement is triggered, environmental protection must be preferred over other matters. This is consistent with the purpose of the EEZ Act which expressly contemplates environmental protection.

This requirement differs from a similar statement in the Fisheries Act 1996: 'the absence of, or any uncertainty in, any information should not be used as a reason for postponing or failing to take any measure to achieve the purpose of the Act'.¹² The purpose of that Act is 'to provide for the utilisation of fisheries resources while ensuring sustainability'. In comparison, the caution requirement in the EEZ Act is clearly heavily weighted towards environmental protection.

This caution requirement in the EEZ Act also differs from the precautionary approach as expressed in international law through Principle 15 of the Rio Declaration 1992: 'in order to protect the environment, the precautionary approach shall be widely applied by States according to their capabilities. Where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as a reason for postponing cost-effective measures to prevent environmental degradation'. Significantly, the EEZ Act does not require 'serious or irreversible adverse effects' before the caution requirement is triggered.¹³

Application of adaptive management

The Trans-Tasman Resources Decision-Making Committee agreed that the Supreme Court's approach is applicable to adaptive management under the EEZ Act.²¹ Trans-Tasman Resources did not propose an adaptive management approach that would enable the activity 'to commence on a small scale or for a short period so that its effects on the environment and existing interests can be monitored'. This was because the applicant considered that 'in order for the operation to be commercially viable, the ability to mine the whole area of the application was required'.²² Instead, Trans-Tasman Resources proposed an approach which involved:²³

- Qualitative environmental performance objectives (specified in the consent conditions)
- Pre-operational baseline monitoring
- A process for developing quantitative 'trigger indicators or values' which would inform compliance with environmental performance objectives
- Operational monitoring
- Management of operations to achieve the identified environmental performance objectives

The Trans-Tasman Resources Decision-Making Committee assessed the approach against the Supreme Court's four-step test (set out above). The Decision-Making Committee concluded that 'the proposed Environmental Performance Objectives do not provide sufficient certainty, clarity or robustness on which to form the foundation of an appropriate adaptive management approach'. The Committee was also concerned 'that, without the option of a scale or staged implementation, the applicant has not provided any compelling adaptive management alternatives should the environmental objectives not be met'.²⁴

International obligations principle

New Zealand does not have full sovereignty beyond 12 nautical miles but relies on rights gained under UNCLOS to regulate activities in the EEZ and continental shelf area covered by the EEZ Act. International obligations are therefore particularly relevant to management of New Zealand's EEZ and continental shelf.

Section 11 of the EEZ Act provides that the Act continues or enables the implementation of New Zealand's international obligations in the marine environment, including those under UNCLOS²⁵ and the Convention on Biological Diversity 1992.²⁶ The section also implies that other relevant international conventions would be relevant including the International Convention for the Prevention of Pollution from Ships 1973 (MARPOL),²⁷ and the Convention on the Prevention of Marine Pollution by Dumping Wastes and Other Matter 1972 (the London Convention).²⁸

EEZ Act, section 11 International obligations

This Act continues or enables the implementation of New Zealand's obligations under various international conventions relating to the marine environment, including —

- (a) the United Nations Convention on the Law of the Sea 1982;
- (b) the Convention on Biological Diversity 1992.

When the EEZ Bill was first introduced to Parliament, it contained a clause which required the Act to be interpreted and applied in accordance with international obligations.²⁹ The clause was deliberately amended due to concerns that international obligations would be utilised to interpret the legislation in a manner that was not considered by Parliament. In addition, it was argued that inclusion of such a clause would create a substantial risk of appeals against decisions under the EEZ Act, on the grounds of inconsistency with international obligations.³⁰

Restrictions on activities

When is a marine consent required?

The EEZ Act prohibits identified activities in the EEZ or continental shelf (see Figure 3.1) unless the activity is permitted by regulations, authorised by a marine consent, or an existing activity exemption applies.

- The construction, placement, alteration, extension, removal, or demolition of a structure on or under the seabed
- The construction, placement, alteration, extension, removal, or demolition of a submarine pipeline on or under the seabed
- The placement, alteration, extension, or removal of a submarine cable on or from the seabed
- The removal of non-living natural material from the seabed or subsoil
- The disturbance of the seabed or subsoil in a manner that is likely to have an adverse effect on the seabed or subsoil
- The deposit of anything or organism in, on, or under the seabed
- The destruction, damage, or disturbance of the seabed or subsoil in a manner that is likely to have an adverse effect on marine species or their habitat
- The construction, mooring or anchoring long-term, placement, alteration, extension, removal, or demolition of a structure or part of a structure
- The causing of vibrations (other than vibrations caused by the normal operation of a ship) in a manner that is likely to have an adverse effect on marine life
- The causing of an explosion

Figure 3.1 Activities which are restricted under section 20 of the EEZ Act³⁴

The Exclusive Economic Zone and Continental Shelf (Environmental Effects) Amendment Act 2013 (EEZ Amendment Act) transfers powers to regulate discharges and dumping (excluding discharges from ships) which are currently regulated under the Maritime Transport Act. This transfer will take effect when regulations are made under the EEZ Act to classify discharge and dumping activities. When the transfer takes effect the new provisions of the EEZ Act will:

- Prohibit discharges of harmful substances into the sea or seabed, unless it is a permitted activity, authorised by a marine consent or an existing activity exemption applies³⁵
- Impose a total prohibition on the dumping of radioactive, toxic or hazardous waste into the sea or on the seabed³⁶
- Prohibit dumping of other waste unless authorised by a marine consent, authorised by an emergency dumping consent, or in accordance with section 248 or 249 of the Maritime Transport Act³⁷
- Regulate burial at sea³⁸

General duties

All persons carrying out activities in the EEZ or continental shelf also have a general duty to avoid, remedy or mitigate the adverse effects of the activity on the environment as well as a duty to provide training, supervision and resources to employees to ensure compliance with the EEZ Act.



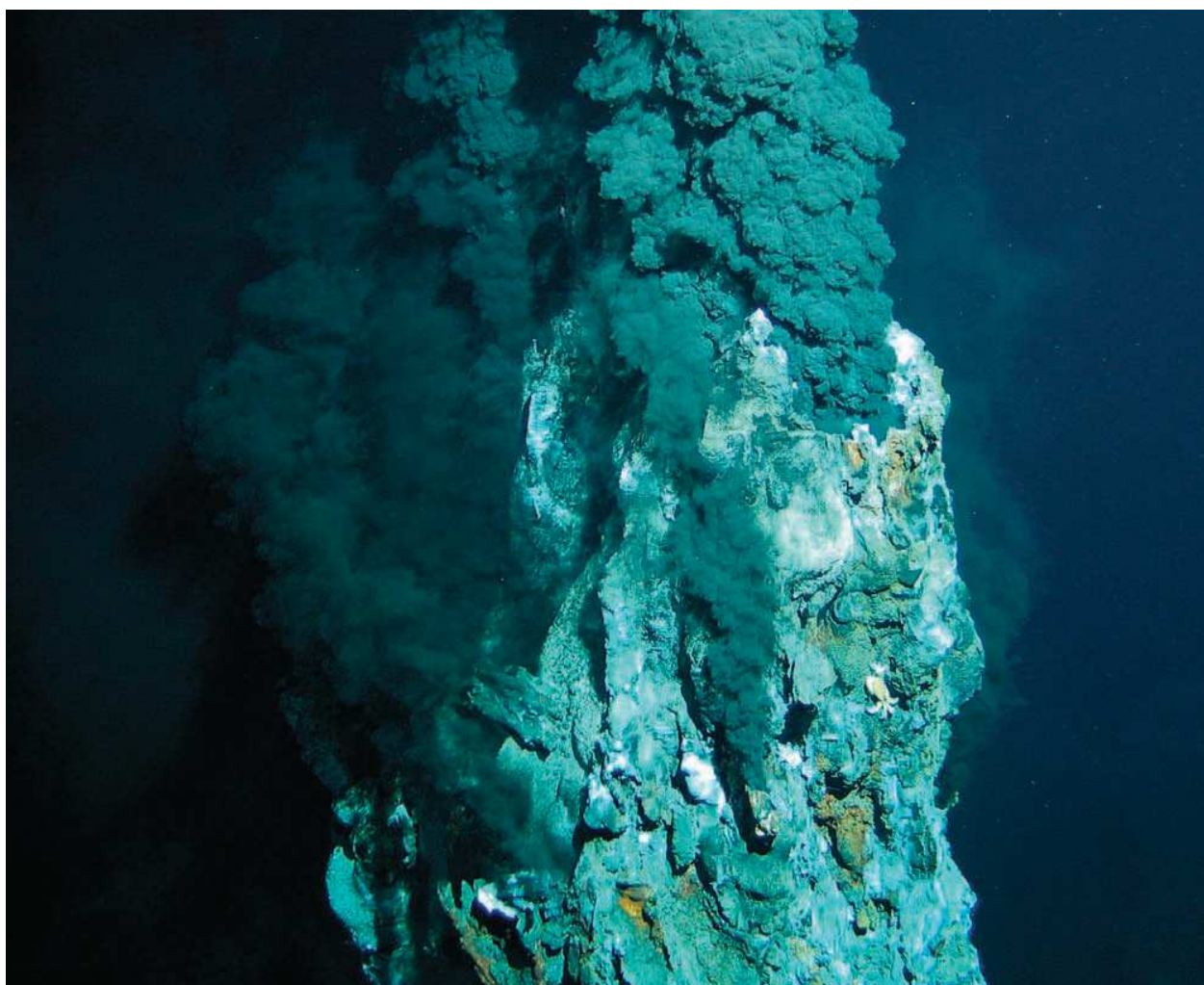
Deep-sea drilling as part of scientific investigations, such as undertaken by this vessel, is a permitted activity within New Zealand's EEZ subject to compliance with a number of conditions (Secretariat of the Pacific Community, 2013a – Photo courtesy of Elaine Baker)

- If the information available is uncertain or inadequate, favour caution and environmental protection
- If favouring caution and environmental protection means that an activity is likely to be prohibited, consider whether providing for an adaptive management approach would allow the activity to be classified as discretionary

To date, regulations have only been promulgated to set fees and charges and to classify activities as permitted or non-notified discretionary (discussed below). At the time of writing, regulations were currently being drafted to classify discharges and dumping activities. However, the broad nature of potential regulations means that they could be used to provide for integrated and strategic management of the marine environment, similar to regional coastal plans prepared under the RMA. There are limitations, as the EEZ Act cannot be used to control all activities in the EEZ, including fishing and maritime transport.

The EEZ Act provides protections for existing and planned activities in the event of an amendment to the regulatory regime. An activity which was lawfully established as a permitted activity under the EEZ Act is classified as an 'existing activity' where the effects of the activity on the environment remain the same or similar in character, intensity and scale. If the regulations change so that the existing activity requires a marine consent, the activity may continue for six months (or a prescribed period) or, if a marine consent is applied for within that six-month period, until the application is determined.⁴⁵ This is similar to the existing activity provision for regional rules in the RMA.

Where new regulations come into force which prohibit an activity which was previously lawfully established as a permitted activity, or was authorised by a marine consent, the activity may continue for the duration of the consent if applicable or for a period prescribed by regulations.



Active hydrothermal chimneys which occur in association with deep-sea venting, such as shown here, are identified as sensitive environments under the EEZ Permitted Activity Regulations (*Secretariat of the Pacific Community, 2013c*)

Permitted activities

Provisions applying to permitted activities are set out in the Exclusive Economic Zone and Continental Shelf (Environmental Effects – Permitted Activities) Regulations 2013 (EEZ Permitted Activities Regulations). The Regulations set out conditions which apply to permitted activities. In general these require the party wishing to undertake the activity to prepare an ‘initial environmental assessment and sensitive environments contingency plan’. This must identify and describe any ‘sensitive environments’ that are likely to exist within the areas where the activity will be undertaken and then to assess ways of reducing impacts on them.

EEZ Permitted Activities Regulations, Schedule 2: Contents of plan

- (1) An initial environmental assessment and sensitive environments contingency plan must—
- (a) describe the activity concerned; and
 - (b) state the co-ordinates of the area within which the activity will be conducted; and
 - (c) describe in general terms (using the best available information) the environment likely to be encountered when the activity is being undertaken; and
 - (d) identify and describe any sensitive environments that are likely to exist within the area; and
 - (e) detail the methods that will be used to undertake the activity; and
 - (f) in relation to any sensitive environment that is likely to be encountered,—
 - (i) assess the feasibility of carrying out the activity in another location; and
 - (ii) assess, in the following order, the feasibility of measures that could be taken to—
 - (A) reduce the amount of contact with the seabed;
 - (B) carry out alternative lower-impact activities;
 - (C) change the methods of operation to lower the impact of the activity on the environment.

The EPA must notify iwi, hapū, customary marine title groups and protected customary rights groups as soon as practicable after receipt of notice that a permitted activity has commenced or has been completed.⁴⁸ The EPA must also monitor permitted activities to determine whether they are being undertaken in accordance with any conditions imposed by the Permitted Activities Regulations.⁴⁹

Sensitive environments

Sensitive environments are identified in Schedule 6 of the Regulations⁵⁰ (see Figure 3.4). These set out indicators for determining if a sensitive environment is present. Sensitive environments were identified for incorporation into the regulations using the MarLIN method⁵¹ developed by the United Kingdom’s Marine Life Information Network. Sensitivity is dependent on the tolerance of a species or habitat to damage from an external factor, and the time taken for its subsequent recovery from the damage sustained. Assessments of tolerance take rarity into account, as the more rare a habitat is, the more an external factor is likely to damage a significant proportion of the habitat.

Habitat definitions were derived by NIWA in 2013 from available scientific literature and the field experience of staff undertaking research in these areas.⁵² If a habitat or species is killed or destroyed by an external factor arising from human activities or natural events, and is expected only to recover over a very long period of time (10–20 years), then it would be considered to be highly sensitive. Similarly, if a habitat or species is adversely affected by an external factor arising from human activities or natural events (damaged) but is expected to recover in a short period of time (one to five years) then it would be considered to be of low sensitivity.



Anemones, which are endemic to a deepsea hydrothermal vent field
(Secretariat of the Pacific Community, 2013a – Photo courtesy NSF Ridge 2000 Program)

Environment	Description
Sea pen field	• Are formed from colonial marine cnidarians • Occur on soft sediments in the deep waters where turbulence is unlikely to dislodge them and where there is a current to ensure a flow of plankton across their feeding polyps • Currently 31 species of sea pens are known from New Zealand waters, although 19 of those have not been formally described • Fauna associated with New Zealand sea pens have not yet been described
Rhodolith beds	• Are formed from free-living calcified red algae which form structurally and functionally complex habitats • Feature high benthic biodiversity supporting many rare and unusual species • Often coincide with productive fisheries and have been identified as important nursery areas for commercial species • Internationally rhodolith beds have been identified as critically important biodiversity hotspots, harbouring high diversity and abundance of marine animals and algae, in comparison with surrounding habitats • Are fragile and slow-growing (0.05–2 millimetres per year) • Very little information exists about the location, extent or ecosystem functioning of rhodolith beds in New Zealand • Are likely to occur in the EEZ at localities characterised by strong currents within the photic zone (layer that receives sunlight) to depths of 200 metres depending on water clarity, particularly around the margins of reefs or elevated banks
Sponge gardens	• Formed from sedentary, filter-feeding metazoans that utilise a single layer of flagellated cells to direct a water current through their bodies for the purposes of feeding and excretion • May occur on hard or soft substrate in many subtidal environments including shallow coastal rocky reefs, seamounts, hydrothermal vent systems and oceanic ridges • Well over 500 sponge species have been formally described from New Zealand waters but there are many more known, with new discoveries every year • Certain areas in New Zealand's EEZ have been identified as sponge 'hotspots', including the Three Kings, North Cape, Spirits Bay and offshore carbonate bryomol gravel banks
Beds of large bivalve molluscs	• Create biogenic structure in what may be an otherwise 'featureless' habitat • Their shells provide a substrate for settlement by organisms such as sponges and bryozoans, and shelter for mobile invertebrates and fishes • Create habitat heterogeneity and have an important role in ecosystem functioning • Influence biogeochemical processes such as regeneration of sediment-associated nutrients, and processing of these nutrients, thus making them available for water-column primary production. Sediment-derived nutrients are considered a major contributor to continental shelf production. • Are generally located at depths of less than 250 metres and are well represented off Northland, off the west coast of the North Island, and off south-eastern and southern-most New Zealand

Seismic surveying

Seismic surveying is permitted if the person undertaking the activity complies with the Department of Conservation's 2013 'Code of Conduct for Minimising Acoustic Disturbance to Marine Mammals from Seismic Survey Operations'.⁵³

The Code of Conduct defines Level 1, Level 2 and Level 3 surveys and requires mitigation actions which are commensurate to the level of risk.

Level 1 surveys involve high-power acoustic sources such as large-scale geophysical investigations employed in oil and gas exploration activities. These are subject to stringent requirements, which include the following:

- Generally, the Department of Conservation must be notified in writing at least three months in advance of the survey commencing
- A Marine Mammals Impact Assessment must be prepared. Particular attention must be given to 'Areas of Ecological Importance', which include marine mammal sanctuaries. Areas of Ecological Importance are identified on the Department of Conservation website.⁵⁴ A core component of the planning process is to limit surveys to the lowest practicable power levels that will achieve the geophysical objectives of the survey. There is no formal approval process but the Director-General will determine if the impact assessment is sufficient for the purposes of the Code of Conduct.
- Two qualified marine mammal observers must be on board the vessel undertaking the survey at all times. At least one must be on watch when the acoustic source is in the water. Seismic survey vessel crew cannot be considered qualified observers.
- Passive acoustic monitoring (which detects vocalising cetaceans) is required, along with two passive acoustic monitoring observers
- The acoustic source cannot be activated until visual observations and passive acoustic monitoring for at least 30 minutes indicates no marine mammals are present
- All sightings of marine mammals must be recorded and reported to the Department of Conservation. Additional measures may be required if the number of cetaceans encountered is greater than predicted.
- Start-up will be delayed or the source shut down if an observer detects a cetacean with a calf within 1.5 kilometres of the source, a 'Species of Concern' (which are listed in Schedule 2 of the Code of Conduct) within 1 kilometre of the source, or any marine mammal within 200 metres of the source



Seismic surveying within the EEZ is permitted if it complies with the Department of Conservation's Code of Conduct. This is designed to avoid adverse effects on marine mammals such as the Hector's dolphins in Akaroa Harbour shown here (Raewyn Peart)

Marine structures

The EEZ regulations permit the maintenance or repair of a marine 'structure' (which includes buildings, equipment and devices but does not include submarine pipelines).⁵⁶ The alteration, extension or removal of a marine structure is also permitted if the person undertaking the activity:

- Notifies the EPA at least 40 working days before the activity commences
- Notifies iwi, hapū, customary marine title groups and protected customary rights groups at least 25 working days before the activity commences
- Provides the EPA with a post-activity report detailing the activity undertaken, the dates, the location, and estimates of the activity's environmental footprint (including an estimate of volume of material removed)

Submarine cables

The EEZ regulations permit the placement of a submarine cable on or under the seabed, or the removal of a submarine cable from the seabed, provided the person undertaking the activity:⁵⁷

- Notifies the EPA at least 40 working days before the activity commences
- Notifies iwi, hapū, customary marine title groups and protected customary rights groups at least 25 working days before the activity commences
- Notifies the EPA within 24 hours of the date on which the person commences the activity
- Notifies the EPA within 24 hours of the date on which the person completes the activity concerned
- Provides the EPA with a post-activity report detailing the activity undertaken, the dates, the location, and estimates of the activity's environmental footprint (including an estimate of volume of material removed)

Marine consent process

There are two marine consent process tracks for discretionary activities: notified and non-notified. The ways in which the public can become involved in a marine consent process are described in Chapter 8: Opportunities for public involvement.

In a notified marine consent process, the public is notified of the application, and is able to make submissions on it. The process involves lodgement, acceptance of the application as complete, public notification, submissions, a hearing and determination of the application (see Figure 3.6). The applicant or submitters may appeal the decision to the High Court on points of law.

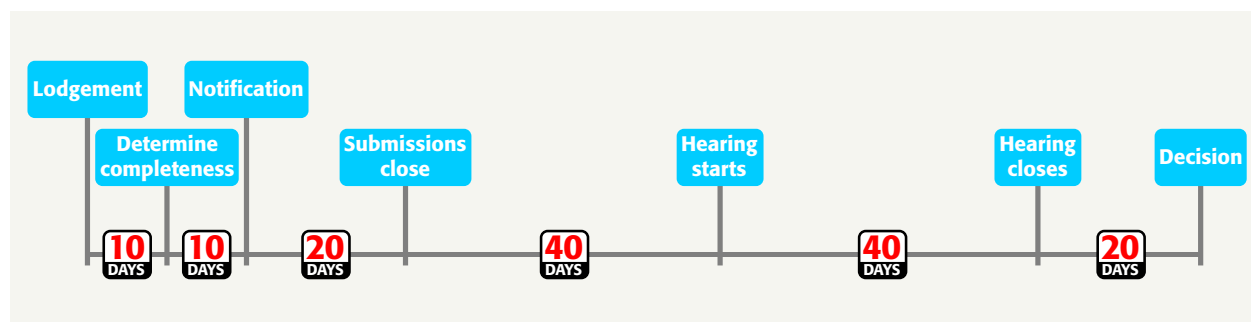


Figure 3.6 Flow chart of marine consent process

In a non-notified marine consent process, the public is not notified of an application, and is unable to make submissions on it. The EPA has up to 50 working days to assess and make its decision on the application, after determining that the application is complete.⁵⁸

Further information

The EPA may request an applicant to provide further information relating to the application. The EPA is required to provide a copy of any further information provided by the applicant to every submitter.

The EPA may commission an independent review of the impact assessment, or seek advice on any other aspect of an application for a marine consent, and must make any reports or advice available to the applicant and submitters.

Application of further information provisions

During the Trans-Tasman Resources marine consent process the EPA made two requests for further information. These can be viewed on the EPA website.⁶⁴ The EPA also commissioned independent technical reviews in relation to 10 topics and independent peer reviews in relation to seven topics. These can also be viewed on the EPA website.⁶⁵ The independent peer reviewers participated in expert conferencing and presented evidence to the Decision-Making Committee.

Notification

Notified discretionary marine consent applications will be publicly notified and any person may make a submission to the EPA within 20 working days of notification.

Non-notified discretionary marine consent applications will not be publicly notified. However, a copy of the application must be provided to iwi authorities, customary marine title groups and protected customary rights groups that the EPA considers may be affected by the application. The EPA may also serve a copy of the application on the following parties:⁶⁶

- Ministers with responsibilities that may be affected by the activity for which consent is sought
- Maritime New Zealand
- Persons with existing interests that may be affected by the application
- Regional councils whose regions may be affected by the application

Pre-hearing dispute resolution

In the notified process, the EPA may request the applicant and submitters to meet to discuss matters in dispute or enter mediation to resolve a dispute on a 'without prejudice' basis. The person who conducts the meeting or mediation must report to the EPA, and the persons who were at the meeting or mediation, on the outcome of the meeting or mediation.

Hearing

Non-notified process

Where an application is processed on a non-notified basis, a hearing may be held if the EPA considers it necessary or desirable, but must be held if the applicant requests a hearing.⁶⁷ If a hearing is held, the applicant may speak and call evidence.

Notified process

Where an application is publicly notified, a hearing may be held if the EPA considers it necessary or desirable, but must be held if the applicant or a submitter requests a hearing. The hearing must commence within 40 working days of the close of submissions. The applicant and submitters must be given at least 20 working days' notice and the hearing must be completed within 40 working days after the first day of the hearing.⁶⁸ The EPA has discretion to extend specified time periods specified in the EEZ Act or in regulations and to waive a failure to comply with service requirements.⁶⁹

The procedure for a hearing is determined by the EPA, but is required to be fair and appropriate in the circumstances. In determining an appropriate and fair procedure for a hearing, the EPA must avoid unnecessary formality, recognise tikanga Māori where appropriate, and receive evidence written or spoken in Māori.

At a hearing, the applicant and submitters who requested to be heard may speak (either personally or through a representative) and call evidence. No person may question a party or witness (i.e. undertake cross-examination) unless the EPA gives permission.

documents under the EEZ Act, and therefore less opportunity for strategic decision-making to provide a framing to consider individual proposals (although regulations to achieve this can be developed).

The term 'rare and vulnerable ecosystems' is not defined in the EEZ Act. This relates to ecosystems, as opposed to individual species. 'Rare' indicates that the ecosystem is not widespread. 'Vulnerable' indicates that an ecosystem may be sensitive to disturbance. The EEZ Act defines 'threatened species' as including 'any species that falls within the definition of threatened species in section 2(1) of the Biosecurity Act 1993, or is declared to be a threatened species or an at-risk species by the Minister of Conservation under section 19'.⁷² On 11 July 2013, the Minister of Conservation declared the marine species listed in the New Zealand Threat Classification System, including dolphin, seabird, fish and invertebrate species, to be 'threatened' for the purpose of the EEZ Act.⁷³

Where a replacement consent is being sought, the EPA must also have regard to the value of the investment in the activity of the existing consent holder. The requirement to 'have regard to' has a similar meaning to 'take into account'.

In considering an application, the EPA must not have regard to trade competition, the effects on climate change of discharging greenhouse gases into the air, or effects on the existing interests of persons who have given written approval to the activity.⁷⁴

Environmental compensation

Current best practice environmental management utilises a 'no net loss' approach and uses environmental compensation to address effects which cannot be avoided, remedied or otherwise mitigated (discussed in Chapter 2: Marine management). Environmental compensation is commonly adopted for large projects consented under the RMA, despite there being no explicit provision for offsetting in that Act. Therefore, while the EEZ Act also does not expressly refer to compensation or offsetting, RMA experience suggests that environmental compensation may be an available approach for addressing effects of activities in the EEZ and continental shelf.

Consent conditions

The conditions that the EPA may impose include:⁷⁵

- Providing a bond for the performance of any condition
- Obtaining and maintaining public liability insurance of a specified value
- Monitoring and reporting on the consent and the effects it authorises
- Appointing an observer to monitor the activity and its effects
- Making records related to the activity available for audit

Marine consent conditions will need to satisfy the Newbury principles (which are applicable to resource consent conditions) to be valid.⁷⁶ These are:⁷⁷

- Principle 1 – conditions must be for resource management purposes
- Principle 2 – conditions must relate to the authorised development
- Principle 3 – conditions must be reasonable

In addition, marine consent conditions will be invalid if they are not certain and enforceable and within the consent authority's power to impose.⁷⁸

Comment on marine consent conditions

Conditions need to have an evidential basis and be able to be reasonably achieved and monitored.⁷⁹ The Trans-Tasman Resources Decision-Making Committee was concerned that the extent of changes made to the proposed conditions throughout the hearing (with the final set provided with the closing submissions) meant that it was uncertain whether there was an evidential base for the conditions and whether they could actually and reasonably be achieved and monitored.

3. To deal with adverse effects on the environment that were not anticipated when the consent was granted or are of a scale or intensity that was not anticipated when the consent was granted
4. If the information provided by the applicant to the EPA contained inaccuracies that materially influenced the decision made on the application and the effects of the activity are such that more appropriate conditions are required
5. If information becomes available to the EPA that was not available when the consent was granted and shows that more appropriate conditions are required
6. If required by an order made following a person being convicted of an offence

The circumstances which allow the EPA to review consent duration or conditions are wider than those under the RMA. In particular, categories 3 and 5 above are not available under the RMA. This reflects the greater level of uncertainty in environmental management in the EEZ and continental shelf.

The EPA may change a condition, cancel a condition or impose a new condition. The EPA may cancel a consent if the review was commenced for any of the reasons identified at 3, 4, 5 or 6 above and the activity has significant adverse effects on the environment or existing interests.⁹¹

The EPA may extend the duration of a consent only if monitoring of the effects shows that the effects are minor or may be avoided, remedied or mitigated by imposing conditions. The EPA may shorten the duration of a consent only if the effects of the activity, or the scale or intensity of the effects, were not anticipated when the consent was granted and shortening the duration is the only way to avoid, remedy or mitigate the effects appropriately. In any case, the duration of the marine consent cannot extend beyond 35 years.⁹²

The consent holder can also request the EPA to change or cancel a condition on marine consent. This application is determined in the same manner as an application for marine consent, unless it is limited to minor matters. It must be publicly notified unless it only affects some of the original submitters or existing interests or relates to minor matters.

Lapse and cancellation

A marine consent lapses on the date specified in the consent or five years after the commencement of the consent unless the consent is given effect to or the EPA grants an extension. A marine consent may be cancelled by the EPA if it has been exercised in the past but has not been exercised during the preceding five years.⁹³

Transfer of marine consents

A marine consent may be transferred to another person, but not to another location.

Reconsenting

If a marine consent is due to expire, and the consent holder applies to the EPA for a new consent for the same activity at least six months prior to the expiry of the existing consent, the consent holder may continue to undertake the activity until the application is granted or declined and all appeals are determined.⁹⁴



Offshore scientific research does not need a marine consent so long as a set of statutory requirements are met (Raewyn Peart)

Endnotes

- 1 www.epa.govt.nz/EEZ
- 2 EEZ Act, section 10
- 3 Adams A, 2012, *Hansard*, 16 August, 2012, 682, NZPD 4492
- 4 *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [86]
- 5 *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [86]
- 6 *Environmental Defence Society Incorporated v The New Zealand King Salmon Company Limited* [2014] NZSC 38 at [25]
- 7 EEZ Act, section 64, defines an adaptive management approach as including 'allowing an activity to commence on a small scale for a short period so that its effects on the environment' can be monitored and 'the activity discontinued, amended or continued'
- 8 EEZ Act, section 61(5)
- 9 *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [117] and [120]
- 10 *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [123]
- 11 <http://www.oxforddictionaries.com/definition/english/favoured?q=favoured>
- 12 Fisheries Act 1996, section 10(d)
- 13 *Trans-Tasman Resource Marine Consent Hearing Transcript*, page 1931
- 14 *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [138]
- 15 *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [139]
- 16 *Golden Bay Marine Farmers v Tasman District Council* W19/2003 at [405]
- 17 *Crest Energy Kaipara Limited v Northland Regional Council* A132/09
- 18 EEZ Act, section 61(3)
- 19 *Sustain Our Sounds Incorporated v The New Zealand King Salmon Company Ltd* (2014) at [129]
- 20 *Sustain Our Sounds Incorporated v The New Zealand King Salmon Company Ltd* (2014) at [133]
- 21 *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [798]
- 22 *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [145]
- 23 *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [793]
- 24 *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [850–851]
- 25 http://www.un.org/depts/los/convention_agreements/convention_overview_convention.htm
- 26 <http://www.cbd.int/>
- 27 [http://www.imo.org/About/Conventions/ListOfConventions/Pages/International-Convention-for-the-Prevention-of-Pollution-from-Ships-\(MARPOL\).aspx](http://www.imo.org/About/Conventions/ListOfConventions/Pages/International-Convention-for-the-Prevention-of-Pollution-from-Ships-(MARPOL).aspx)
- 28 <http://www.imo.org/OurWork/Environment/LCLP/Pages/default.aspx>
- 29 When the Bill was introduced to Parliament, clause 11 read, 'This Act must be interpreted, and all persons performing functions and duties or exercising powers under it must act, consistently with New Zealand's obligations under the [Law of the Sea] Convention'
- 30 Departmental report on the Exclusive Economic Zone and Continental Shelf (Environmental Effects) Bill, page 35: http://www.parliament.nz/resource/en-nz/50SCLGE_ADV_00DBHOH_BILL11023_1_A226517/b43ae930f23894c57e87ad2b413a73d4b8a38e10
- 31 Exclusive Economic Zone and Continental Shelf (Environmental Effects – Permitted Activities) Regulations 2013
- 32 *Trans-Tasman Resources Marine Consent Decision* (June 2014), at [95]
- 33 *Trans-Tasman Resources Marine Consent Decision* (June 2014), at [594]
- 34 EEZ Amendment Act, section 20
- 35 EEZ Amendment Act, sections 20B and 20C
- 36 EEZ Amendment Act, sections 20E and 20F
- 37 EEZ Amendment Act, section 20G
- 38 EEZ Amendment Act, section 20J
- 39 EEZ Amendment Act, sections 29 and 29D
- 40 EEZ Amendment Act, section 28
- 41 EEZ Act, section 27
- 42 EEZ Amendment Act, section 32
- 43 EEZ Amendment Act, section 33
- 44 EEZ Amendment Act, section 34
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4

Kaitiakitanga



Ngāti Paoa waka at Karaka Bay, Auckland (Raewyn Peart)

Introduction

Iwi nationally are involved in a range of commercial sectors including tourism, property development, seafood and forestry. Māori own around half of the aquaculture industry and about one-fifth of the fishing industry. Some have interests in other marine ventures, such as marinas and marine tourism.

Such developments can be entirely consistent with kaitiakitanga, both in terms of protection of the environment, and in its utilisation. Often tensions arise within iwi regarding development proposals. Whereas a mainstream developer will endeavour to meet legislative constraints, an iwi will also need to ensure that its development aspirations meet its beneficiaries' expectations of commercial gain as well as the requirements of kaitiakitanga. This can produce a far more stringent set of conditions than other developers face.

Scope of kaitiakitanga

The connection between land management and impacts on the marine environment are critical to coastal Māori. Gathering kaimoana, both for sustenance and to provide for visitors, is an integral component of Māori life. When these resources are compromised by inappropriate development, a way of life is also compromised. 'Kaitiakitanga' is now part of the vocabulary of environmental managers in New Zealand, but is often not well understood.

There are statutory definitions of kaitiakitanga. However, unlike other terminology in environmental law, there is little case law or published best practice to guide practitioners in its application. The following description acknowledges the statutory definitions, but provides an understanding within a broader cultural context.¹



The Ngai Tai Umupuia Te Waka Totara Trust exercising kaitiakitanga at Umupuia Beach, Auckland (Raewyn Peart)

Iwi as marine resource managers

In 2010, Māori were estimated to control up to 37 per cent of fisheries quota.² In addition, they own around 50 per cent of the aquaculture industry and this proportion is likely to increase as more space for aquaculture is created. Three iwi/Māori-owned (or part-owned) companies are among the five largest fishing companies in the country; Aotearoa Fisheries Limited,³ Sealord Group Limited (50 per cent Māori ownership through Aotearoa Fisheries Limited) and Ngāi Tahu Fisheries Settlement Limited (100 per cent Ngāi Tahu-owned).⁴

There are 57 iwi (or groups of iwi⁵) recognised in the Māori Fisheries Act 2004. The distribution of fisheries assets to these groups is currently being implemented by Te Ohu Kai Moana.

Relevant legislative provisions

Many pieces of legislation which apply to the marine area have general provisions of relevance to the exercise of kaitiakitanga.

Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012

The EEZ Act does not contain legislative recognition of kaitiakitanga. The Act states that its provision for a Māori Advisory Committee, iwi feedback on regulations, consideration of existing interests, and notification of consent applications to iwi ensures the legislation gives effect to the principles of the Treaty of Waitangi (described more fully in Chapter 3: Exclusive Economic Zone legislation).⁶

The Māori Advisory Committee provides advice to the EPA on matters relating to policy, process and decisions of the EPA. The advice and assistance is given from a Māori perspective. Iwi authorities (as well as the public, regional councils and persons whose existing interests are likely to be affected) are notified of proposed regulations and given adequate time and opportunity to comment on them.

Iwi authorities, customary marine title groups⁷ and protected customary rights groups⁸ are to be notified of a marine consent application if the EPA considers they may be affected. Full rights of participation (along with everyone else) allow kaitiakitanga to be addressed in submissions and at hearings.

Affected iwi authorities, customary marine title groups and protected customary rights groups are to be notified of a marine consent application which is for a non-notified activity. Ministers, Maritime New Zealand, persons with existing interests that may be affected, and affected regional councils may also be notified. However, such notification does not include the right to make a submission on the application. Unless the EPA specifically requires information from groups notified (under section 44), there is limited opportunity for these Māori groups to give feedback, potentially inhibiting the ability of Māori to undertake their kaitiakitanga responsibilities.

When considering an application for a marine consent, the EPA must take into account any effects on existing interests of allowing the activity. This is particularly relevant to iwi fishing interests. The EPA must also have regard to any advice received from the Māori Advisory Committee.

Resource Management Act

The RMA includes a requirement for decision-makers to have particular regard to kaitiakitanga⁹ and this provides a tool to assist with providing for 'the relationship of Māori and their culture and traditions with their ancestral lands, water, sites, wāhi tapu, and other taonga' under section 6(e). In addition, section 8 requires decision-makers to 'take into account' the principles of the Treaty of Waitangi.

RMA, Section 8, Treaty of Waitangi

In achieving the purpose of this Act, all persons exercising functions and powers under it, in relation to managing the use, development, and protection of natural and physical resources, shall take into account the principles of the Treaty of Waitangi (Te Tiriti o Waitangi).

The legislative recognition of kaitiakitanga in the RMA provides, in principle, an opportunity for the implementation of traditional practices as well as for greater public appreciation of their nature. Many tangata whenua resource management/kaitiaki units have

Customary commercial fishing rights

Māori have a long history in commercial fishing. They were trading in fish domestically prior to European arrival, and were actively involved in commercial fishing for provision to European explorers and settlers from Captain Cook's time onwards. Under the Treaty of Waitangi, Māori were guaranteed 'undisturbed possession' of their fisheries until they wished to dispose of them to the Crown.¹⁰ Subsequent fisheries legislation and management regimes failed to uphold this guarantee. Over time, Māori found themselves largely dispossessed of their fisheries through depletion, habitat degradation and finally through the introduction of the Quota Management System (QMS) in the early 1980s which took away commercial fishing rights from part-time fishers, many of whom were Māori.

This brought the issue of Māori fishing rights to a head. The implementation of the QMS led to particular concern by Māori that rights to fisheries guaranteed under the Treaty of Waitangi were being alienated by the Crown. In 1987 the High Court granted claimants relief through an injunction on the further implementation of the QMS pending settlement of the claims.

At that time, the Waitangi Tribunal was already considering claims in respect of fisheries resources and the 1988 Muriwhenua report found that the Crown was in breach of Treaty obligations. Significantly, the Tribunal found that the QMS was a fit and proper system for the management of fisheries, but that allocation of rights had not recognised the Māori interest.

This finding led to the interim settlement of claims legislated in the Māori Fisheries Act 1989, which provided for 10 per cent of all existing quota to be provided to Māori by the Crown. As a result, government bought back 10 per cent of the quota shares it had given to fishers under the QMS, and transferred these to the Treaty of Waitangi Fisheries Commission, for the benefit of Māori.

A further report by the Tribunal on claims by Ngāi Tahu found a more extensive 'development interest' was inherent in the Treaty provisions and history of Māori fishing. This led to further negotiations between Māori and the Crown, culminating in the full and final settlement of commercial claims to fisheries in 1992. This involved government giving Māori a cash settlement that was used to buy half of New Zealand's biggest fishing company, Sealord. The government also agreed to give Māori 20 per cent of the commercial quota shares of any new species brought into the QMS.¹¹

The result of these settlements is that Māori customary commercial fishing rights are managed under the QMS in the same way as other commercial fishing interests.¹²

Customary non-commercial fishing rights

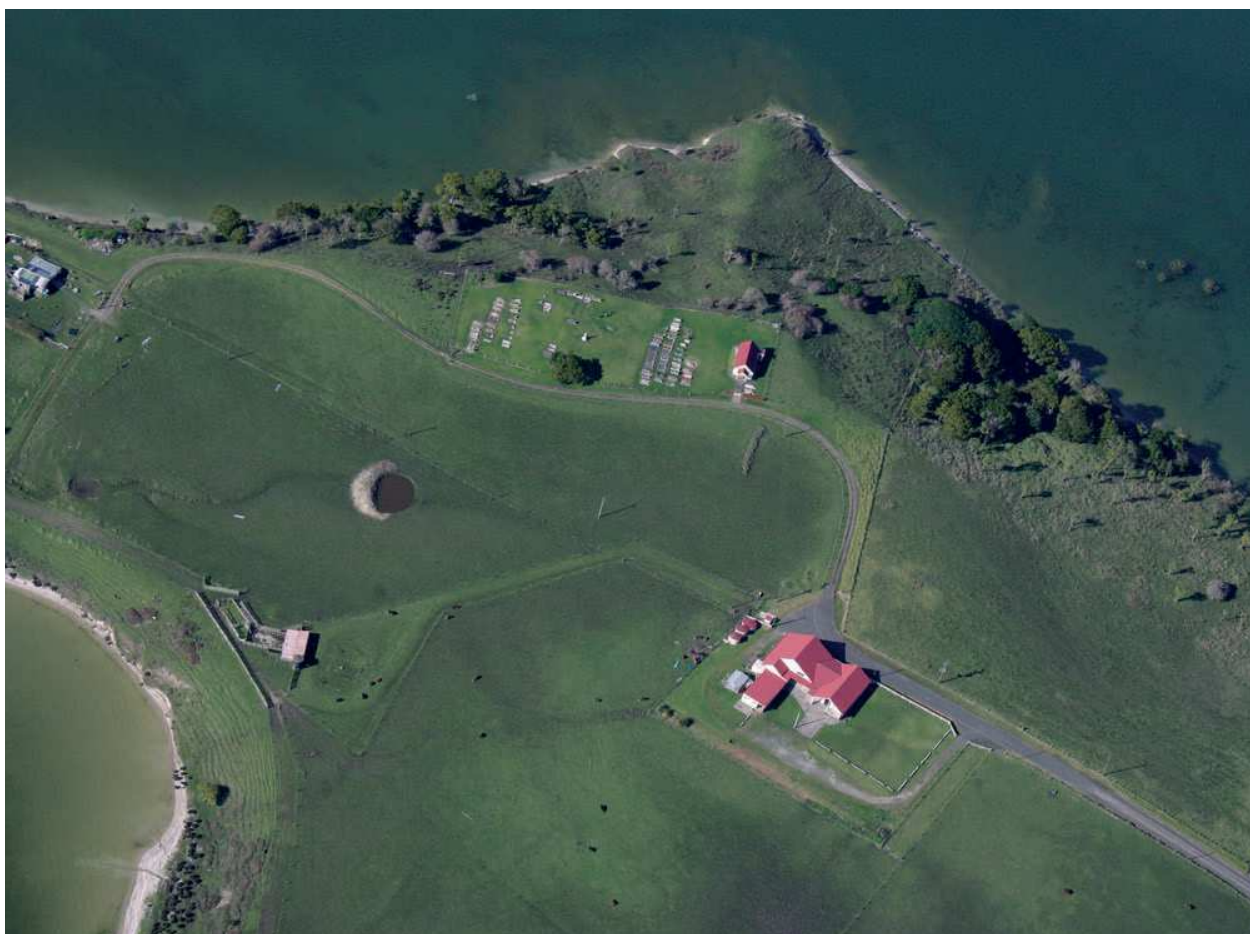
Provision for Māori customary non-commercial fishing rights has been less well defined. The Waitangi Tribunal reports contain useful information on traditional fishing practices and associated rights. However there has as yet been no systematic investigation into the nature and extent of non-commercial customary fishing rights.

The legislation providing for the Sealords Deal also addressed non-commercial rights and interests to some extent. It provided that:¹³

- Māori non-commercial fishing claims would continue to give rise to Treaty obligations on the Crown
- The Minister would 'develop policies to help recognise use and management practices of Māori in the exercise of non-commercial fishing rights'
- The Minister should recommend the making of regulations 'to recognise and provide for customary food gathering by Māori and the special relationship between tangata whenua and those places which are of customary food gathering importance (including tauranga ika and mahinga mataitai), to the extent that such food gathering is neither commercial in any way nor for pecuniary gain or trade'

The Sealord legislation also provided that 'the rights or interests of Māori in non-commercial fishing' shall have no legal effect except to the extent that they are provided for in regulations made under the Fisheries Act. As a result, Māori rights to customary non-commercial harvest are now limited by regulatory provisions.

'Maori customary non-commercial fishing interests' are allowed for under the Fisheries Act 1996 when setting the total allowable commercial catch.¹⁴ But the main mechanism for providing for such interests is through the provisions of the Fisheries (Kaimoana Customary Fishing) Regulations 1998 and the Fisheries (South Island Customary Fishing) Regulations 1999. These provide for Māori to manage 'customary food gathering' through the appointment of kaitiaki who can issue permits to authorise the harvest of fish in their area. Implementation of these regulations is still incomplete around the country and much of the current Māori non-commercial fishing activity is undertaken under the recreational fishing regulations (described in Chapter 16: Marine recreation and tourism). The fisheries regulations also provide for the establishment of mātaihai reserves which are discussed below.



Much Māori settlement was closely related to the marine environment, as shown here by the Hukatere Marae on the Kirikiri inlet, Kaipara Harbour (Kevin L Jones)

In response to the ongoing debate a review was conducted which led to the enactment of the Marine and Coastal Area (Takutai Moana) Act 2011. The new legislation defines a 'common marine and coastal area' which includes the marine and coastal area, excluding existing freehold title and areas owned by the Crown as conservation areas, national parks or reserves. It states that the common marine and coastal area has a 'special status' and that neither the Crown nor any other person owns, or is capable of owning it.¹⁶ Every person has the right to enter, pass over, and engage in recreational activities in the common marine and coastal area.¹⁷

Marine and Coastal Area (Takutai Moana) Act, section 9, interpretation

Marine and coastal area:

- (a) means the area that is bounded,—
 - (i) on the landward side, by the line of mean high-water springs; and
 - (ii) on the seaward side, by the outer limits of the territorial sea; and
- (b) includes the beds of rivers that are part of the coastal marine area (within the meaning of the Resource Management Act 1991); and
- (c) includes the airspace above, and the water space (but not the water) above, the areas described in paragraphs (a) and (b); and
- (d) includes the subsoil, bedrock, and other matter under the areas described in paragraphs (a) and (b)

Marine management tools

Mātaítai reserves

Mātaítai reserves can be declared under the Fisheries (Kaimoana Customary Fishing) Regulations or the Fisheries (South Island Customary Fishing) Regulations on application by tangata whenua. These are areas which are recognised traditional fishing grounds, with which tangata whenua have a special relationship. The reserve is managed by tangata tiaki/kaitiaki chosen by the tangata whenua. The tangata tiaki/kaitiaki can restrict or prohibit fishing in the mātaítai reserve by recommending bylaws. Although the bylaws apply to any person fishing in the area, fishing for the purposes of sustaining the functions of a marae may continue if authorised by the tangata tiaki/kaitiaki.

There is a clear process for establishing mātaítai, with an application being lodged by the tangata whenua or tangata tiaki/kaitiaki (who has been confirmed by the Minister) with the Ministry for Primary Industries. Before approving the application, the Minister must be satisfied that a special relationship exists between the tangata whenua and the proposed reserve, that the general management aims are consistent with the sustainable use of fisheries resources in the area, and that the area is an identified traditional fishing ground and is of a size which can be effectively managed by tangata whenua. In addition, the Minister must also be satisfied that the reserve will not unreasonably affect the ability of the local community to take fish for non-commercial purposes or prevent persons taking their quota or annual catch entitlement within the quota management area for that species.

Once a mātaítai reserve is established, commercial fishing is not allowed unless it is re-instated by regulation. The regulations must be recommended by the tangata tiaki/kaitiaki, and apply to specified species, restricted by quantity or time period. Māori and non-Māori may fish in mātaítai reserves.

Taiāpure and rāhui

Part 9 of the Fisheries Act also creates opportunities for the practical application of kaitiakitanga through the establishment of taiāpure and the temporary closure of fishing areas (rāhui). These recognise and make provision for the use and management practices of tangata whenua.



Maketū Estuary is part of a taiāpure established in 1996 (Raewyn Peart)

Mount Maunganui Mātaitai Reserve

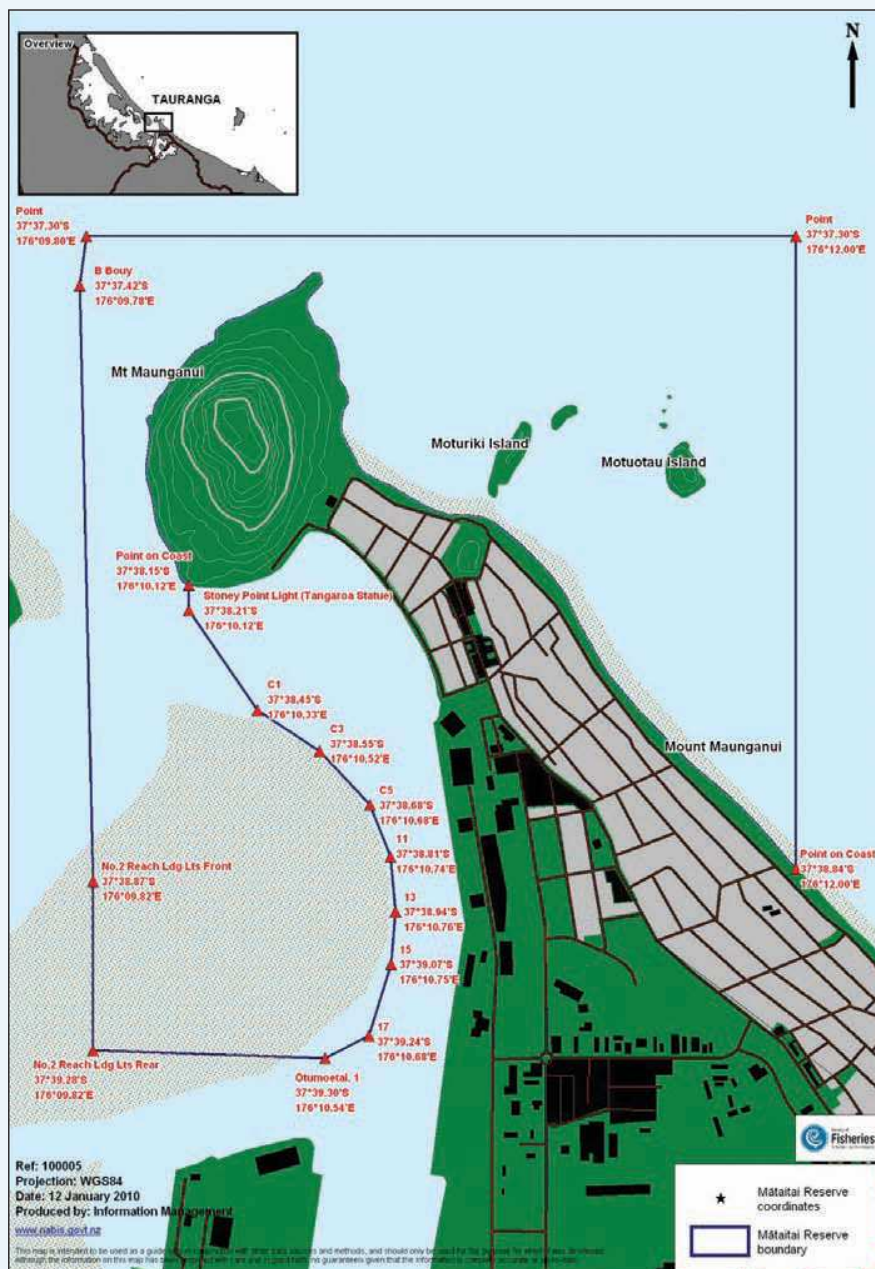
The Mount Maunganui Mātaitai Reserve, established in 1998, covers the waters surrounding Mount Maunganui and part of Tauranga Harbour. It recognises the strong and enduring connection that local hapū have with this area.¹⁹

The local Tauranga Moana, Ngaiterangi, Ngāti Ranginui and Ngāti Pukenga advise the Ministry for Primary Industries directly on how best to manage fishing in the reserve. Commercial fishing is banned within the 6 square kilometre reserve area, but recreational and customary fishing is still allowed, and recreational fishing does not require a permit from local Māori.

This mātaitai reserve is in place to allow the Tauranga Moana tribal groups to more effectively manage customary fishing in important traditional fishing grounds. For example, a temporary closure was put in place during the late 2000s for the taking of green-lipped mussels between Moturiki and Motuotau Islands to assist with the recovery of the population.

This initiative has helped improve public understanding of the relationship of local iwi with the area and enables tangata whenua to more effectively manage customary fishing through marine management tools. It demonstrates the need to be cognisant of wider community desires for use and management of a marine area and has offered the opportunity for public education about kaitiakitanga.

The grounding of the *Rena* in 2011 provided a valuable opportunity for testing the protection framework that has been put in place at the Mount Maunganui Mātaitai reserve. The *Rena* disaster is discussed as a case study in Chapter 15: Vessels. The response to clean up the oil that had arrived on the shores of the Mātaitai reserve was swift because of the known high values at the site. The site is now listed within the 'Bay of Plenty Marine Oil Spill Contingency Plan 2014' as a 'Site of cultural and spiritual significance to iwi', with these resources being identified as 'At Risk'.²⁰ The 'Rena Kaitiaki Impact Assessment' prepared in 2012 to provide the *Rena* Long-Term Environmental Recovery Team with a tangata whenua perspective on the impacts of the disaster noted that 'the oil and debris that came ashore in these areas is still having an effect on their confidence in the health of the kaimoana'.²¹



Boundary of Mount Maunganui Mātaitai Reserve (Ministry of Fisheries)²²

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Endnotes

- 1 <http://www.teohu.maori.nz/>
- 2 Business and Economic Research Limited, 2010
- 3 Aotearoa Fisheries business comprises Moana Pacific Fisheries, OPC Fish and Lobster, Prepared Foods, Pacific Marine Farms and Kia Ora Seafoods
- 4 Inns J, 2013
- 5 Some groups of iwi, such as the 12 Hauraki iwi or the 11 Te Arawa waka iwi, have a single entity to represent them identified in statute
- 6 EEZ Act, section 12
- 7 A group to which a customary marine title order applies. A customary marine title order means an order of the Court granted in recognition of a customary marine title of a customary marine title group. Customary marine title exists in a specified area if the applicant group holds the specified area in accordance with tikanga, and has exclusively used and occupied the area from 1840 to the present day without substantial interruption, or received the area after 1840 through a customary transfer. Marine and Coastal Area (Takutai Moana) Act 2011, section 9
- 8 A group to which a protected customary rights order applies. A protected customary rights order means an order of the Court granted in recognition of the protected customary rights of a protected customary rights group. A protected customary right is a right that has been exercised since 1840, which continues to be exercised in a particular area in accordance with tikanga by the applicant group (the way it is exercised may evolve over time), and is not extinguished as a matter of law. Marine and Coastal Area (Takutai Moana) Act 2011, section 9
- 9 RMA, section 7(a)
- 10 <http://www.fish.govt.nz/en-nz/Commercial/About+the+Fishing+Industry/default.htm>
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- 12 Meredith P, 2012
- 13 Treaty of Waitangi (Fisheries Claims) Settlement Act 1992, section 10
- 14 Fisheries Act 1996 section 21(1)
- 15 *Attorney-General v Ngati Apa* [2003] 3 NZLR
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- 19 <http://www.fish.govt.nz/en-nz/Archive/News/Press+Releases+2008/August08/Mt+Maunganui+and+Tauranga+Harbour+mataitai.htm>
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- 21 http://www.renarecovery.org.nz/media/24749/mauao_papamoa.pdf
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- 26 <http://www.port-tauranga.co.nz/Media%20Room%20and%20Publications/News%20Archive/Port-of-Tauranga-gains-approval-for-harbour-dredging/>
- 27 Bay of Plenty Regional Council, 2014

5

Marine biosecurity



Invasive Mediterranean fan worm (*Sarah Faulwetter*)

Introduction

As an island nation in a global economy, New Zealand faces increasing threats to its marine biodiversity, ecosystems, public health and economy from unwanted pests and diseases. Biosecurity is a component of national security and is essential to protecting New Zealand's key economic and environmental assets.¹

Exotic organisms can compete with, or prey on, other species, change habitats and damage ecosystems. They can affect recreational and customary values, and impact on the economy. They do this by competing with economically important native species and reducing overall biodiversity. Recent incursions of invasive marine aquatic species are highlighting the difficulties associated with managing aquatic pests once they reach our shores.

National leadership for marine biosecurity is the responsibility of the Ministry for Primary Industries, with responsibility for regional leadership sitting with regional councils. The Ministry's emphasis is on prevention, detection and immediate response. Other agencies, including regional councils and industry, are playing an increasingly active role in biosecurity management including undertaking longer-term management of incursions.



Invasive *Eudistoma elongatum* (sea squirt) on the Sandspit estuary in Auckland (Roger Grace)

Invasive marine species

Marine pests are species that are not indigenous to New Zealand which have been introduced to our waters by human activities (and less commonly through natural dispersal) and which have the potential to significantly damage marine ecosystems and species.

Many marine invasive species have become established in New Zealand. Some have been in the country for a long time, such as the Pacific oyster. Others are more recent arrivals, such as *Undaria pinnatifida* (Japanese kelp). Both the Pacific oyster and Japanese kelp are now well established and have been adopted into the country's aquaculture industry.

In 2008, 288 non-indigenous species were recorded.² These include algae (both large and microscopic), barnacles, crabs, bryozoans, sea squirts, mussels and other molluscs. Figure 5.1 shows the cumulative number of marine non-indigenous species discovered in New Zealand over time. It is worth noting that at least part of the significant rise in the number of detections between 2001 and 2010 is likely due to the active surveillance programmes put in place during that period, and greater awareness of the negative impacts of marine pests.

Surveillance programme (both discussed below), as well as NIWA's Marine Invasive Taxonomic Service (which identifies marine specimens on behalf of the Ministry) and other verified observations. This portal plays an important role in distributing information about current knowledge on non-native marine organisms in New Zealand.

Additional information on marine pests can be found at the Marine Biosecurity Porthole website.⁷



Styela clava, which is well established in New Zealand, competing with native species and fouling vessels and equipment
(Jeannine Fischer, Ministry for Primary Industries)

Impacts

Invasive marine pests can cause a number of negative environmental impacts. They often compete with native species for space and food, and some introduced species appear to be more aggressive competitors than native ones. By smothering areas of seafloor, some pests fundamentally change habitats.⁸ Figure 5.2 provides a summary of some of the key impacts from particular species.

Species	Status in New Zealand	Impacts
Clubbed tunicate/ Leathery sea squirt <i>Styela clava</i>	Established	• Fast-growing and forms dense groups • Filter feeder that competes with native and aquaculture species for food (planktonic larvae) and space • Nuisance fouler of vessels, aquaculture and fishing equipment and other artificial structures
Mediterranean fan worm <i>Sabella spallanzanii</i>	Established	• Forms dense groups • Competes with native species for food and space • Negative impact on the establishment of new generations of some species, and on nutrient flow in the water column • Dense beds foul fishing equipment and aquaculture structures
Japanese kelp <i>Undaria pinnatifida</i>	Established	• Rapidly forms dense stands that overgrow and exclude native seaweed species • Nuisance fouling causes problems and increased costs for aquaculture

Invasion pathways

Pathways are human activities that, intentionally or unintentionally, may move a harmful organism from one place to another.¹⁰ The Biosecurity Act defines a 'pathway' as 'movement that is of goods or craft out of, into, or through a particular place in New Zealand or a particular kind of place in New Zealand, and has the potential to spread harmful organisms'.¹¹

There are many marine pathways, but the most pressing issue facing marine biosecurity is increasing international shipping and the consequential transfer of non-indigenous species from one country to another via vessels.

Spread of Whangamatā sea squirt

The Whangamatā sea squirt (*Didemnum vexillum*) is a marine pest that was first seen in Whangamatā in 2001. It has quickly established itself in the Marlborough Sounds, Port Nelson, Golden Bay, Whangamatā and Tauranga. It is a leathery or spongy, light mustard-coloured sea squirt. It can look like wax dripping over a structure such as a rope or mussel line and can quickly smother sea life. The Whangamatā sea squirt can be spread over long distances by infected vessel hulls and marine equipment and as fragments in ballast water.¹²

Biofouling

Biofouling risk occurs when marine organisms attach themselves to the hulls of vessels overseas and are then transported into New Zealand waters on the vessel. Hull fouling is the process where organisms accumulate on a surface in the sea. In the initial stages, organic material sticks to the hull and is rapidly colonised by bacteria and microalgae forming a 'slime layer'. This layer is then colonised by larger organisms such as macroalgae, barnacles, bryozoans and tube worms, creating more complex fouling. Over time, as the complexity of fouling increases, habitat is created for other organisms such as isopods and crabs. As a result, invasive species can be transported around the world by vessels with fouled hulls. Many of these species are relatively benign, but some have caused major impacts. For example, the sea squirt *Didemnum vexillum* arrived in Picton on a vessel hull and has become a major marine pest in the Marlborough Sounds. Biofouling is the most common cause of marine pest invasions. Large structures which are towed into New Zealand waters, such as drilling rigs, can also create biofouling risks. Biofouling of vessels and structures (both international and domestic) moving around the country is also responsible for the spread of pests around New Zealand once they arrive here.



Undaria, shown here on a mooring line, has been spread around the country on fouled hulls
(Kath Blakemore, Department of Conservation)

Spread of *Styela* through biofouling on recreational vessels

In November 2013 a boat infested with the unwanted marine sea squirt *Styela clava* was found berthed in Tauranga Harbour. It was the second boat with the sea squirt attached to its hull that had been found in the harbour. One of the boats was moored in Pilot Bay and the other was berthed at Bridge Marina. Both boats had come from Auckland, where the sea squirt is established, without prior cleaning of their hulls.¹³

Spread of *Undaria* to Fiordland

The Asian seaweed *Undaria pinnatifida* has established in many parts of New Zealand over the last 20 years. However, much of Fiordland has remained free from this invasive marine pest. *Undaria* is an highly invasive seaweed that rapidly overruns native species, altering marine ecosystems. In April 2010 a single mature *Undaria* plant was found on a line mooring a barge to the shoreline in the remote Sunday Cove in Breaksea Sound.¹⁴ Fiordland is a high-value area and the presence of *Undaria* in Fiordland is considered a threat to this globally unique marine environment. As a result of the discovery, and following consultation with stakeholders and tangata whenua, a response involving the Ministry for Primary Industries, the Department of Conservation and Environment Southland (lead agency) was initiated to eliminate *Undaria* from Fiordland. Joint agency decision-making is undertaken at both the strategic and operational level, with an equal three-way split of operational costs.

The Biosecurity Act requires masters of craft arriving from overseas to give notice of when and where they will enter New Zealand so as to prevent uncleared goods leaving the vessel without authorisation from an inspector. The Biosecurity Standard 'Requirements for Vessels Arriving in New Zealand'¹⁵ sets out additional rules.

Part 4 of the Biosecurity Act deals with surveillance and prevention and its purpose is 'to provide for the continuous monitoring of New Zealand's status in regard to pests and unwanted organisms'. Surveillance is essential for detecting pests and diseases before they become established in New Zealand. Early detection minimises the impacts of newly-introduced marine pests and diseases, and is especially important because the opportunity to respond and eradicate diminishes as more time elapses before they are detected.

Part 4 of the Act seeks to promote early detection through placing duties on all persons to undertake timely reporting of organisms not normally seen in New Zealand.¹⁶ It also provides legal powers to enable the Ministry for Primary Industries to gather information on organisms' presence, to prevent them from spreading and to enable their identification. The Ministry undertakes several national surveillance programmes to aid early detection.

Ministry for Primary Industries surveillance programme

Biological baseline surveys for non-indigenous marine species were completed at 16 major ports and marinas of international entry between 2001 and 2007. An additional eight secondary ports and pristine locations have also been surveyed. The Ministry for Primary Industries now maintains a regular 'targeted surveillance programme' which surveys 11 high-risk locations biannually, in winter and summer, for a suite of target organisms.

Pest management is dealt with under Part 5 of the Biosecurity Act and its purpose is 'to provide for the eradication or effective management of harmful organisms that are present in New Zealand'. The Minister is currently developing a national policy direction to guide the implementation of these functions. Its purpose is to ensure that pest management activities provide the best use of available resources for New Zealand's best interests. Part 5 of the Act sets out the process for the development of a range of different pest management plans.



Mediterranean fan worm *Sabella spallanzanii* (Polychaeta) (Sarah Faulwetter)

Type	Circumstances	Lead intervention decision-maker responsible for bringing parties with the necessary powers, functions and resources together	Reason for role
	Risk to any national or regional value associated with intra-regional movement of vectors (e.g. of structures, equipment and vessels)	Regional councils	Have regional capacity and powers to act in the public interest
	Risk to any national or regional value associated with development of marinas, wharves, jetties and moorings and the ongoing maintenance of such facilities		Have powers under the RMA (e.g. can include conditions in resource consents)
	Risk to any national or regional value associated with dumping of organic material from vessels (within the 12 nautical mile limit and on land)		Administer the Resource Management (Marine Pollution) Regulations 1998
	Risk to any national or regional value associated with dumping of organic material from vessels and offshore installations in the EEZ (from the 12 to 200 mile nautical limit)	Maritime New Zealand	Has authority and responsibility in the EEZ under the Maritime Transport Act
Site/place management (Management to protect values of specific places)	Marine reserves, marine parts of wildlife management reserves and sanctuaries, reserves and national parks administered by Department of Conservation	Department of Conservation	Administers these protected areas under the Marine Reserves Act, Wildlife Act, Marine Mammals Protection Act, Reserves Act and National Parks Act
	Coastal marine areas of the Subantarctic Islands and Kermadec Islands	Minister of Conservation	Minister of Conservation has the responsibilities, functions and powers of a regional council of section 30(1)(d) under the RMA for these areas. The department may act on behalf of the Minister
	Marine protected areas administered by bodies other than the Department of Conservation	The primary administering body with the necessary powers	Marine protected areas can be established under various statutes with potentially multiple administering agencies. Some agencies will have the necessary administering powers and functions and others will not.

Figure 5.4 highlights the different ways marine pests are managed in New Zealand during the different stages of entry and incursion.

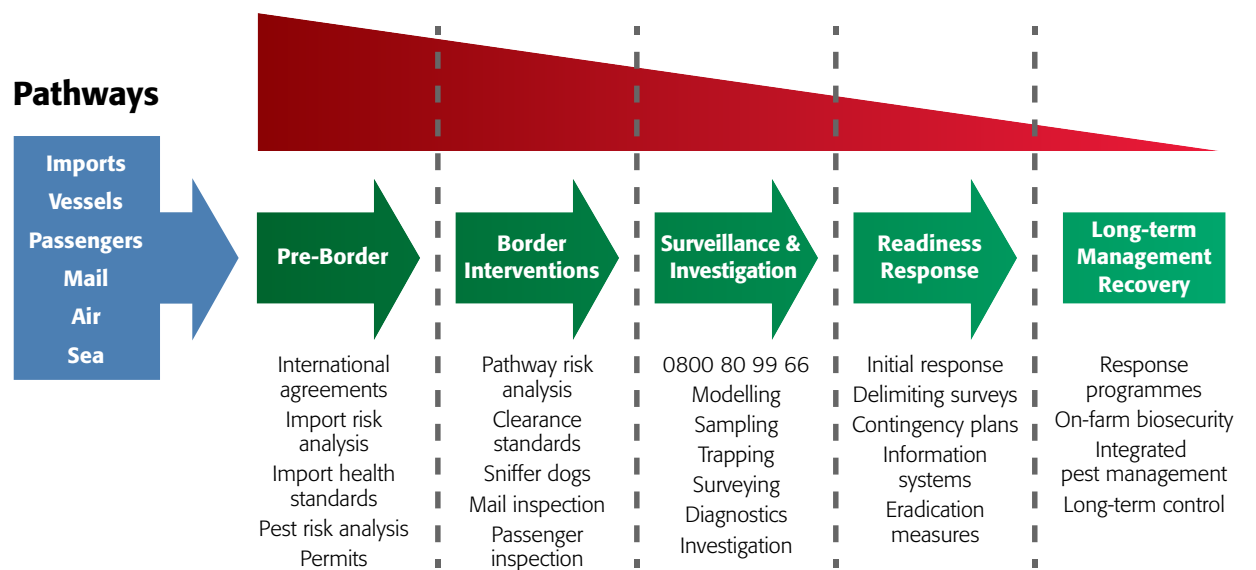


Figure 5.4 Tools used to manage marine pests (Ministry for Primary Industries)

Preventing arrival

The first line of defence is the prevention of new marine pests arriving into New Zealand waters. This requires both an international and a domestic approach. New Zealand is working with international organisations to develop measures for minimising the transfer of marine pests. There are also a number of unilateral tools which are used to prevent the entry of unwanted pests into New Zealand, including import health standards, risk analysis and craft risk management standards.

Import health standards

Import health standards apply to the import of items into New Zealand that pose a biosecurity threat. The standards seek to mitigate the risks associated with bringing items into New Zealand. They set out the requirements that must be met in the exporting country, during transit and during importation, before biosecurity clearance can be given.¹⁹

Risk analysis

An analysis of risk is the first stage in the development of import health standards. It involves the identification of pests and diseases that might be associated with a good, the likelihood of entry and establishment in New Zealand, and the potential impacts on the economy, environment and human health. Risk analysis is still an evolving discipline, but it is now an accepted basis under the World Trade Organization Sanitary and Phytosanitary Agreement for establishing international trading standards, where safety aspects of animal and plant products are concerned. New Zealand is one of the countries leading the way in establishing risk analysis procedures internationally.²⁰

Ships' ballast water

The 'Import Health Standard for Ships' Ballast Water from all Countries' was issued in 2005. It applies to ballast water loaded within the territorial waters of a country other than New Zealand and intended for discharge in New Zealand waters. It outlines the requirements for any vessel discharging ballast water in New Zealand waters, including four options for vessels:²¹

- The ballast water must have been exchanged en route to New Zealand in an area free from coastal influences, preferably 200 nautical miles from the nearest land and in water over 200 metres in depth (this requirement is based on the assumption that species adapted to living in surface waters of the open sea are relatively unlikely to establish in New Zealand's coastal environment); or

Identifying incursions

Surveillance for new marine pests

Surveillance is the process of systematically collecting, analysing and interpreting information about the presence or absence of unwanted organisms.²⁵ The Ministry for Primary Industries' marine biosecurity surveillance programme falls into three main groups: baseline, targeted and passive (see Figure 5.5).

Programme	Involves	Current examples
Historical – Baseline surveillance	- Obtaining baseline information at highest risk sites - Providing a benchmark against which the effectiveness of border control and other management practices can be measured	- Baseline surveillance for marine pests between 2001–03 and 2005 - Baseline surveillance has been completed on non-indigenous marine species at 16 major ports and marinas of international entry - An additional eight secondary ports and hubs have also been surveyed
Current – Targeted surveillance	- Focusing on a specific organism or group of related organisms - Can be designed as an ongoing programme or as a one-off survey - Maintain a watch for non-target, new-to-New Zealand organisms	- Five marine pests are currently targeted (Chinese mitten crab, European shore crab, Asian clam, <i>Caulerpa taxifolia</i> and Northern Pacific seastar) - Targeted surveillance programmes at 11 high-risk locations, biannually, for a range of target organisms
Current – Passive surveillance	- Asking all persons in New Zealand (including public and industry) to keep watch for unwanted pests and diseases - Investigating notifications of suspected unwanted pests and diseases - Monitoring and analysing trends in information relating to pests and diseases	- Targeted communication material such as lists of notifiable pests, diseases and unwanted organisms, and guides for vessel owners circulated to raise awareness and enhance passive surveillance - The Exotic Pest and Disease Hotline (0800 80 99 66) - Incursion Investigators

Figure 5.5 Marine biosecurity surveillance programmes

Marine investigations

An incursion is where an organism not previously known in New Zealand has been found and is considered likely to be part of a self-sustaining or breeding population. Re-invasion of a species that has already been eradicated or controlled is considered a new incursion.

The Ministry for Primary Industries has a dedicated team which follows up on suspected new marine pests. In 2013 there were 24 marine investigations carried out.²⁶ The incursion team is notified of suspected pests in the marine environment from a number of different sources, including scientists, surveillance programme laboratories, fishers, marine farmers, the general public, other parts of the Ministry, local government and other specialist groups. A notification usually comes through the Exotic Pests and Diseases Hotline which then passes the information on to the appropriate duty incursion investigator. In some cases, investigators will require samples to be submitted by the informant, and these will be sent to the Marine Invasive Taxonomic Service for species identification. The Investigation and Diagnostic Centre is responsible for investigating any suspected unwanted organism.

Response to *Pyura* incursion³³

In 2007 large ascidians (*Pyura doppelgangera*) were found covering some rocks in the Far North. By May 2009 these ascidians were covering a much wider area than the original detection site. Using the Biosecurity Response Knowledge Base, the Ministry for Primary Industries undertook an assessment of the potential impact of the invasion to New Zealand's core values. This indicated medium-level risks to the economy, the environment, social and cultural well-being and Māori values. The values at risk were assessed as outlined in Figure 5.6.

	Impact
Economic	• Medium risk to the aquaculture industry, in particular mussel farming • Approximately 80 per cent of the mussel spat that supplies the mussel industry is collected from the Ninety Mile Beach area, which is in the vicinity of the known location of the risk organism • The origin of the mussel spat supply is unknown, as is the contribution of intertidal mussel populations in the Far North to spat supply • The ascidian can form large, dense colonies, crowding out adult mussels in rocky intertidal areas • The local spat-collecting economy could also be impacted • <i>Pyura</i> could inadvertently be spread to mussel farms via the seaweed bearing the mussel spat
Environmental	• Medium risk to native marine species and ecological processes • In Chile, a similar non-indigenous <i>Pyura</i> species is now the dominant species in the mid-to-lower intertidal area with the native mussel confined to the mid-to-upper intertidal area • Observations in the Far North showed that <i>Pyura</i> was the dominant organism in some rocky intertidal areas, suggesting that it had the potential to cause local-scale ecological disturbance to rocky intertidal communities, including competing with mussels
Social and cultural well-being	• Medium risk • Recreational mussel gathering may be affected if <i>Pyura</i> displaces or significantly reduces mussel beds • Aesthetic values may be affected as <i>Pyura</i> forms dense mats, changing the intertidal landscape • On the other hand, the sea squirt may be used as fish bait, as is currently the case in Australia and South Africa for related species
Māori values	• Medium risk • Iwi in the Far North rely on locally harvested shellfish • Cape Reinga is spiritually significant and the proximity of the detection of <i>Pyura</i> is of concern to iwi

Figure 5.6 Impact assessment of the *Pyura doppelgangera* invasion

A survey was undertaken to determine the extent of the infestation. The survey showed *Pyura* was present at many locations on the west and eastern coasts of the Far North, suggesting the species had been present for some time prior to the initial detection. The Ministry decided that eradication could not be achieved, given the widespread nature of the incursion. However, local iwi and stakeholders were very concerned about the marine pest, and the Ministry decided to trial the feasibility of local elimination at three sites in the Far North beginning in 2010.

The field team comprised local iwi, who received training in marine biosecurity and field survey methods. The results of the trial suggested that periodic clearances of the ascidian could suppress the populations at specific sites. The Ministry, the Department of Conservation and Northland Regional Council contributed resources. The Te Hiku o te Ika Fisheries Forum, comprising representatives of mandated iwi organisations, participated throughout the trial. The Forum developed a marine biosecurity strategy for the Far North and is currently working with the Northland Regional Council to develop a long-term management plan for this marine pest.



Pyura has been found in the Houhora Harbour as well as in numerous other locations in the Far North (Raewyn Peart)

Region	Approach
Bay of Plenty	• A scientist conducting survey work in Tauranga Harbour detected a single <i>Sabella</i> on the rocky substrate of Pilot Bay and notified the Ministry in September 2013 • The Bay of Plenty Regional Council and the Ministry jointly responded to this detection and a delimiting survey was carried out at all high-risk sites in the harbour, including in the vicinity of the detection, moored vessels and moorings, wharves at the port and the two marinas during the next two months • A small number of <i>Sabella</i> were found in one of the two marinas and a single <i>Sabella</i> was found on the hull of a yacht moored in the vicinity of the original detection. The owner was contacted and voluntarily agreed to have the yacht urgently slipped and defouled. • During the Marine High Risk Site Surveillance programme in May 2014, a small number of <i>Sabella</i> were detected in the two marinas • The regional council quickly responded the following week, again with support from the Ministry, by surveying both marinas • Further <i>Sabella</i> were found in low numbers • The recent detection of <i>Sabella</i> suggests that some were most likely missed during the first survey, particularly where water visibility was low, as there are high levels of biofouling species in the marinas and the fan worms are small in size • The regional council has developed a marine biosecurity management plan to respond to future incursions of marine pests and to carry out surveillance programmes within the council's harbours
Nelson	• In November 2013, a single <i>Sabella</i> was found in the Nelson Marina during the Marine High Risk Site Surveillance programme • The Nelson City Council and the Ministry jointly responded and implemented a delimiting survey of the entire marina and other high-risk sites in and around the port during the same month as the initial detection • This survey found and removed 12 <i>Sabella</i> of varying tube lengths • The agencies decided to undertake a second survey prior to the 2014 winter, in case some animals were missed, which proved to be a prudent measure as a further 22 fan worms were found • The council has decided to continue surveillance for the fan worm over the next year. The total number of fan worms collected (less than 40), combined with the relatively confined area they were found (the central area of the marina), gives confidence that this incursion can be suppressed.
Marlborough	• In February 2014, a research scientist surveying fouled vessels in Waikawa Bay, Marlborough Sounds collected <i>Sabella</i> from a heavily fouled yacht on a mooring • The Marlborough District Council and the Ministry agreed to jointly respond to the incursion • The Council arranged to get the vessel slipped and defouled within two weeks of detection, with the agreement of the owners • Tracing showed the yacht had relocated from Auckland to Waikawa Bay approximately one to two years previously, suggesting that it had probably become infested in Auckland • A delimiting survey was undertaken in early April and all high-risk areas, including the Waikawa Marina, were inspected with no further <i>Sabella</i> found • The Marine High Risk Site Surveillance programme included the Waikawa Marina during June as part of its usual programme, with no detection of <i>Sabella</i> • Targeted surveillance for <i>Sabella</i> will search all the high-risk sites later in 2014

Figure 5.7 Regional responses to *Sabella* incursions

The four most recent incursions demonstrate that the goal of local elimination is more likely to be successful if surveys and removal of the *Sabella* are undertaken soon after detection and prior to reproductive maturity. However, the risk of incursions from newly-arriving infested vessels remains. Large numbers of mature *Sabella* were recently removed from the hull of a coastal freighter that travels between New Zealand ports. A combination of pathway management, readiness and response will be necessary to ensure *Sabella* does not spread and establish into new areas.

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6

Marine protected areas



Diver enjoying school of demoiselle, Tuhua (Mayor Island), Bay of Plenty (Daniel Sharp)

Introduction

The identification of spatial marine areas is a useful approach to help preserve marine biodiversity as well as to maintain the overall health of the marine environment. Such areas commonly fall under the umbrella term 'marine protected areas'. This term is not currently defined in statute in New Zealand. However, the 'Marine Protected Areas Policy Statement and Implementation Plan' prepared jointly by the Department of Conservation and the former Ministry of Fisheries in 2005 has defined a marine protected area as being:

An area of the marine environment especially dedicated to, or achieving, through adequate protection, the maintenance and/or recovery of biological diversity at the habitat and ecosystem level in a healthy functioning state.¹

Government has adopted the objective of establishing a network of marine protected areas. In 2000 the Government released the New Zealand Biodiversity Strategy in order to fulfil, in part, the country's commitments under the Convention on Biological Diversity. As an action to achieve the objective of protecting marine habitats and ecosystems, the Biodiversity Strategy identifies 'a target of protecting 10 percent of New Zealand's marine environment by 2010 in view of establishing a network of representative protected marine areas'.

Marine protected areas can be designed to protect complete habitats and ecosystems within an area, or can be more specifically tailored to protect specific habitats such as seamounts. Spatial areas can also be used to protect specific species such as marine mammals or seabirds, although such areas are not included within the definition of marine protected areas set out in the Government's policy statement.



The Te Matuku Marine Reserve protects a diverse range of estuarine habitats and is important habitat for seabirds (Raewyn Peart)

Potential benefits of marine protected areas

Marine protected areas can have a range of ecological, social, cultural and economic benefits which are summarised in Figure 6.1. The extent to which these benefits can be achieved in practice is largely dependent on the location, design and ongoing management of the areas.

- Protection of biodiversity
- Increased productivity
- Increased resilience of the marine environment to damage from human activity, thereby safeguarding ecosystem services
- Provision of 'benchmarks' against which the impacts of human activities in other parts of the marine environment can be compared
- Improved fisheries through spillover of eggs, larvae and adult fish
- Protection of geological processes
- Protection of cultural values
- Increased opportunities for tourism and recreational activities
- Maintenance of New Zealand's 'green' international image
- Increased opportunities for education and science (this is the thrust of the Marine Reserves Act)

Figure 6.1 Potential benefits of marine protected areas

The spillover of harvestable species from reserves is evident from the presence of fishers around the edges of the protected areas. For example, cray pots often line the reserve boundaries around the marine reserve at Leigh.⁸ The likelihood of such spillover depends on the mobility and habits of the species concerned, conditions inside and outside the marine protected area, and the area's design (e.g. longer edges enhance the likelihood of spillover).⁹

Marine reserves can support denser populations of larger fish for resident species that are heavily fished.¹⁰ A recent study of three marine reserves in the Greater Hauraki Gulf (at Leigh, Tāwharanui and Hahei) found that there were on average 13 times more snapper above the legal size limit within the reserves than outside them.¹¹ This has important implications for reproduction. Larger fish can produce far more eggs, per unit of body weight, than smaller fish. In addition, there is evidence that the larvae produced by larger fish have better survival rates.¹²

Great Barrier Reef Marine Park larvae spillover

A genetic study of adult and juvenile coral trout and stripey snapper within the Great Barrier Reef Marine Park demonstrated that the reserves contributed a greater proportion of the juvenile fish in fished areas than their size would indicate. The six reserves in the Keppel Islands comprised around 28 per cent of the reef habitat in the region but supplied around 50 per cent of the total recruited fish. The scientists concluded 'our findings indicate that adults in reserves were making a large contribution to the replenishment of populations on both reserve and fished reefs'.¹³ Recent research at the Cape Rodney-Okakari Point Marine Reserve supports these findings, indicating that snapper larvae from the reserve make a strong contribution to surrounding fished stocks.¹⁴

Protection of geological features and processes

Marine protected areas can be used to protect geological features and processes identified as particularly unique or typical. This may be closely interlinked with the protection of the biodiversity they support. For example, deepsea vents are valued both as rare geological features and for the unique ecological communities which are dependent on them.

Protection of cultural values

The protection of marine space can be of considerable cultural value. For example, the establishment of Te Whanganui A Hei – (Cathedral Cove) Marine Reserve was supported by Ngāti Hei, because the area included tapu sites associated with tribal massacre. Ngāti Hei believed that the establishment of the reserve enhanced their mana. Likewise, the Sugar Loaf Islands marine protected area includes tapu sites for at-sea burials.

Te Tapuwae O Rongokako Marine Reserve

The local hapū at Whangara (Ngāti Konohi) supported the establishment of Te Tapuwae O Rongokako Marine Reserve on the Gisborne coast, and joined with the Department of Conservation to make the application. The reserve was seen as a valuable way to protect kaimoana and give back to Tangaroa.¹⁵

Increased recreational and tourism opportunities

Studies of the impacts of marine reserves in New Zealand have shown a range of tourism and recreation benefits. For example, the establishment of the Tonga Island Marine Reserve adjacent to the Abel Tasman National Park increased the recreational value of the area and, in turn, attracted greater numbers of visitors. In response, new accommodation enterprises have established, and there are growing numbers of water taxi operators. There has also been a significant growth in the number of kayak companies, from the two operating prior to the establishment of the marine reserve to at least 13 currently.¹⁶

Economic benefits of marine reserves

Significant economic benefits have been experienced from the Cape Rodney-Okakari Point Marine Reserve. An economic impact analysis undertaken in 2008 identified an estimated 375,000 annual visits to the reserve and the contribution of \$18.6 million a year into the local economy. This was compared to the cost to the Department of Conservation of managing the reserve of around \$70,000 per year.¹⁷

Type of protection	Key features
Area-based fisheries restrictions <i>Fisheries Act 1996</i>	No specific purpose stated in the Act, but they can be used to avoid, remedy or mitigate any adverse effects of fishing on the aquatic environment Spatial fishing restrictions can be created within the territorial sea and EEZ Created by the Minister for Primary Industries by <i>Gazette</i> notice or Governor-General by Order-in-Council Can restrict or prohibit certain types of fishing activity Managed by the Ministry for Primary Industries
Areas of significant conservation value <i>Resource Management Act 1991</i>	Can cover any area within the territorial sea Created by inclusion of the area in a regional coastal plan with associated rules Managed by the regional council which enforces rules in the regional coastal plan and conditions on resource consents
Areas to be avoided by shipping <i>Maritime Transport Act 1994</i>	Purpose is to protect sensitive marine environments from the risks, principally of marine oil spills, posed by shipping operations Excludes ships in transit from entering defined marine areas Draft rules are initiated by Maritime New Zealand and approved by the Minister of Transport Rules are enforced by Maritime New Zealand
Protected areas <i>Submarine Cables and Pipelines Protection Act 1996</i>	Purpose is to protect submarine cables and pipelines Excludes fishing and anchoring within the protected area Enforced by Maritime New Zealand
Special legislation	Can be tailored to specific issues and conditions Initiated by Member of Parliament (usually the majority party) Approved by Parliament

Figure 6.2 Overview of mechanisms to spatially protect marine areas



Marcaciones Point, part of the Te Awaatu Channel (The Gut) Marine Reserve in Doubtful Sound which protects significant rock wall and deep-sea habitats (Raewyn Peart)

- An application to create a marine reserve is lodged with the Director-General of Conservation
- Notice of intention to apply to create a marine reserve must be publicly notified in a range of newspapers
- Notice in writing of the application must also be given to owners of land adjoining the proposed reserve and other public bodies
- Members of the public have two months in which to lodge a written objection to the proposal with the Director-General of Conservation
- The applicant is given three months to answer the objections in writing
- The Director-General forwards the objections and answers received to the Minister of Conservation
- The Minister of Conservation decides whether any objection should be upheld. If the objection is upheld the marine reserve cannot be declared.
- The Minister of Conservation then decides if the reserve would be in the best interests of scientific study, if it would be for the benefit of the public and if it would be expedient. There are no appeal rights against the Minister's decision although it can be judicially reviewed.
- If the Minister decides to create the marine reserve, concurrence is sought from the Ministers of Transport and Primary Industries
- If concurrence is granted the Minister of Conservation recommends that the Governor-General makes an Order-in-Council creating the reserve

Figure 6.3 Process to establish a marine reserve

A list of current and proposed marine reserves, and information about them, can be found at the Department of Conservation's website.²⁰

The Marine Reserves Act is currently under review and is likely to be replaced with legislation which will enable the creation of broader types of marine protected areas (rather than just marine reserves), improve the process for their creation, and enable fully protected areas to be created within the EEZ.²¹



The Mata Manua (Pollen Island) Marine Reserve in the Waitematā harbour has extensive shellbanks and is an important feeding ground for seabirds. The North-Western Motorway runs through the reserve (Raewyn Peart)

Marine parks

Two marine parks were established under the now repealed Harbours Act 1950 and regulations under the Fisheries Act. These were the Tawharanui Marine Park in the northern part of the Auckland region and the Mimiwhangata Marine Park to the north. The Tawharanui Marine Park was declared a marine reserve in 2011. The fisheries regulations, which apply to the Mimiwhangata Marine Park, exclude all commercial and some recreational fishing. A third marine park was set up around the Sugar Loaf Islands off New Plymouth (which is now the Ngā Motu/Sugar Loaf Islands Marine Protected Area – see further discussion below).

A fourth marine park has been established under its own special legislation, the Hauraki Gulf Marine Park Act. This, amongst other things, establishes a Hauraki Gulf Marine Park which primarily consists of reserve land, Ramsar wetlands, foreshore and seabed owned by the Crown, marine reserves and seawater (but not marine life). Provisions in the legislation provide for council-owned and privately-owned reserve land to be added to the marine park with the consent of the owner. Taiāpure and mātaītai reserves may also be added to the park with the consent of the Minister for Primary Industries and the committee of management or tangata whenua.

Anyone administering an area within the Park must recognise and give effect to its purpose which is identified in the Act as:

1. To recognise and protect in perpetuity the international and national significance of the land and the natural and historic resources within the Park
2. To protect in perpetuity and for the benefit, use, and enjoyment of the people and communities of the Gulf and New Zealand, the natural and historic resources of the Park including scenery, ecological systems, or natural features that are so beautiful, unique, or scientifically important to be of national significance, for their intrinsic worth
3. To recognise and have particular regard to the historic, traditional, cultural, and spiritual relationship of tangata whenua with the Hauraki Gulf, its islands and coastal areas, and the natural and historic resources of the Park
4. To sustain the life-supporting capacity of the soil, air, water, and ecosystems of the Gulf in the Park

The legislation also establishes the Hauraki Gulf Forum and provides a set of management objectives which apply to the Gulf's coastal marine area, islands and catchments. These are described more fully in Chapter 2: Marine management.



Anchor Bay, part of the Tāwharanui Marine Reserve which was established in 2011, protects extensive rocky reef systems
(Raewyn Peart)

It is also possible for any person to lodge a private plan change to insert provisions into regional coastal plans identifying significant conservation areas and providing for their protection. Private plan changes go through a similar process, although they can be rejected by the council on a number of grounds.

Fishing methods, which have an impact on the marine environment, can potentially be restricted in regional coastal plans in order to promote sustainable management. Regional coastal plans cannot, however, 'control the taking, allocation or enhancement of fisheries resources for the purpose of managing fishing or fisheries resources controlled under the Fisheries Act 1996'.

Areas to be avoided by shipping

Areas to be avoided by shipping can be established under the Maritime Transport Act and are discussed in Chapter 15: Vessels

Cable protection areas

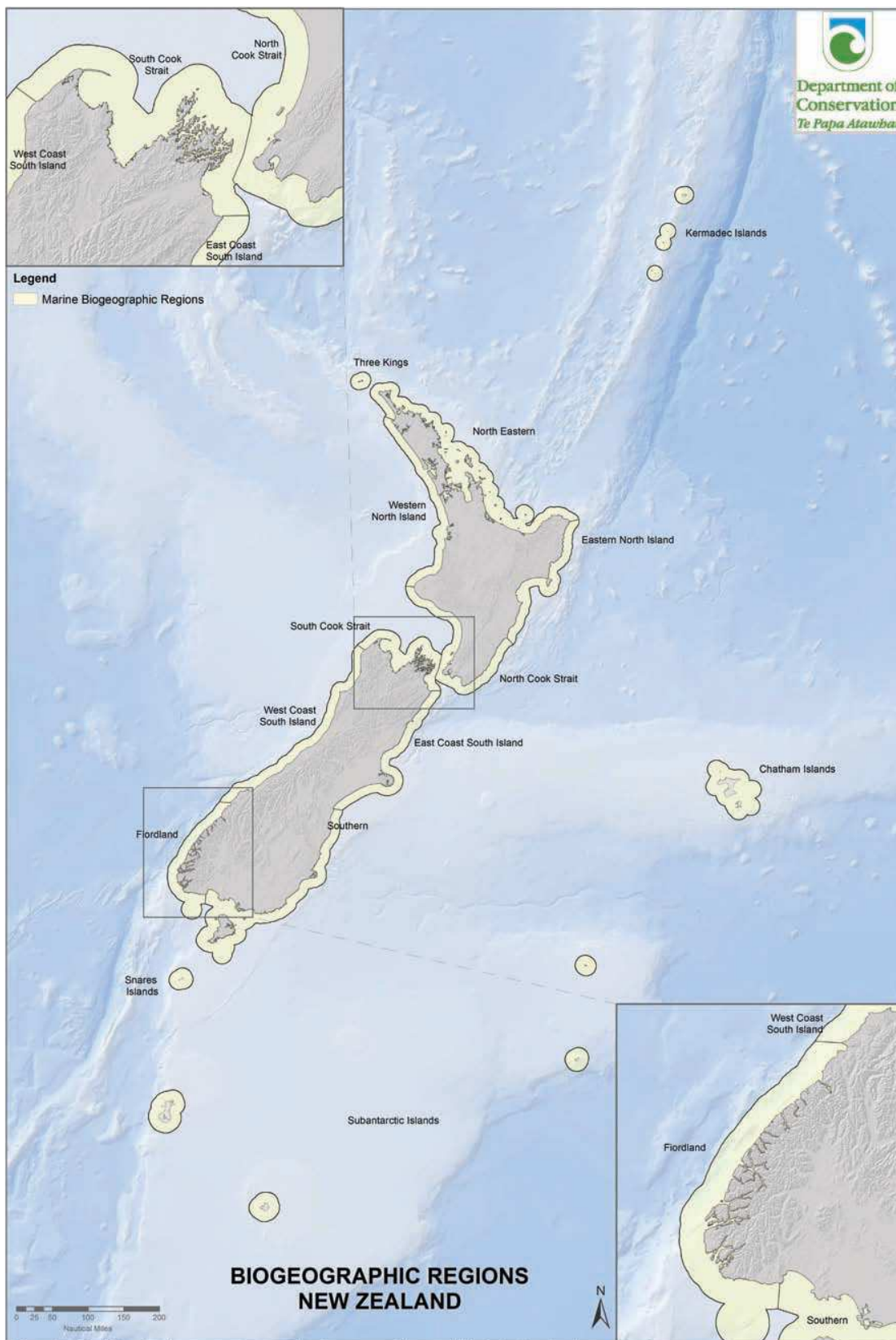
Protected areas can be declared under section 12 of the Submarine Cables and Pipelines Protection Act (see Figure 6.5 for an example). All fishing operations and anchoring are prohibited in such areas. This exclusion is not for environmental purposes, but is intended to protect cables and pipelines from damage. However, it may have the indirect effect of restricting activities which could also impact on biodiversity within the area.



Figure 6.5 Cable protection zones in the Hauraki Gulf (*Auckland Council*)

Wildlife management reserves

Reserves can be established under the Wildlife Act and can potentially extend into the marine area although this has yet to occur. However, co-locating wildlife and marine reserves can create land-to-the-sea protection. For example, the Westhaven (Te Tai Tapu) Marine Reserve is adjacent to the Westhaven (Whanganui) Wildlife Management Reserve. Effectively the estuary at Westhaven is therefore protected both as marine reserve and as a wildlife reserve. The marine reserve covers 536 hectares of tidal sandflats and channels within Whanganui Inlet, on the western coast of Golden Bay, and the wildlife management reserve covers over 2,112 hectares of tidal sandflats and channels not included in the marine reserve.²²



This product is for informational purposes and may not have been prepared for, or suitable for legal, engineering, or surveying purposes.

Figure 6.6 New Zealand's coastal biogeographic regions and marine protected areas (Department of Conservation)

Area	Involves
Identifying potential MPAs	• Protect whole habitats and ecosystems • Be of sufficient size to provide for the maintenance of populations. Fewer larger areas are to be preferred than numerous smaller areas • Maximise and enhance linkages among individual and groups of protected areas • Include differences in habitats and ecosystems that cover both latitudinal and longitudinal or cross-shelf ranges (i.e. from the intertidal zone to deep offshore waters) • Consider adjacent uses which may affect the protected areas • Use simple boundaries and minimise boundary to area ratios
Selecting candidate MPAs – Primary considerations	• Protect the full range of marine habitats and ecosystems • Consider information on cultural use and values and Treaty settlement obligations • Consider information on social and economic interests and users • Minimise adverse impacts on existing users and Treaty settlement obligations
Selecting candidate MPAs – Secondary considerations	• Protect multiple habitats within each protected area • Have fewer larger rather than numerous smaller protected areas • Consider activities such coastal structures, dredging or dumping sites which may negatively impact on the site • Align where possible with other protected areas • Provide, where possible, replication of habitats and ecosystems

Figure 6.8 Guidelines for selecting potential protected areas

At the time of writing two fora had concluded for the West Coast South Island and Subantarctic biogeographic regions. A forum for the Otago coastline was announced in March 2014. The Hauraki Gulf Marine Spatial Planning process (called Seachange) is expected to develop recommendations for marine protected areas in the Hauraki Gulf (see Chapter 7: Marine spatial planning). An initiative led by Ngāti Kuri, known as Te Korowai o Te Tai o Marokura or Kaikōura Coastal Marine Guardians, worked in a collaborative manner to develop the Kaikōura Marine Strategy. This is in the process of being implemented through the Kaikōura (Te Tai-o-Marokura) Marine Management Act.

The timetable for progressing the development of marine protection plans in other areas is uncertain. A review of spatial marine protection in New Zealand, the operation of planning fora, and recommendations for reform can be found in the Environmental Defence Society publication *Safeguarding Our Oceans*.²⁷

Elements of best practice

The IUCN World Commission on Protected Areas has produced a document that represents a synthesis of the expertise, knowledge and views of world-leading experts in marine protected area network design and implementation. The following elements of best practice have been drawn from this document. To maximise its effectiveness, a network needs to include a group of marine protected areas that are carefully designed to function collectively and synergistically as an ecological network. This includes the following design considerations:²⁸

- Does the MPA network protect some of all the habitat types and habitats linked by species life cycle patterns found in a biogeographic region?
- Does the MPA network include both transition zones between biogeographic areas and core zones within each biogeographic region?
- Does the MPA network include representation of ecosystem processes as well as habitat types?

Cape Rodney-Okakari Point Marine Reserve (Goat Island)

New Zealand's first marine reserve was established near Leigh in May 1977. The impetus behind the creation of a 'protected area in the sea' came from University of Auckland scientists who had established a marine laboratory in the vicinity during 1962. The scientists soon came to realise that, without some form of official protection, scientists would be unable to carry out many useful experiments. This was because their marine 'subjects' were being harvested by fishers.

The marine reserve was first suggested by Professor Val Chapman in 1965, but could only be created after the Marine Reserves Act was passed in 1971. The reserve itself extends 5 kilometres along the coast from Cape Rodney to Okakari Point, and 800 metres out to sea from the high water mark, encompassing Te Hāwera-a-Maki/Goat Island. The reserve has clear, warm water and many rocky reefs.²⁹

The reserve has now been in place for 38 years. It has enabled scientists to learn much about the dynamics of rocky reef communities and the impacts of fishing.



The University of Auckland's marine laboratory established near Leigh in 1962 provided the impetus for New Zealand's first marine reserve legislation and the creation of the country's first marine reserve. Scientists based at the laboratory have undertaken extensive research within the reserve which has significantly contributed to our understanding of rocky reef communities (*Raewyn Peart*)

Crayfish were one of the first harvested species to show a rapid recovery within the reserve with numbers increasing five-fold within four years. In addition, red moki soon tripled in number. However, initially there was little change in snapper numbers, although individual fish were larger than in adjacent fished areas.

One important area of research has been on the phenomenon called 'urchin barrens'. These bare-looking areas, where the kelp forest had been lost, were extensive when the reserve was created. By 2006, the area of kelp forest within the reserve had more than doubled when compared to 30 years earlier. In addition, the area of urchin barren had declined from 31 to one per cent.³⁰



A wide range of marine species can be viewed underwater at the Cape Rodney-Okakari Point Marine Reserve as shown on this noticeboard (*Raewyn Peart*)

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7

Marine spatial planning¹



Tutukaka, Northland (Craig Potton)

Introduction

For some decades now, marine managers have realised that case-by-case management of activities impacting on the marine area is ineffective in protecting the health of marine ecosystems. This is because the marine environment is much more highly interconnected than land, and impacts in one area can have flow-on effects elsewhere. In addition, marine management is typically split between numerous agencies, which make decisions in isolation of each other, and so fail to adequately address interconnections and resulting cumulative effects.

As the marine area becomes more utilised, conflicts between activities are becoming more acute. This has been increasingly evident in New Zealand, with recent conflicts between marine protection and fisheries interests; between fishing and mining activity; and between aquaculture and landscape protection. Such poorly managed conflicts create cost and uncertainty for all parties and the environment.

Marine spatial planning is a rational and strategic approach which can be used to proactively plan for the future use of the marine environment. At its heart is a concern to protect the underlying ecological backbone or productivity of the marine area, but it also seeks to reduce conflict and maximise synergies, providing greater certainty on where marine activities can and cannot locate.

Marine spatial planning is now a well-used tool internationally and it is a key element of marine management in the 21st century. The first marine spatial plan in New Zealand is currently being prepared for the Hauraki Gulf.²

What is marine spatial planning?

Marine spatial planning seeks to provide greater direction on how defined areas of marine space, including coastal and offshore areas, are to be managed in order to meet desired societal outcomes. Several definitions have been proposed for this approach. In 2010, a group of 21 scientists posited the following definition, which focuses on using marine spatial planning as a tool to implement ecosystems-based management:³

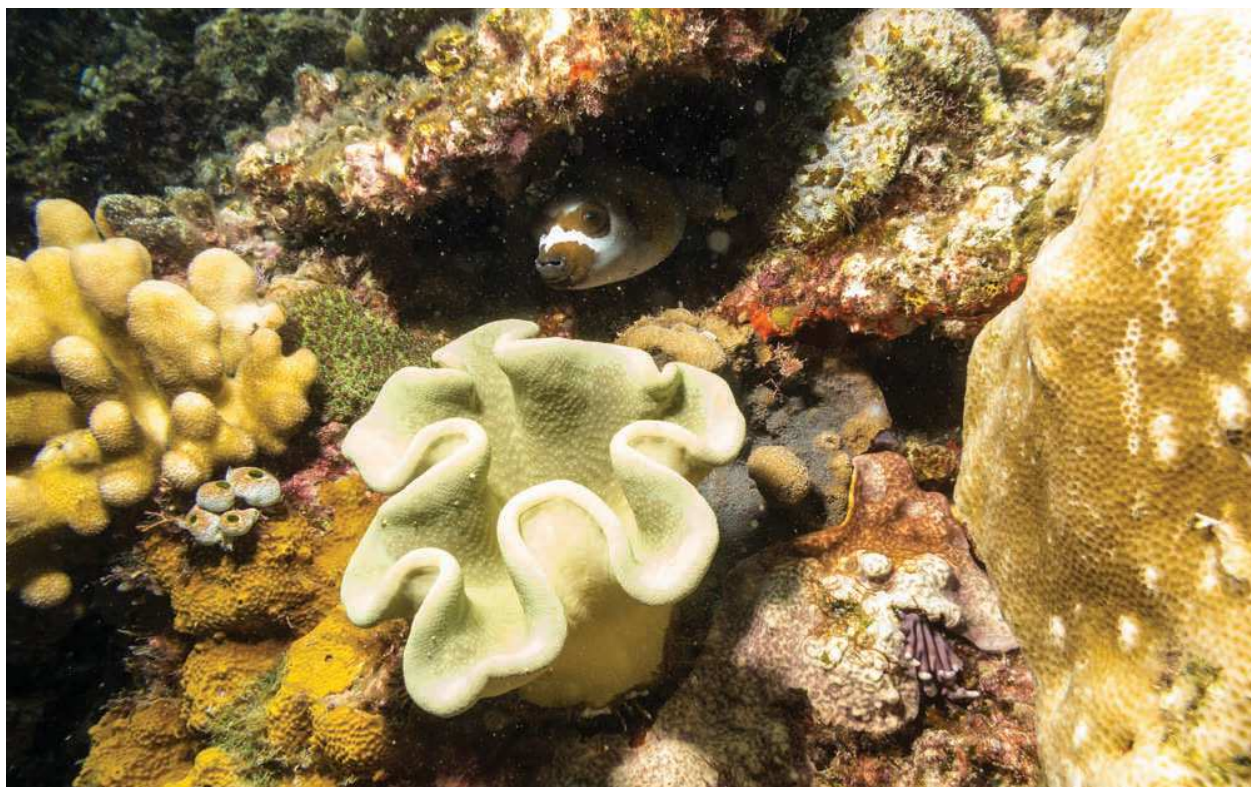
Ecosystem-based MSP [marine spatial planning] is an integrated planning framework that informs the spatial distribution of activities in and on the ocean in order to support current and future uses of ocean ecosystems and maintain the delivery of valuable ecosystem services for future generations in a way that meets ecological, economic, and social objectives.



Spatial planning for the marine environment, such as the Hauraki Gulf shown here, is much more challenging than on land
(Raewyn Peart)

The United Kingdom Marine and Coastal Access Act 2009 provides a legal framework for spatial planning in that jurisdiction. This includes the development of a marine policy statement and a series of regional marine plans covering coastal and offshore areas. Most licensing decisions are required to be made in accordance with these documents.

In New Zealand, there is no legislative provision for integrated marine spatial planning. The RMA does provide for partial spatial planning through the preparation of regional coastal plans, but these do not address fishing, marine reserves or marine mammal sanctuaries. Despite this legislative vacuum, New Zealand's first marine spatial planning process has commenced in the Hauraki Gulf. Called 'Sea Change – Tai Timu Tai Pari', the process has adopted a collaborative, stakeholder-led, non-statutory process for the development of the plan, which is expected to emerge later in 2015 (discussed further in a case study below).



Australia's Great Barrier Reef was one of the first areas where spatial planning was applied to the marine area (Raewyn Peart)

Potential benefits of marine spatial planning

There are many potential benefits from the adoption of marine spatial planning. They include:⁶

- Application of an ecosystems approach to the management of human activities through safeguarding important marine ecological processes and the overall resilience of the marine system
- Provision of a strategic, integrated and forward-looking framework for all uses of the sea which takes into account environmental as well as cultural, social and economic objectives
- Identification, conservation or restoration of important components of coastal and marine ecosystems
- Allocation of space in a rational manner which minimises conflicts of interest and maximises synergies across sectors
- Management of cumulative impacts over space and time
- Provision of greater certainty for marine users

Marine spatial planning can also help create a better linkage between science and management. The process of mapping physical and ecological information involves the collation and application of existing data to relevant management issues. It also enables information gaps to be identified and future research to be prioritised.

managed for specific ecological purposes, and areas suitable for specific activities such as fishing, mining, oil and gas and aquaculture

- Opportunities for restoration where the marine environment has been degraded in order to restore its productivity, diversity and resilience
- Targets to be achieved, monitoring and review: to make it clear what is expected to be achieved, to keep track of progress, and to review the effectiveness of the plan so it can be refined and improved

Elements of best practice

- Include iwi, user groups and conservationists at the forefront of the process
- Ensure all statutory agencies which will need to implement the plan are fully engaged from its inception
- Be strategic and focus on what is important, taking a long-term perspective. Many problems have taken decades to develop and will take time to resolve.
- Include an historical perspective to address the 'sliding baseline' problem where the extent of degradation is underestimated due to it spanning more than one human lifetime
- Plan for the future rather than the past and anticipate future pressures and uses
- Focus on identifying opportunities which serve to both protect the environment and support economic, social and cultural well-being
- Use science to inform but not drive the process
- Integrate mātauranga Māori into the understanding of the issues and potential solutions
- Start with an in-depth understanding of the ecological backbone of the marine area, and what areas, habitats and species are important to its ongoing productivity, diversity and health
- Plan for other activities within the framework provided by the ecological backbone
- Be creative and consider a range of tools to achieve desired outcomes, rather than relying solely on current regulatory tools

Marine spatial planning is a well-proven approach to achieving better management of marine space. It has the potential to assist with the effective integrated management of New Zealand's seas, especially in high-value areas that are facing competing uses such as the Hauraki Gulf, the Marlborough Sounds, Tasman and Golden Bays, the Bay of Islands and the Chatham Rise.



Areas facing competing uses, such as the Bay of Islands shown here, could benefit from marine spatial planning (Raewyn Peart)

A governance entity was established to oversee the project with members consisting of representatives of the statutory bodies involved in managing the Gulf and an equal number of manawhenua representatives, thereby putting in place a co-governance structure.

The plan itself is being developed by a stakeholder working group with representatives from commercial fishing, recreational fishing, farming, aquaculture, infrastructure, community and environmental interests. The group meets monthly and an independent chair was appointed to lead the collaborative process. Six roundtables have been established to focus the plan development work on key elements of the overall picture. The topics for the roundtables are fish stocks, water quality and catchments, aquaculture, biodiversity and biosecurity, accessible Gulf and Gulf infrastructure. These groups include several stakeholder working-group members and others representing a wider range of stakeholder interests.

An extensive public process has been running alongside the stakeholder working group. This has involved public meetings, 25 'listening posts' in communities around the Gulf which totaled more than 250 participants, a web-based use and values survey, a presence at marine-related events, and an active website and email updating programme.

Spatial data sets have been assembled on a web-based tool called SeaSketch for easy accessibility. The software enables users to test and communicate spatially referenced ideas. A technical team consisting of agency staff has also been assembled to support the stakeholder working group and topic roundtables and to access science as requested.

The stakeholder working group first met in late 2013 and is tasked with producing a plan by mid-2015. The plan will then go out for public comment before adoption by the governance group and implementation by the requisite statutory agencies.

The Hauraki Gulf Forum's triennial state of the environment reports and recent stock take of economic activity show there are issues and opportunities not being addressed through traditional policy and planning approaches. Inviting the Gulf's communities and users to prepare the plan will help generate new approaches and solutions, in accordance with the Hauraki Gulf Marine Park Act. (Hauraki Gulf Forum Chair John Tregidga)¹⁰



The Hauraki Gulf marine spatial plan will consider the harvesting of seafood, such as the gathering of cockles shown here at Kawakawa Bay, Auckland (Raewyn Peart)

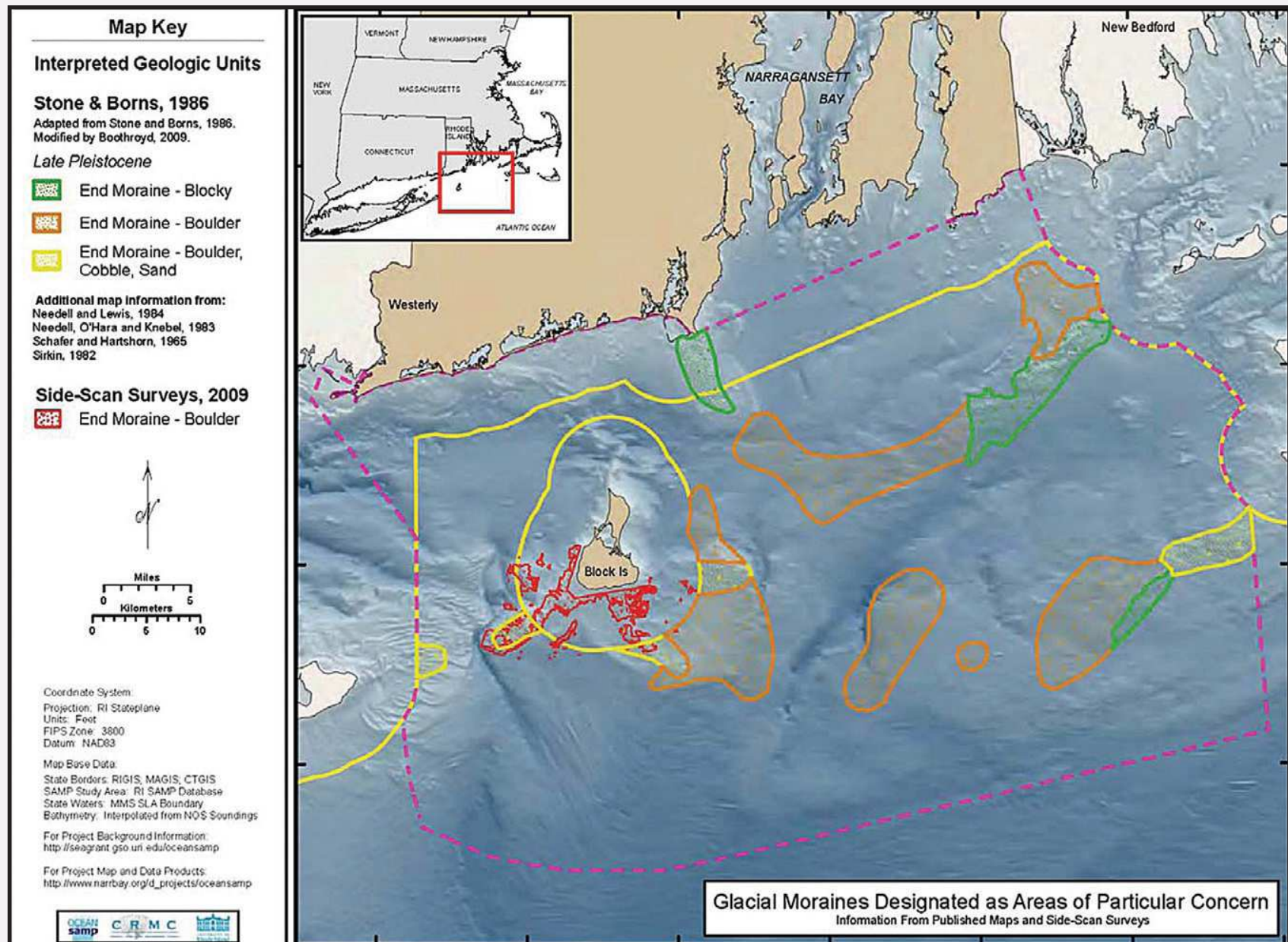


Figure 7.1 Moraines protected as Areas of Particular Concern (Rhode Island Coastal Resources Management Council, 2010)

Opportunities for public involvement



Introduction

There are many opportunities for members of the public to become involved in statutory processes regarding environmental outcomes for the marine area. Some of these processes involve establishing policies that govern decision-making. Others involve developing rules and regulations that apply to activities in the marine environment. Yet other processes involve considering specific proposals to undertake marine activities. In Section Two, these processes are discussed where they relate to specific activities.

Because there are so many different acts applying to the marine area, which are administered by a range of statutory authorities, there is little consistency in the processes that apply to different management jurisdictions. In some areas, such as under the RMA, there are quite extensive rights of public participation. In other areas, such as under the Fisheries Act, consultation rights are focused on directly affected parties, although the Ministry for Primary Industries will typically provide the opportunity for written public submissions to be made before delivering final advice to the Minister who then makes a decision. Where there are no rights of submission or appeal, a statutory decision may still be able to be challenged by initiating judicial review proceedings in the High Court.



A party landing at Motuihe Wharf to visit and enjoy Motuihe Island, Hauraki Gulf (Raewyn Peart)

Policy/plans	Application	Opportunity for public involvement
No legislative basis		
Fisheries standards	All fisheries	• No statutory rights to participate and the standards are non-statutory • In practice, written submissions are sought on draft standards
Recommendations for marine protected areas in biogeographic regions	Marine areas in territorial sea and EEZ	• No statutory rights to participate and the process is non-statutory • In practice, stakeholders will be represented on a marine protection planning forum and written submissions will be sought on the forum's recommendations

Figure 8.1 Summary of opportunities for public participation in the preparation of policies and plans affecting the offshore environment

RMA policies and plans

There is a hierarchy of policies and plans that guides decision-making under the RMA, and members of the public have the opportunity to influence the content of these documents. At the national level is the NZCPS; the National Policy Statement for Freshwater Management which governs freshwater bodies, many of which feed into the marine environment; and the National Policy Statement for Renewable Electricity Generation, which includes provisions on tidal, wave and ocean current resources. There is also the possibility of other national policy statements being prepared which affect the marine environment.

Members of the public must be given the opportunity to make submissions on proposed national policy statements. If a Board of Inquiry process is adopted by the Minister for the Environment (or the Minister of Conservation for a NZCPS), submitters will also have the opportunity to present their case (including submissions and evidence) at a public hearing. The Board of Inquiry then makes recommendations to the relevant Minister, who makes the final decision. There are no appeal rights against the relevant Minister's decision on national policy statements, and the only redress is through judicial review proceedings in the High Court.

At the regional level, regional policy statements, regional coastal plans and regional plans are prepared. At the district level, district plans are prepared. These documents typically contain objectives, policies, rules, methods and anticipated environmental results which guide decision-making on resource consent applications.

There is a very open process for public participation in council policy and plan-making under the RMA (as shown in Figure 8.2). Councils are required to consult with a range of parties prior to proposed plans and policy statements being formally notified, including Ministers of the Crown, local authorities and iwi. Councils also commonly consult with a range of other stakeholders. Increasingly, councils are issuing draft policy statements and plans, and providing members of the public with an opportunity to give feedback prior to formal notification of the documents. This provides an additional opportunity to have input into the preparation of the document outside of the statutory process.



Members of the public have the opportunity to submit on regional coastal plans which help determine where marine activities, such as the moorings, jetties and reclamations shown here in Lyttelton, can be located (Raewyn Peart)

Other parties can become party to any appeal (referred to as section 274 parties) only if they can show that they fall within one of the following categories:

- They have lodged a submission about the subject matter of the proceedings at the local authority stage
- They have an interest in the proceedings greater than the public generally (excluding trade competitors). This will normally include neighbours, downstream resource users and iwi authorities. It is uncertain whether public interest groups will be able to become parties under this category as well.

Decisions of the Environment Court can be appealed to the High Court on questions of law, but not on the merits.

Conservation strategies and plans

Under the Conservation Act, the Minister of Conservation can prepare general policy and conservation management strategies and plans. These guide the Department of Conservation's marine protection as well as other work. The public process for preparing each of these documents is similar. Members of the public must be given the opportunity to make written submissions on draft documents. Submitters must then also be given the opportunity to present their submissions to the Director-General of Conservation in person (for submissions on general policy) or to a meeting consisting of representatives of the Director-General and affected conservation boards (for strategies and plans).



Members of the public have the opportunity to make written submissions on conservation management plans which guide day-to-day management of marine reserves including the Tāwharanui Marine Reserve shown here (Raewyn Peart)

Population management plans

Population management plans can be prepared by the Minister of Conservation for marine mammals under the Marine Mammals Protection Act and for other wildlife such as seabirds under the Wildlife Act. Members of the public must be given the opportunity to make written submissions on the draft plan and also to present their submissions in person to the Director-General or his or her representatives. No such plans have yet been completed.

Marine protected areas

The government's 'Marine Protection Areas Policy Statement and Implementation Plan' is a non-statutory document that provides for the establishment of regional marine protection planning forums which prepare recommendations for marine protected areas within biogeographic regions. The forums are made up of local stakeholder representatives. The recommendations of the forums have no legislative basis. There are consequently no statutory rights for the public to participate in forum deliberations. However, in practice, members of the public will be given the opportunity to make written submissions on the proposed recommendations. In addition, the recommendations may be implemented through the processes for marine reserve creation under the Marine Reserves Act described below and these provide the opportunity for public objections. Alternatively they may be implemented more directly by special legislation. This is discussed further in Chapter 6: Marine protected areas.

National environmental standards

National environmental standards are regulations issued under the RMA which prescribe technical standards, methods and other requirements for environmental matters. Regional councils and territorial authorities must enforce these standards (or they can enforce stricter or weaker standards when the standard provides for this). In this way, national environmental standards can be used to ensure that consistent minimum standards are applied to management of the marine environment.

There is no prescribed process for the preparation of national standards other than that the public must be given adequate time and opportunity to comment. Usually written submissions will be sought on a proposed standard before it is adopted, but there will not be a public hearing to consider submissions.

Under the RMA, plans (but not regional policy statements) contain rules as well as objectives and policies, and the opportunity to participate in plan-making processes has been described in the previous section.

Exclusive Economic Zone regulations

There is no prescribed process for the preparation of EEZ regulations other than that the public, iwi authorities, regional councils, and persons whose existing interests are likely to be affected must be notified of the proposed subject matter of the regulations and must be given adequate time and opportunity to comment. Usually written submissions will be sought on regulations before they are adopted, but there will not be a public hearing to consider submissions.

Fisheries sustainability measures

For the setting of sustainability measures under the Fisheries Act there are no statutory rights of public participation. Sustainability measures include setting the total allowable catch, regulatory measures such as restricting fishing methods or locations, and measures to manage the effects of fishing-related mortality on protected species.

The Minister must consult with representatives of groups having an interest in the stock or the effects of fishing on the aquatic environment in the area concerned. This will normally include commercial and recreational fishing interests and environmental organisations. The Minister must also provide for the 'input and participation' of tangata whenua.⁶

When setting or varying the total allowable commercial catch, the Minister must consult with representatives of groups having an interest including Māori, environmental, commercial and recreational interests.⁷

In practice, the Ministry for Primary Industries makes available an initial advice paper and will accept written submissions from any party before finalising its advice to the Minister. If you have an interest in particular fisheries sustainability matters, contact staff at the Ministry so they can add you to the relevant circulation lists.



There is usually an opportunity for members of the public to make a submission on proposals to change the total allowable commercial catch of fish stocks such as snapper (Daniel Sharp)

Deciding individual proposals

Application	Area	Opportunity for public involvement
Resource consent	Activity within the catchment or territorial sea	• Statutory right to participate only where application publicly notified (or where there is limited notification to affected persons) • Written submissions on application • Presentation of submissions and evidence to council hearing (where no direct referral or call-in) • Appeal to Environment Court – unless a nationally significant proposal decided by the Board of Inquiry • Presentation of submissions and evidence to Environment Court hearing (or board of inquiry for some call-ins) • Appeal to High Court on points of law
Marine consent – unless classified as non-notified	Activities within the EEZ and extended continental shelf	• Written submissions on application • Presentation of submissions and evidence to hearing • Appeal to High Court on points of law

Figure 8.4 Summary of opportunities to participate in the determination of individual applications affecting the marine environment

Resource consents

The opportunities to become involved in decisions about individual proposals for activities in the marine area are much more limited than the opportunities provided for in the development of policies and rules.

Under the RMA there are broad rights of participation in resource consents, when they are publicly notified, and these usually include being heard at a council-initiated hearing, appealing the council's decision to the Environment Court and being heard again, and appealing the Environment Court's decision to the High Court on points of law only.

There are no appeal rights against a council's decision not to notify a resource consent and the only redress is through initiating judicial review proceedings in the High Court. In cases where the council decides not to publicly notify the application, it may still be given limited notification. This means that parties who are directly affected by the proposal, and who have not given consent, will have the opportunity to lodge a submission and subsequent appeals. In 2012/13 only around five per cent of resource consent applications were publicly notified, and the small proportion of notified consents has steadily decreased since 2007/08.⁸

Matters can be directly referred to the Environment Court for decision on the request of the applicant. In this case there will be no hearing by the council in the first instance. The decision of the Environment Court can be appealed to the High Court on points of law only.

Matters of national importance can be 'called in' by the Minister for the Environment (or Minister of Conservation for applications within the coastal marine area). In addition, applicants can choose to lodge the application with the EPA. The Minister will then decide if it is nationally significant, and it also can refer it to the Environment Court or Board of Inquiry for decision. In the case of a call in there will also be no council hearing in the first instance, but the matter will be processed by the EPA and determined either by the Environment Court or by a Board of Inquiry appointed by the Minister. Decisions can be appealed to the High Court on points of law only.



Anyone can make a submission and be heard on a notified resource consent application, such as for the canal development shown here at Marsden Cove, Whangarei (Raewyn Peart)

Stage	Detail
Hearing	
Hearing procedure	Marine consent hearings are required to avoid unnecessary formality. At the time of writing, only one marine consent application has been heard so a 'normal' hearing procedure cannot yet be identified. The Decision-Making Committee will produce a document setting out Hearing Procedures in advance of the hearing. It is expected that cross-examination of expert witnesses will generally be permitted but only the Committee will ask questions of submitters.
Preparation of submissions to present at the hearing	• Always make your submissions available in written form. You will need to bring sufficient copies for all members of the Decision-Making Committee, the applicant, and the EPA. • It is useful to identify ahead of time who will be on the hearing body, and their backgrounds and likely points of view, so that you can target your submissions accordingly • The purpose of your submissions is to persuade the Decision-Making Committee to agree with your point of view. Avoid being argumentative or badgering. • Avoid personally attacking the applicant, expert witnesses, the Decision-Making Committee or the EPA. This is inappropriate and counterproductive. • Clearly state your concerns and provide information to back up your point of view • Be concise and to the point. Avoid being verbose. Do not get sidetracked with irrelevant matters, allegations or conspiracy theories. • Make sure that all the material presented to support your submission is consistent • Where possible illustrate your points with maps, photographs and illustrations • Always use words that you understand and customarily use so that your presentation flows naturally
Structure of submissions	• Start by introducing yourself and/or your organisation. Provide a brief description of your background, and your organisation's objectives, activities and membership base. • Where you have made efforts to resolve your concerns directly with the applicant, state this, and the reasons why no resolution was achieved (without directly attacking the bona fides of the applicant) • State concisely the issues which are of concern to you. The Decision-Making Committees has powers to limit oral presentations both in terms of content and time. • Where there is more than one person involved in presenting your case, state who will be giving submissions or evidence and on what issues. Each issue of concern should have a separate section in your submission. This should state in more detail the nature of your concern, provide supporting information to support your concern, and clearly state what action you would like the Decision-Making Committee to take in order to address the concern. • Include a brief concluding section which summarises the essence of your concerns and action sought • Thank the Decision-Making Committee for taking the time to hear and consider your submissions
Sources of supporting information	Supporting information for issues of concern can include: • Drawing the Decision-Making Committee's attention to relevant parts of the EEZ Act, EEZ Regulations, and other statutes and instruments relevant to marine management (such as the RMA and NZCPS) • Identifying any relevant research and reports which support your concerns • Describing any local knowledge that you have about the environment and the impacts of the proposed activity on it • Describing practical cases or examples which illustrate your main points • You may wish to use a lawyer to support your case and call expert witnesses relevant to your specific matters of concern (e.g. marine ecologist, noise specialist and/or planner)
Presenting your submissions	• Read from your written submissions but, where appropriate, ad-lib relevant comments • Speak slowly and clearly and try to vary your tone to keep up interest • Where possible illustrate points through reference to photographs, maps, slides and other pictorial material. Make sure that all members of the Decision-Making Committee have copies. It can be useful to take a large map or diagram, which can be placed on a display board, to illustrate points during your presentation. • Try to avoid repeating what others have already said. Even if you have included such material in the written version of the submissions, you can skip over these sections, and politely inform the hearing body that you are leaving out sections to avoid repetition. • There is no charge for presenting oral submissions to a Decision-Making Committee

Reporting offences

Another way in which members of the public can initiate action to improve the health of the marine environment is through reporting issues of concern to the relevant authorities, particularly where there is suspicion of an offence being committed. Before reporting the issue, it is useful to record as much relevant information as possible, including taking photographs or filming a video. Most authorities have hotlines and after hours numbers and these are often prominently displayed on their websites.

- Pollution incidents and suspected illegal structures or unlawful mangrove removal in the marine area should be reported to the regional council
- Suspected illegal fishing or shellfish harvesting outside marine reserves should be reported to the Ministry for Primary Industries
- Suspected illegal fishing or shellfish harvesting inside marine reserves should be reported to the Department of Conservation
- Harm to protected species such as marine mammals and seabirds should be reported to the Department of Conservation
- Suspected new alien marine species should be reported to the Ministry for Primary Industries

Most regional and territorial authorities have a 24 hour pollution hotline, which can be called if pollution of the catchment occurs, and which can be found on council websites. Pollution response teams respond to reports of pollution events, and aim to put an immediate halt to the discharge, ensure that polluted waters are cleaned up and collect evidence of the pollution incident.

When an incident occurs, the regional council can send a warning letter setting out how the perpetrator has breached either the rules of a regional plan or the RMA. Then the council can issue an abatement notice requiring the perpetrator to stop the offending activity occurring or take some action to ensure compliance. The council can issue infringement fines of up to \$1,000 or prosecute the offender. The maximum penalties for offenders are up to two years in prison or a fine of up to \$300,000, with a further \$10,000 for every day that the offence continues. The maximum penalty for a commercial organisation is \$600,000, with a further \$10,000 for every day that the offence continues. The Environment Court can issue an enforcement order requiring the offender to comply with the RMA.¹³



Unlawful mangrove removal, such as shown here at Whangamatā, should be reported to the regional council (Raewyn Peart)

Judicial review

Judicial review involves challenging the exercise of a statutory discretion. The challenge cannot be based on the merits of the decision, but only on the basis that it was unlawful, that it was unreasonable or that the decision-making process did not follow the proper procedure. Judicial review has been used to challenge decisions such as the setting of the Total Allowable Commercial Catch under the Fisheries Act, the decline of a marine reserve application under the Marine Reserves Act, and the failure to notify resource consents under the RMA.

In order to overturn a decision on the basis that it is unlawful, it will normally be necessary to establish one of the following grounds:¹⁴

- The decision was motivated by an improper purpose, which was not contemplated by the legislation
- The decision-maker took into account irrelevant matters
- The decision-maker failed to take into account relevant matters that were mandatory under the legislation
- The decision was made under an invalid delegation
- The decision-maker made an error of law
- The decision-maker had insufficient information on which to base the decision

It is also possible to challenge a decision on the basis that it was unreasonable. This is a hard test to meet and involves establishing that the decision was 'so unreasonable that no reasonable [decision-maker] could ever come to it'.¹⁵

Another ground on which a decision may be set aside is on the basis that the procedure which the decision-maker followed did not accord with the principles of natural justice, including acting in good faith and listening fairly to both sides.

Tairua Marina

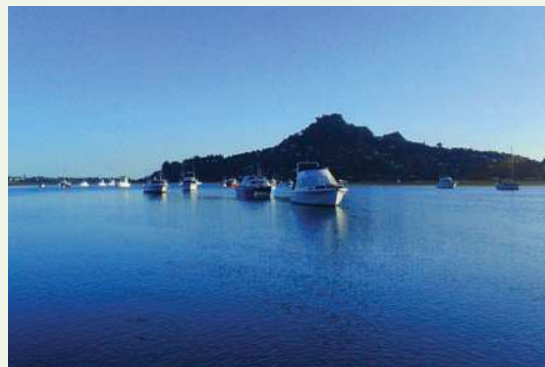
In the small coastal community of Tairua, on the Coromandel's east coast, a 253 berth marina was proposed in the early 2000s. The development required the dredging of Paku Bay to enable sizeable vessels to access the marina, and the disposal of 270,000 cubic metres of sediment into a reclamation of 4.6 hectares of seabed and foreshore. Ongoing dredging was also required to maintain the channels.

Some local residents were concerned about the potential negative impacts of the proposed large-scale marina development on their community and the local marine environment. The Guardians of Paku Bay Association was a community group borne out of a desire to protect these values. The first gathering, after invitation for submissions was announced, attracted just 23 people, 12 of whom formed the committee. Ten weeks later the group had grown to 140, and five years after that it had reached 425.¹⁹ Bi-monthly meetings were held throughout the process to keep the community, including tangata whenua and iwi with whom a good relationship was formed, informed of progress.

In 2004 a redesign of the proposal was presented in a resource consent application. The main differences were:²⁰

- The reduced scale of reclamation and increased beach width
- A beach renourishment area along the northern foreshore area
- Change to the northern edge of the inner breakwater
- Realignment and deepening of the Graham's Stream diversion channel
- A proposed bird roost

The Waikato Regional Council and Thames-Coromandel District Council refused the resource consents. In 2005 the applicants appealed these decisions to Environment Court, where the Court supported the councils' decisions to decline the application. This decision was then appealed to the High Court in 2006, where the appeal was dismissed.²¹



A local community group played an important role in reducing the size of the marina proposed for Tairua Harbour (Daniel Sharp)

Four years later a new application was lodged with the councils. This application included a significantly reduced footprint, which provided 95 berths and 40,000 cubic metres of capital dredging. The application was approved. The Guardians of Paku Bay Association and Tairua Environmental Society appealed this decision. The appeal was heard by the Environment Court in 2010. At the hearing a number of experts presented evidence about the impacts on coastal processes, water quality, marine ecology, avian ecology, landscape/natural character/visual amenity and cultural values. The Environment Court confirmed the councils' decisions to grant consent.²²

In 2011 the applicant for the marina sought costs against the Guardians of Paku Bay Association through the Environment Court. The Court stated that the hearing was one in a long series of hearings related to the use of the Tairua Harbour, in which parties had relatively entrenched views. The Court concluded that this was a clear case where the public and participatory nature of the RMA was intended to be achieved, and in which costs should not be awarded. Accordingly, costs were to lie where they fell.²³ In 2012 the Guardians of Paku Bay Association appealed the Environment Court decisions to the High Court, where the appeal was dismissed.²⁴

The tireless work of some members of the local community ensured that enough funds were raised to supplement the funds received from the Ministry of the Environment's Environmental Legal Assistance Fund in order to cover legal costs. This was done through raffles, garage sales and art auctions. The media was used as a way to communicate to residents and ratepayers about the development proposed. To raise awareness, pamphlets detailing the proposal and issues were dropped to all homes, petitions were started and bumper stickers circulated. Over 200 letters to the local newspaper editor opposing the proposal were published. The group engaged with industry professionals, both individuals as well as recognised bodies, as sources of knowledge and advice.

Whilst not able to prevent the development, the local community's fight against the proposal was instrumental in reducing the footprint of the marina to a quarter of what was originally proposed.²⁵ The experience they gathered throughout the process has enabled them to form partnerships and provide support to others with similar environmental objectives.

An underwater photograph showing a large, dense school of blue maomao fish swimming in clear blue water. To the left, a coral reef is visible with various colorful corals in shades of orange, red, and purple. The fish are silvery-blue with a slight yellowish tint on their sides. The scene is brightly lit, suggesting a shallow depth.

Section Two

Management of activities

Blue maomao, Poor Knights Islands *(Daniel Sharp)*

9

Management of fisheries



Fishing boat tied up at the Mangonui wharf, Northland (Raewyn Peart)

Introduction

New Zealand's commercial fishing sector has changed significantly over the last 40 years. Up to the 1970s, the domestic industry was almost entirely made up of relatively small inshore vessels, manned by owner-operators under a licence system. The country's offshore waters were being fished by overseas vessels coming from nations such as Japan, Korea, Taiwan and Russia. The declaration of the 200 nautical mile EEZ in 1977, followed by the property-rights-based QMS in 1986, fundamentally changed the sector. The largest-volume fisheries are now offshore and they are dominated by a small number of New Zealand companies.¹ Due to the settlement of fisheries claims under the Treaty of Waitangi Māori have a major stake in New Zealand's commercial fisheries industry (see Chapter 4: Kaitiakitanga).

Overview of fishing industry

- Eight fishing companies provide 80 per cent of production by weight
- There are a large number of equally important medium and smaller fishing operations, with the majority of these being inshore
- Approximately 2,200 individuals and companies own quota in New Zealand fisheries
- The estimated value of the total quota is \$3.5 billion
- Over 1,500 commercial fishing vessels are registered in New Zealand²

Commercial fishers target well over 100 different wild species, which include finfish, squid, shellfish, lobster, crabs, sharks, sea cucumbers and seaweeds. The most valuable species is rock lobster, with \$223 million worth exported in 2012, followed by hoki (\$195 million) and squid (\$86 million).³ In terms of tonnage, the highest is hoki with over 130,106 tonnes caught in the 2011–12 year, followed by jack mackerel (40,261 tonnes) and squid (35,207 tonnes).⁴

There are three main types of commercial fisheries within New Zealand's waters:

- Inshore (coastal) fisheries – these are located out to the edge of the continental shelf in relatively shallow waters up to 200 metres deep. They target a wide range of species such as paua, rock lobster, snapper, terakihi, scallops and red cod flatfish
- Deepwater fisheries – these are located within the deeper waters of the EEZ and include many of the commercially important species such as squid, hoki, orange roughy, ling, hake and jack mackerel
- Migratory species fisheries – these are target species that travel into and beyond New Zealand's waters such as tunas and swordfish

Much of the deepwater commercial fishing effort is concentrated off the South Island. In 2010, approximately 60 per cent of the fish harvest came from the Chatham Rise and Subantarctic areas, including the main hake, hoki, ling, silver warehou, squid, orange roughy and oreo fisheries. A further 30 per cent was caught off the country's west coast, mainly when some stocks spawn there in winter and spring, including hake, hoki, ling and silver warehou.⁵

Coastal fishing occurs all around New Zealand, including around the Chatham Islands, with varying importance regionally and with differences in catch and method importance depending on area.

Figure 9.1 shows the general spatial pattern of commercial trawl activity.



Snapper being packed at the Leigh Fisheries factory at Leigh, Auckland
(Raewyn Peart)

In terms of industry structure, Seafood New Zealand provides overarching representation of the commercial fishing sector. It provides economic information and advice to the sector, co-ordinates industry resources, and seeks to enhance the industry's profile in the community. Currently, most commercial fisheries in New Zealand are represented under five sector representative entities, which are (in alphabetical order):

- Aquaculture New Zealand
- Deepwater Group Limited
- Fisheries Inshore New Zealand Limited
- Paua Industry Council Limited
- Rock Lobster Industry Council

For information on farmed fish species please see Chapter 10: Aquaculture.



Snapper caught by longline in the Hauraki Gulf, ready for packing and sale (*Raewyn Peart*)

Of particular environmental concern is the capture of protected species such as marine mammals, seabirds and turtles in fishing gear. Dolphins, fur seals and sea lions can become entrapped in fishing nets, with some being released successfully and others drowning. Seabirds are attracted to baited hooks in longline fisheries and can get hooked or tangled in the line and drown. Others can become entangled in trawling gear, both the net and the cables towing the net. A high proportion of marine mammal and seabird species in New Zealand are vulnerable or endangered. All except black-backed gulls have protected status.

Impacts on seabed habitats

Fishing methods which involve equipment contacting the seabed can impact on benthic habitat and the organisms which live there. Bottom trawling and dredging are the methods which have the most impacts on the seafloor. Dredging is considered more destructive to the seabed in the localised areas where it occurs. Trawling has a higher cumulative impact within New Zealand's marine area, as it is undertaken over much larger areas. During 2008, 85,222 square kilometres of seabed were trawled, which equates to around a third of New Zealand's land area, and two per cent of the country's entire marine environment.⁸ Other methods which involve the use of equipment on the seabed include seining, longlining and potting, although the effects are significantly less than those caused by dredging and trawling.

Disruption of marine ecosystems

The reduction of targeted fish populations may have significant ecological flow-on effects. These impacts are not well understood. Large-scale purse seining of inshore species such as kahawai, jack mackerel and trevally may have reduced food for seabirds such as white-fronted terns and Hutton's and fluttering shearwaters which typically feed in association with schooling fish. These larger schooling fish drive the smaller fish to the surface where they can be captured by these seabirds.

Squid is an important part of the diet of many bird species, toothed whales, dolphins, seals and sea lions. Harvesting large quantities of squid may have an impact on these species, but the effects are poorly understood.⁹

Reductions in targeted fish populations can impact on the entire food web. When top predators are removed, their prey species may increase in number, in turn increasing predation further down the food web. This is thought to be the cause of 'kina barrens'.¹⁰



Seabirds often rely on larger schooling fish to drive smaller fish near to the surface where they are accessible, such as in this 'boil up' in the Hauraki Gulf (Raewyn Peart)

Collecting and diving

Divers gather kina and paua by hand for commercial sale. Cockles are harvested by hand using mechanical digging and raking. This has minimal impacts other than on the targeted species. It enables only target species to be harvested and undersized specimens to be left undisturbed. Mechanical methods of collection are more indiscriminate and can impact on habitats and other organisms living on the seafloor.

Squid jigging

Jigging for squid involves using powerful lights, which illuminate seawater at night and attract the prey of and the squid themselves to the area under the boat. Lures are attached at short intervals along lengths of line that are dropped over the side of the fishing boat. The lures are 'jigged' up and down by mechanically-operated pulleys. These entice the squid which are caught on the hooks and hauled onboard. This is a relatively low-impact fishing method that only catches the targeted species. If lines or jigs are lost at sea, they can become a hazard for other marine species, which may become entangled in the fishing equipment.

Trapping and potting

Traps and pots are used to catch rock lobster, blue cod, octopus and crabs. They are usually constructed from steel mesh. A baited pot is lowered to the seafloor and connected by a line and float. The targeted species are attracted to the bait and enter the pot or trap. Once inside they are unable to escape. The pots are lifted by winch back onto the boat and the catch removed. This is also a relatively low-impact fishing method, although fragile species located on the seafloor such as soft corals, sponges and bryozoans can be damaged if heavy pots are dropped on top of them. Undersized and non-target specimens can be returned to the sea relatively unharmed. Some seals and shags can drown as a result of being caught in rock lobster pots. Marine mammals can become entangled in pot lines.



Crayfish are commonly caught by the use of pots (Daniel Sharp)

Trolling

Trolling is mainly used to catch species such as tuna and yellowtail kingfish. A boat slowly drags a set of hooked lures through the water. Weights can be attached to the lines so that they sink in order to target fish at lower depths. This is a low-impact and selective method of fishing. Few non-target species are caught and it has no impact on benthic habitats. Occasionally fur seals and seabirds can become hooked on the lures but this is a rare occurrence.

Longlining

Longlining is used to catch a number of species, including:

- High-value inshore species such as snapper and groper
- Deeper-living species such as ling and bluenose
- Migratory species such as yellowfin and southern bluefin tuna

Longlining involves setting a long main line, which has numerous branch lines or snoods connected to it, each containing a baited hook. Longlining can be used to target fish near the surface, in mid-water or on the bottom. Longlines (between 80 to 150 metres deep) are used to target tuna and other large schooling and migratory species. They can be many kilometres long and each line can have thousands of hooks. Surface longlines are held up by a series of floats. The line is gradually let out as the fishing boat slowly moves ahead. The entire process of setting and hauling in a surface longline can take up to 12 hours. Bottom longlines are used to target the likes of ling, snapper, hāpuku, bluenose and school shark. They are shorter than the surface lines and are

Trawling

Trawling catches the greatest quantity of fish in New Zealand and the majority of tonnage in deepwater fisheries. It involves one or occasionally two boats towing a net through the sea. Steel cables attach the net to the specially designed trawling vessel. The mouth of the net is spread open by two doors. The fish enter the net through the mouth and are forced down to the smaller cod end. The mesh size helps to avoid catching small fish, although as the net fills with fish the meshes can get blocked by fish already in the net.

Trawling can be undertaken in mid-water or along the seafloor. Bottom trawling is used to catch species such as orange roughy, ling, hake, squid and scampi. The net is often protected along the leading edge by rubber rollers. Mid-water trawling is also used to catch a range of species including hoki, jack mackerel and southern blue whiting.

This is a relatively indiscriminate method of fishing and can result in a large amount of bycatch. Fur seals, sea lions, dolphins and seabirds can become caught in the trawling equipment and drown. Fur seals, sea lions and common dolphins often deliberately swim into trawl nets to feed on the fish which have been caught.

Bottom trawling disturbs the seabed, and can damage or destroy the organisms and communities that live there. Heavy chains or rollers dragged along the seafloor crush many benthic organisms including sponges, bryozoans, seaweeds and soft corals. In addition, such equipment can flatten rocky and biogenic outcrops which provide refuge for juvenile species of fish. Trawling can also suspend sediment in the water column which can disrupt filter feeders.

Repeated trawling of an area can reduce its overall diversity and productivity. Recent research comparing trawled and untrawled areas around Separation Point, between Tasman and Golden Bays, identified significant differences. The trawled area showed a reduction in the size, biomass and productivity of marine life in the area, with the original shell-gravel substrate transformed to silt-mud, and larger long-lived species replaced by smaller opportunistic species.¹¹

A risk assessment exercise looking at activities within the EEZ and extended continental shelf was completed for the Ministry for the Environment by NIWA in 2012. This identified bottom trawling as creating the third most significant risk to New Zealand's marine environment after ocean acidification and increasing sea temperatures from climate change.¹²

Dredging

Dredging is used to harvest oysters and scallops. It involves dropping a steel dredge to the seafloor and then pulling it along the sea bed. Dredging can also be harmful to benthic habitats. The heavy dredge scrapes the seafloor, crushing benthic organisms and flattening the seabed profile. It also suspends sediment in the water column, which reduces light levels, smothers remaining benthic organisms, and clogs up the apparatus of filter feeders. The dredge can also damage a significant proportion of the target species, such as scallops.



Deepwater trawling vessel berthed at Nelson (Raewyn Peart)



Scallop-dredging vessel, Whitianga (Raewyn Peart)

Decision-making framework

Fisheries Act

The Fisheries Act governs fisheries management throughout New Zealand's territorial sea and EEZ. The purpose of the Act is 'to provide for the utilisation of fisheries resources while ensuring sustainability'.¹³ There are three main categories of fishing activity which are managed under the legislation: commercial, customary non-commercial (discussed in Chapter 4: Kaitiakitanga) and recreational (discussed in Chapter 16: Marine recreation and tourism).

Fishing activity is treated as commercial if the fish are taken for the purpose of sale.¹⁴ Most commercial fisheries are managed under the QMS which first requires the identification of a fish stock. A fish stock may include a single species such as snapper, or occasionally several species as is the case with the flatfish stock, which includes eight different species. A stock comprises the population of such a species or species group within a defined area called a 'quota management area'. A single species may be managed as a number of stocks. There are generally between one and 10 quota management areas per species.

Quota management areas

The boundaries of these areas differ between species and are designed to facilitate the management of each stock. For example, the quota management area for Flatfish 1 (FLA1) extends over the entire northern half of the North Island, whereas the quota management area for Snapper 1 (SNA1) extends only over the northern east coast (see Figure 9.4).

Once a quota management area is established, it can only be changed with the agreement of the owners of at least 75 per cent of the affected quota, or if the Minister is satisfied that it is necessary to ensure sustainability.

Total allowable catch

Under section 13 of the Act, the Minister is required to set a total allowable catch that maintains each fish stock at or above a level that can produce the 'maximum sustainable yield'. Maximum sustainable yield is defined in section 2 of the Fisheries Act as:

the greatest yield that can be achieved over time while maintaining the stock's productive capacity, having regard to population dynamics of the stock and any environmental factors that influence the stock.

The total allowable catch indicates the maximum amount of fish which can be taken by the combined commercial, customary and recreational fishing effort each year from specific stocks, as well as making an allowance for other sources of fishing-related mortality including illegal fishing.

Where a fish stock is below the estimated size required to generate its maximum sustainable yield (called B_{MSY}), the Minister is required to set a total allowable catch which will enable the stock to increase to a level at or above B_{MSY} . The time period within which the stock should be able to rebuild to this level is undefined, but is to be a 'period appropriate to the stock and its biological characteristics' having regard to relevant 'social, cultural and economic factors'.

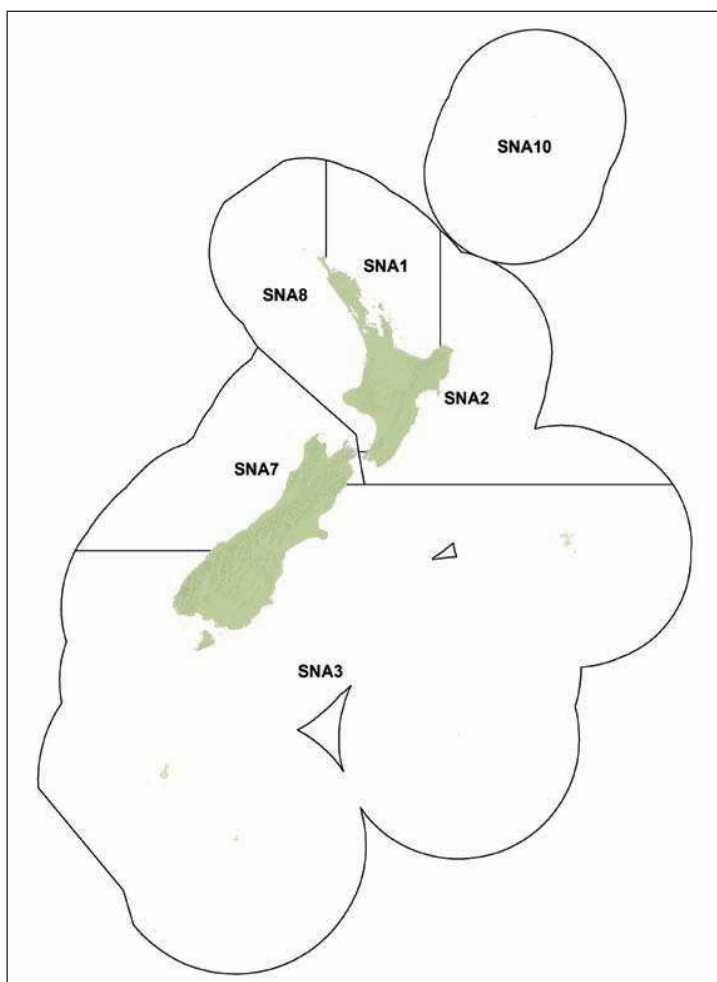


Figure 9.4 Map of quota management areas for snapper
(Ministry for Primary Industries)

entitlements are not held. When fishers have insufficient catch entitlements to cover their catch, they are required to pay to the Ministry for Primary Industries the deemed value of the excess fish. The level at which deemed values is set is important. If they are too high, they will encourage fishers to (illegally) discard excess fish. However, if they are too low, they will fail to provide an incentive for fishers to acquire sufficient catch entitlements, or to keep their catch within their allocation, and therefore their combined catch within the total allowable commercial catch.

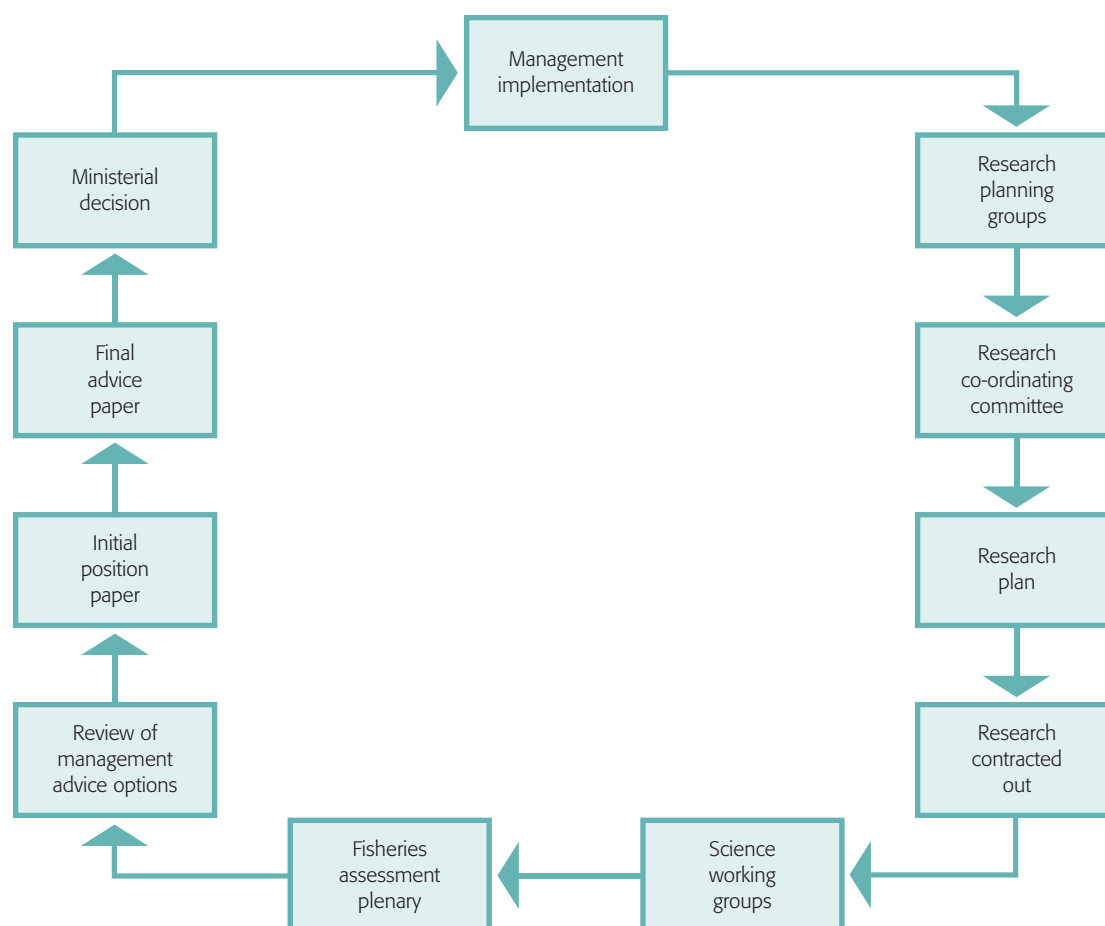


Figure 9.5 Management framework for commercial fishing

Management processes

National Fisheries Plans

The Fisheries Act enables the Minister to approve fisheries plans, which must be taken into account by the Minister when setting sustainability measures. The Act is silent on who may prepare a fisheries plan, but in recent years the Ministry has taken on this function. The plans are not mandatory and may relate to fish stocks, fishing years, or areas.

The Act provides a broad statutory framework for the preparation of fisheries plans:

- Plans may relate to fish stocks, fishing years and/or areas
- Plans may include fisheries management objectives, strategies to achieve the objectives, and/or performance criteria to measure the achievement of the objectives amongst other things
- Plans are prepared within the overriding purpose of the Act which is 'to provide for the utilisation of fisheries resources while ensuring sustainability'¹⁹

The role of the working groups is advisory only. Membership of the working groups is open to all interested parties and usually includes a range of representatives of fishing interests and environmental groups.

Several working groups focus on the assessment of fish stocks. They each prepare a report, which includes information on fishing interests in the stock, sustainability issues and estimates of stock size. These reports are incorporated into a fisheries assessment plenary report.²⁴ Two plenary reports are produced each year:²⁵

- The May Fisheries Assessment Plenary Report summarises fishery, biological, stock assessment and stock status information for 82 of New Zealand's commercial fish species or species groups in a series of working group or plenary reports. Each species or species group is split into 1–10 stocks for management purposes.
- A mid-year plenary is produced each November for species that operate on different management cycles and includes 18 working group and plenary summaries for highly migratory species, Antarctic toothfish, rock lobster, scallops and dredge oysters

More recently, working groups have focused on protected species and environmental effects and contributing to the Aquatic Environment and Biodiversity Annual Review. This provides a summary of interactions between fishing and the marine environment and other relevant information for fisheries management. The review also assesses the current status against specified targets or limits.²⁶

Fisheries management advice

Every year the Minister reviews a small number of fish stocks and makes a decision on whether to change the management measures in place. This can include the amount of fish that can be harvested and any restrictions on bycatch and the manner and locations in which fishing is undertaken. The processes to undertake these reviews are called 'sustainability rounds'.

Currently there are two sustainability rounds a year (called the April or October round), resulting in new management measures being in place by 1 April or 1 October. Of the stocks that are reviewed each year, most are reviewed in the October round, but rock lobster, southern blue whiting and some shellfish species are reviewed in the April round. Ministry officials undertake a priority-setting process that determines which stocks will be included in the sustainability round.



Management decisions which affect fishing activity, such as the set netting shown here at Ōmaha beach, result from the result from the sustainability and/or regulatory rounds undertaken and/or Regulatory Rounds undertaken by the Ministry of Primary Industries (Raewyn Peart)

- 'Where existing mitigation measures are impractical or of limited effectiveness in reducing the mortality of New Zealand seabirds, new or improved mitigation measures have been sought and where identified are under development for all priority fisheries or fishing methods...'

The development of this National Plan of Action has been a collaborative effort involving the Ministry for Primary Industries, the Department of Conservation, the fishing industry, tangata whenua and environmental groups. A risk assessment approach is being used to guide management priorities, an online database provides more detail about seabird captures, and an advisory group monitors the implementation of the Plan.

Iwi and recreational fora

In order to develop a closer working relationship with some key groups involved in fisheries management, the Ministry for Primary Industries has facilitated the establishment of a number of iwi and recreational fora. These fora meet regularly (every three months), and provide the opportunity for the participants to provide feedback to Ministry staff, and also for the Ministry to update participants on current and upcoming issues and processes. There is also an annual national meeting of all regional recreational fora.

Observer programme

The Ministry for Primary Industries' observer programme fulfills an important role in fisheries management through the collection of data, assisting with stock assessment, and monitoring of the environmental impacts of fishing. Observers are trained using the NZQA-approved Observer Qualification. Data is captured using electronic laptops or tablets. When observers return from trips, the information can be downloaded and reviewed during debriefing.

The Department of Conservation is a partner in the Ministry's observer programme with fisheries observers acting as the Department's 'eyes and ears at sea'. This work is an important part of the Conservation Services Programme, which is managed by the Department and which aims to ensure that 'commercial fishing is undertaken in a manner that does not compromise the protection and recovery of protected species in New Zealand fisheries waters'.³¹ The Programme monitors the impacts of commercial fishing on protected species, researches endangered species populations, and identifies ways to limit bycatch. The commercial fishing industry is legally required to contribute to the costs of research provided by this programme. These levies have helped to fund a number of measures, including the development of potential mitigation devices, provision of advisory officers and the development of bycatch management measures.³²



The Conservation Services Programme aims to ensure that commercial fishing is undertaken in a manner that does not compromise protected species, such as the common dolphins shown here (*Raewyn Peart*)

Issue	Best practice considerations
Targeted fish stock	• Is the size of the exploited population such that it retains its productivity, provides margins of safety for error and uncertainty, and retains its capacity for yields over the long term? • If the fish stock is depleted, is it able to recover, and if so are management measures in place to enable its recovery within a reasonable time frame? • Is fishing conducted in a manner to ensure that the age or genetic structure, or sex composition, of the stock is not changed to a degree that impairs reproductive capability?
Bycatch species	• Are any threatened or endangered species affected by the fishery such as sea lions, Hector's and Maui's dolphins and seabirds? If so, are effective measures in place to ensure that any negative effects on populations are mitigated? • Are any other protected species affected by the fishery such as seals, dolphins, whales or seabirds? If so, are effective measures in place to ensure that the effects are minimised? • Is there any significant bycatch of untargeted species including juveniles? If so, are effective measures in place to reduce this? • Does management ensure recovery of depleted populations? • Are fisheries impacts on bycatch species reversible over a reasonable time period (20–30 years)?
Associated or dependent species	• Is the fishery depleting food sources important to other species such as seabirds or marine mammals and is this having a detrimental effect on these populations? • Does management ensure recovery of depleted populations? • Are fisheries impacts on associated and dependent species reversible over a reasonable time period (20–30 years)?
Biodiversity	• Is the fishing activity threatening marine biological diversity at the genetic, species or population levels? • Is the fishing activity enabling the natural functional relationships among species to be retained? Is it leading to trophic cascades or ecosystem state changes? • Is the natural structure and function of the marine ecosystem being maintained, including its biodiversity and productivity?
Habitats/ ecosystem approach	• Are habitats of particular significance for fisheries management well-managed or protected, including addressing land-based impacts? • Is fishing activity impacting on habitat important for juvenile fish? • Are changes to the marine habitats caused by fishing activity reversible over a reasonable time period (20–30 years)? • Are sufficient representative and unique benthic habitats protected from trawling and dredging activity, including vulnerable marine ecosystems? • Is fishing activity reducing the resilience of marine ecosystems to climate change-related impacts? If so, are effective measures in place to reduce these impacts?
Information and uncertainty	• Has the decision been based on the best available information and has the adequacy and reliability of the information been assessed? (see Fisheries Act, section 10) • Has a recent assessment of the biological status of the fish stock and impacts of the fishery been conducted? • Is sufficient research being undertaken into the environmental impacts of the fisheries to enable informed management decisions to be made? • Is sufficient research being undertaken to understand the important trophic linkages between the fish stock and other marine species? • Where information is inadequate, has a precautionary approach been adopted? • Has uncertainty in the available information not prevented action being taken when the sustainability of the fisheries resource or the health of the aquatic environment is at risk?

- **Tori lines** – a set of streamers are hung off a line located above the fishing lines or nets. This is designed to scare seabirds away from baited hooks or dangerous cables.
- **Blue-dyed bait** – research has shown that birds are less interested in blue-dyed bait
- **Weighted lines** – this helps the bait to sink fast meaning the birds have less time to get tangled up in the hooks or lines
- **Night setting** – there are usually less feeding birds around at night. However, there are exceptions, such as black petrels feeding on bioluminescent squid during darkness hours.
- **Bird baffle** – booms are attached to the back of a vessel on either side and swing plastic tubes or cones that are tied onto the booms, essentially scaring the birds away
- **Move to another fishing spot** – some commercial skippers have found it is best to stop setting lines and to move to another spot, especially if the birds are hungry and/or at-risk species

The development of vessel-specific seabird risk management plans can help fishers to consider the overall risk to seabirds of the vessel's operation and to identify the most effective ways of mitigating the risks.

There are some mandatory government measures in place requiring the use of certain seabird mitigation technologies. For example, since 2006 regulations have been in force requiring fishing boats over 28 metres in length to carry and deploy a seabird scaring device if using a trawl net, but currently no regulations apply to smaller inshore trawlers.³⁶

A unique conservation partnership called Southern Seabird Solutions is an example of some of the collaborative innovative work underway between a wide range of stakeholders to reduce the impact on seabirds from commercial longlining. This example is discussed in the case studies below.

Reducing marine mammals bycatch

The most effective way of reducing bycatch of marine mammals is to avoid using set and trawl nets within their habitats. Set nets are a particular risk to coastal dolphins including Hector's, Maui's and common dolphins. While less of a risk than set nets to some species, trawl nets are a particular risk to Hector's and Maui's dolphins, common dolphins and New Zealand sea lions. Set net prohibitions have been put in place along areas of the coast to protect Hector's and Maui's dolphins.³⁷ There has also been a recent prohibition on set netting attached to the marine mammal sanctuary located off the Taranaki Bight.³⁸

Voluntary marine mammal operational procedures

The Deepwater Group represents the owners of quota for deepwater and middle-depth fish species in New Zealand. The group has developed a set of voluntary operational procedures which relate to marine mammals. These are given to all skippers of vessels involved in the jack mackerel fishery (where there is a particular conflict with common dolphins) and annual briefings are carried out to reinforce the message. The recommended procedures include:³⁹

- The officer on watch confirms that the area is visibly clear of dolphins before the net is deployed
- If the vessel turns during trawling, it must haul the doors to the surface so the end of the net is closed before making the turn
- Nets are not to be launched or retrieved between 02.30 and 04.30 hours
- A dolphin dissuasive device (acoustic pinger) is to be deployed on night tows

Experiments with acoustic alarms on nets overseas showed promising results initially, but have not yet proved to be practically applicable. Other measures that can help to reduce, but not eliminate, the risk of marine mammal bycatch include:

- Increasing the depth of the trawl net in the case of conflicts with common dolphins
- Adopting new technologies to enable animals to escape from trawl nets, such as the Sea Lion Exclusion Device (see Figure 9.9)
- Avoiding dumping offal when the net is being shot (launched) or hauled⁴⁰
- Keeping watch for marine mammals, and if any are sighted, not deploying a net and leaving the area
- Hauling the net in as quickly as possible
- Recovering any accidentally lost fishing gear⁴¹

- Avoiding trawling in areas which have not already been impacted
- Minimising the size of the trawl footprint
- Increasing the time period between repeated trawling of the same area to enable recovery
- Excluding trawling activity from vulnerable marine ecosystems and areas of ecological benthic importance such as coral and sponge gardens, shellfish beds, seagrass beds, hard and shelly substrates, seamounts and underwater features
- Actively developing new trawl technology which reduces benthic impacts

Within the EEZ, 17 areas around underwater features (including some surrounding seamounts) and 17 Benthic Protection Areas have been closed to trawling to protect benthic habitats. Other forms of fishing are allowed in these areas.

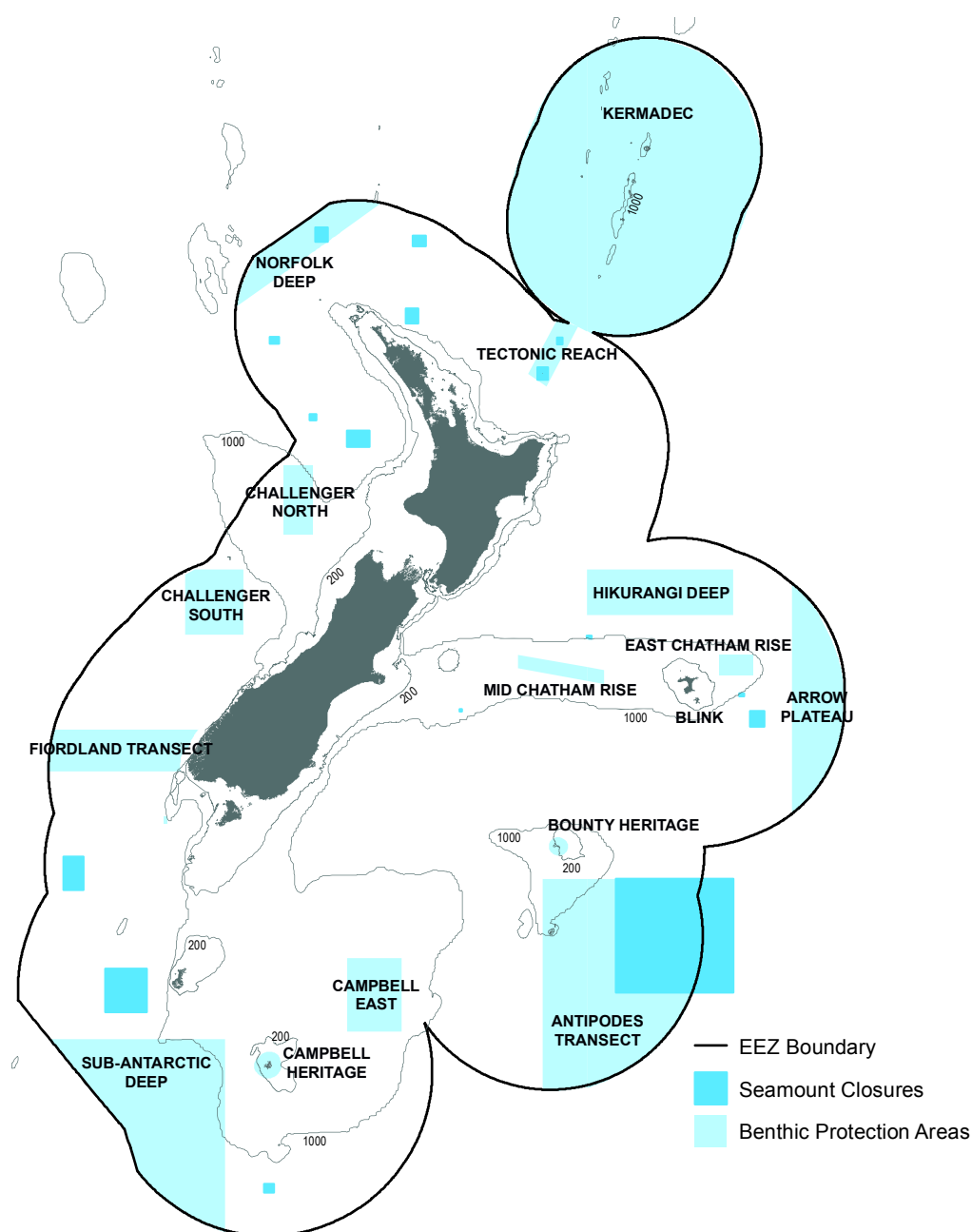


Figure 9.10 Benthic protection areas (Deepwater Group Ltd, 2014)

Southern Seabird Solutions

In July 2002, Southern Seabird Solutions was formed at a workshop attended by parties with a shared interest in the development and implementation of sustainable and responsible fishing practices, in recognition of the need to improve management of the impact from commercial and recreational fisheries on seabirds.

The concept was officially launched in November 2003 as a registered charitable trust, in partnership with the seafood industry, Department of Conservation, Ministry for Primary Industries, WWF New Zealand and Te Ohu Kaimoana. Representatives from these groups make up the 15 person board of trustees and management committee. Its aim is to work with commercial fishing fleet crew and skippers, as well as recreational anglers, to minimise the harm caused to seabirds through fishing. The initiative has generated much support from some environmental groups, ecotourism operators, fishing gear retailers and seabird ecologists.



Flesh-footed shearwaters, shown here near the Mokohinau Islands in the Hauraki Gulf, are susceptible to being caught on longline hooks (Raewyn Peart)

Underpinned by a motto of 'conservation through cooperation', Southern Seabird Solutions empowers and encourages new attitudes and behaviours amongst fishers, and advocates the adoption of practical fishing practices to reduce accidental seabird capture. The group acknowledges that there is no single solution to the problem, and it works collaboratively with partners and supporters to educate and share ideas through workshops and awareness campaigns. Fishing technology designers help to refine solutions based on feedback from industry professionals and those whose hands are literally on deck.

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Southern Seabird Solutions offers a range of useful tools and resources to professional skippers and crews including:⁴⁵

- Inshore finfishing and seabird workshops – these are tailored to fit the local fleet and help crew learn about seabird lifecycles and best practice measures. In 2012 there were over 160 skippers and crew involved in these workshops.
- Commercial fishing and seabirds mitigation DVDs (including: *Sharing Worlds* – seabirds and fishing Otago/Southland; *Sharing the Hauraki Gulf* – fishers and black petrel; and *Seabird Smart Fishing* – deepwater trawl and longline)
- Seabird safe release (from hook events) poster and guidelines
- International mitigation mentor programme – including 'cross pollination' of ideas with South America and South Africa and some skipper swaps with South American crews
- 'New Zealand Seabirds Long Haul Experts' booklet – providing a range of facts about different seabirds found in New Zealand⁴⁶
- Seabird Smart Awards – recognising people from the industry who demonstrate passion and commitment for a seabird smart fishing culture



Southern Seabird Solutions is keen to see New Zealand fishers become world leaders in the development and use of mitigation measures to avoid seabird mortality from fishing. It also hopes to be able to transfer technology and knowledge between countries, and to be part of a movement that engages fishers, governments, agencies and individuals worldwide.

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